Material Science

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1 Introduction to Material Science

Historical Perspective

- Most people probably do not realise how deeply materials permeate the society. Every aspect of one's daily life is impacted by materials in some way. This includes how people get about, what they wear, how they communicate, how they have fun, and even how their food is grown and prepared.
- The availability of appropriate materials has been crucial to the creation of numerous technologies that improve the quality of life. Understanding how a certain material works have led to incremental technological improvements.
- It is the degree of material sophistication that has been used to categorise early civilisations known as the Stone, Bronze, and Iron Age.
- Cars, for instance, would not exist if cheap steel had not been readily available or if a suitable alternative had not existed. Modern gadgets depend heavily on semiconducting materials for several critical components.

Material Science

- Material science deals with investigating the relationship between the structure and properties of the material. In contrast, material engineering deals with designing structures by using materials on the basis of structure property correlations.
- The study of how the composition and atomic arrangement of matter affects its behaviour is the focus of material science, an area of engineering. Its goal is to create novel materials and enhance current ones so that they may be used in engineering applications.

Material

- It is the stuff with the help of which one can design something. It consists of matter.
- Every engineer and scientist in practise, whether they are specialised in mechanical, automobile, civil, metallurgy, chemical, or electrical fields, eventually face a design challenge with materials. A transmission gear, constructing a superstructure, or a computer chip are all possible examples. Those who study and develop materials are called material scientists and engineers.
- When dealing with a materials issue, it is often a matter of narrowing down the hundreds of potential solutions to find the one that is best suited. The ultimate verdict is often based on various factors.
- A classic example includes the relationship between strength and ductility. Typically, a material with great strength would have low ductility. In such a circumstance, it may be important to settle on a middle ground between two or more desirable qualities. The needed qualities of the materials are determined by the in-service circumstances, thus characterising them is the first step. Materials with the maximal or optimum combination of characteristics are very uncommon. As a result, this could be essential to sacrifice one quality for another.
- Degradation of material qualities throughout the service process is a second factor to consider when making a final choice. Exposure to high temperatures or corrosive conditions, for instance, may significantly diminish mechanical strength.

Classification of Materials

- Engineering materials can be classified as follows.
  - Metals and alloys
  - Ceramics and glasses
  - Organic polymers
  - Composites
- While certain substances do not fit neatly into any category, the vast majority of substances can be categorised.
- Nano engineered materials, bio materials, semiconductors, and smart materials are

all examples of sophisticated materials employed in technological contexts.

- Metals have a distinct look and feel, can be permanently reshaped, and are excellent thermal and electrical conductors. Alloys are made by combining several metals.
- Organic polymers have a greater level of plasticity, are lightweight, and chemically inert.
- Ceramics are inorganic materials that are not metallic; they are brittle and have excellent insulating characteristics, both thermally and electrically.

Structure

- It is the arrangement of components. The structure of the materials can be categorised in the following ways.
- Macrostructure: it can be observed with a naked eye or under low level magnification. The internal symmetry of atoms can reflect in the macrostructure of material.
- Microstructure: it is the structure which is observed with the help of an optical microscope without loss of information. With an optical microscope magnification of about a 1500 times linear is possible. The structures observed by a microscope have a very high magnification and resolution than the optical microscope is known or substructure. With an electron microscope magnification, for example, of 1000000 times linear is possible.
- Crystal structure: it deals with the arrangement of atoms in crystalline solids. The arrangement of atoms in a crystal can be described in terms of a unit cell.
- A unit cell is defined as the smallest group of atoms by repeating it in all the directions, a crystal is formed.
- Electronic structure: electronic structure deals with the arrangement of electrons in various orbits of the atom (valence orbit).
- Nuclear structure: it is studied with the help of spectroscopic techniques such as given below.
    - Nuclear magnetic resonance (NMR)
    - Mossbauer studies

Property

- It is defined as the material trait in terms of kind and magnitude of response to an externally imposed stimulus (excitation).
- Material properties can be of the following types.
    - Mechanical
    - Electric
    - Magnetic
    - Thermal
    - Optical
    - Deteriorative

Semiconductors

- Semiconductors are materials that lie between those that are electrically conductive, namely metals and metallic alloys, and insulators, namely ceramics and polymers with characteristics that are somewhere in the middle.
- These materials' electrical properties are very susceptible to the effects of even trace amounts of impurity atoms, whose quantities may be manipulated in exceedingly fine spatial domains.

New and Improved Materials Known as Advanced Materials

- These high performance materials are either newly created or old materials with improved qualities. They may be made from any material including metals, ceramics, and polymers, and are often rather pricey.
- Electronic devices such as camcorders, DVD players, computers, fibre-optic devices, aeroplanes, and military rocketry are all examples of high technology as they all work or function utilising very elaborate and sophisticated concepts.

Smart or Intelligent Materials

- Many of the technologies people use today will be profoundly impacted by the newly formed field of 'smart' also fondly known as 'intelligent' materials.
- Smart materials have the ability to perceive and react to varying environmental conditions in predetermined ways, characteristics shared with live creatures.

Practice Exercise

1. Consider the following statements. **[UPSC-ESE 2018]**

I. Material science deals with the strength and stiffness behavior of components (buildings, machines, vehicle facilities) based on their response to imposed stresses (forces, moments, torque, etc.).

II. Material properties are dependent on their microstructure and response to force fields and surface interaction.

Which of the above statements is/are correct?

(A) I only
(B) II only
(C) Both I and II
(D) None of the above

2. Consider the following statement regarding the material.

I. In microstructure external forms of the crystal may reflect the internal symmetry of the atom.

II. Electronic structure is an arrangement of different electrons in orbits of the atom.

III. A crystal is formed by a non-repeating cell in a single direction.

IV. Nuclear structure can be studied by Mossbauer studies.

Which of the above statements are correct?

(A) I, II, and III
(B) I, II, and IV
(C) I, III, and IV
(D) II, III, and IV

Answer Key

1. (B) **2.** (B)

2 Atomic Structure and Chemical Bonding

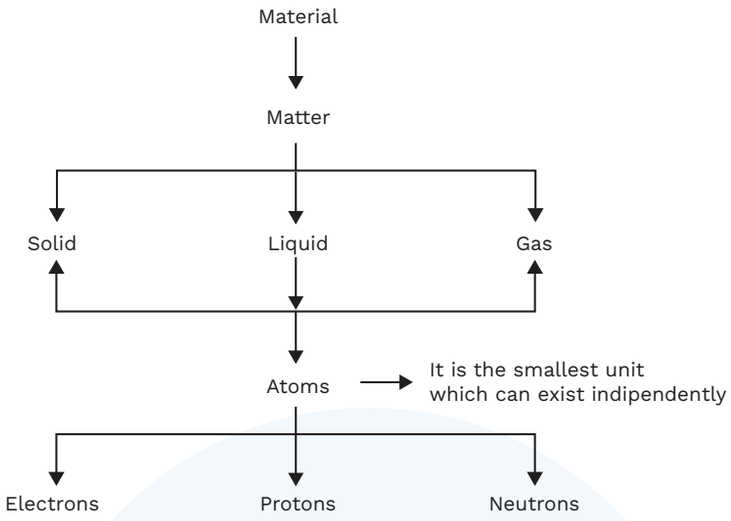


Fig. 2.1

Charge of an electron = -1.6×10^{-19} C
Charge of a proton = $+1.6 \times 10^{-19}$ C

Atomic Number (Z)

Z = number of electrons or number of protons in a neutral form
Mass of a proton = 1.66×10^{-27} kg = Mass of neutron
Mass of an electron = 9.1×10^{-31} kg = $\frac{1}{1836}$ times mass of the proton

Atomic Weight

It is the standard unit of measurement for the average relative weight of an atom relative to the mass of one atom of carbon-12 (C-12) and is measured in atomic mass unit (amu).

Atomic Mass Number (A)

A = Number of protons + Number of electrons

Atomic Mass Unit

It is defined as $\frac{1}{12}$th of the mass of an atom of C-12
C-12 = 12 amu
$6p^+ + 6no + 6e^- = 12$ amu
Proton = Neutron

$6x + 6x + \frac{6x}{1836} = 12$ amu
$12x = 12$ amu
$x = 1$ amu
1 amu $\approx 1.66 \times 10^{-27}$ kg

Mole

- It is the SI unit for the mass of substance. One mole is the amount of the substance that contains as many particles, or entities as there are atoms in 12 gm of the carbon-12 isotope.
- The mass of one mole of a substance in grams is called molar mass.
  1 mole = 6.023×10^{23} atoms

Avogadro's Number

Avogadro proposed that equal volumes of gases at the same temperature and pressure should contain equal number of molecules. The number of entities or atoms in a mole is known as Avogadro's number.
At constant temperature and pressure
$v \propto n$
$N_A = 6.023 \times 10^{23}$ atoms/mol

Elements

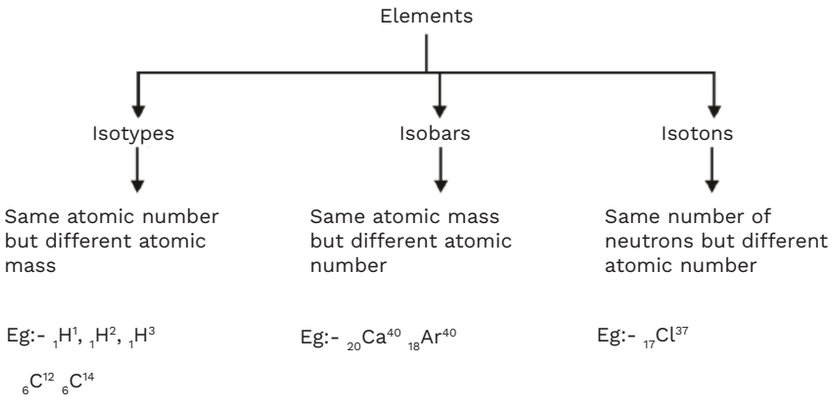


Fig. 2.2

Comparison between Alpha (α), Beta (β), and Gamma Rays (γ) Rays

Alpha Rays

- The alpha particle has a mass of 4 amu and is made up of two protons and two neutrons. Therefore, it has the largest Ionisation potential and is a positively charged particle i.e., having a charge of +2e.
- Heavier elements are the sources of ∝-particle emission. It moves at a slower rate and has little penetrating ability. Its penetrating capacity is so weak that it cannot even go through the superficial layers of human skin.

Beta Rays

- Those are some high energy electrons coming out of the nucleus. To put it simply, this particle has a negative charge (–e).
- The potential for ionisation is smaller than that of ∝-particles. The speed of travel for beta rays is close to that of light.
- These particles can penetrate lighter substances than the ∝-particle. These are capable of readily piercing human flesh but are stopped by even a thin metal sheet.

Gamma Rays

- High frequency electromagnetic waves and gamma rays have no mass or electric charge. The ionisation potential of these is relatively low.
- The penetrating strength of gamma rays is the greatest, according to the literature. That hefty coating of concrete is no match for them, and these move at the speed of light.

Atomic Theories

Dalton's Atomic Theory

- Atoms, which are tiny particles, compose all matter. In other words, atoms have neither negative nor positive states and cannot be formed or destroyed.
- Atoms of various elements mix in a pre-determined ratio to produce compounds stated by the law of chemical composition. Atoms of the same element are chemically and physically similar to one another, whereas atoms of other elements have distinct masses and other characteristics.

Thomson's Atomic Model

JJ Thomson hypothesised that an atom is spherical and is packed with positively charged stuff dispersed evenly with sufficient electrons embedded in it. Therefore, there is no charge on an atom.

Rutherford's Atomic Model

According to this theory, the nucleus is an incredibly tiny area that contains the vast majority of an atom's mass and is positively charged. Rutherford suggested the nuclear model of the atom based on the results of his renowned α-particles scattering experiment. Similar to the orbits of the planets around the sun, the electrons travel in a circular route around the nucleus.

Bohr's Atomic Model

- The followings are the presumptions upon which Bohr's hypothesis rests.
- The majority of an atom's mass is located in its nucleus, which is made up of protons and neutrons. The electrons also follow similar circular orbits around the nucleus.
- Energy is lost as radiation whenever an electron moves from a higher energy orbit to a lower energy orbit. The same is true when the electron moves in the other direction. The difference in the energies of two orbits reflects the quantity of energy transferred or absorbed.

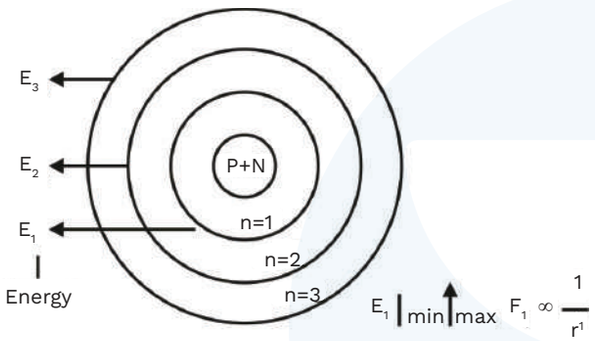


Fig. 2.3

- Only, in particular, in non-radiating orbits, known as stationary orbits, can electrons rotate. When electrons orbit along these paths, no energy is dissipated in the process. If the electron's angular momentum is not an integral multiple of $\frac{h}{2\pi}$, where h is the Plank's constant, then Bohr's result states that the electron cannot be in that orbit.

$$mvr = \frac{nh}{2\pi}$$

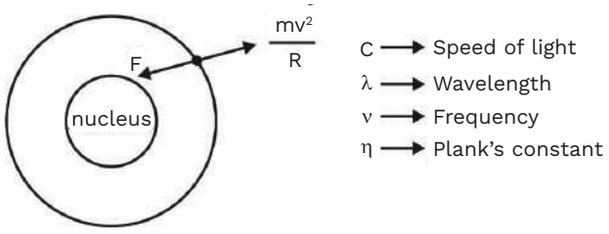


Fig. 2.4

Electron Energies Through Bohr's Model

$$F = \frac{1}{4\pi E_0} \cdot \frac{q_1 \cdot q_2}{r^2}$$

$$E = \frac{F}{q} \text{ and } E = -\frac{dV}{dx}$$

$\int Dv = -\int E. - Dv$

Potential energy (PE) = q V

Kinetic energy (KE) = $\frac{1}{2}$ mv^2

$$E_n = \frac{-13.56z^2}{n^2} \text{ ev (electron volt)}$$

n = 1, 2, 3, ...

1 ev = 1.6×10^{-19} joule

Z = atomic number of the element

ev is a unit of energy

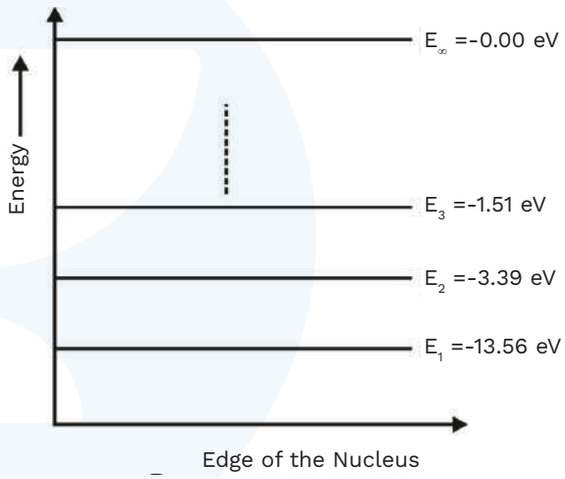


Fig. 2.5

Bohr's Orbital Radius

$$R_n = \frac{0.529n^2}{Z} \text{ A°}$$

A° = Angstrom

1 A° = 10^{-10} m

Some Significant Points

- An electron circling extremely near to the nucleus in the first shell is closely bonded to the nucleus and contains a very minimal amount of energy. So, it would be impossible to eliminate these electrons from their orbit. Also, the total energy of an electron is said to increase proportionately with the distance that separates it from the nucleus.

- But an electron in the outermost cell known as valence orbit is only weakly connected to the nucleus and has a lot of energy on its own. So, these electrons may be readily thrown out of their orbit. Because of this, valence electrons play an important role in chemical reactions and chemical bonds.

Sommerfeld's Model

According to Sommerfeld, the electron follows an elliptical orbit as it revolves. When an electron travels in an elliptical orbit, its speed fluctuates at various points in the path, leading to a corresponding shift in the electron's mass. To manage these two factors, the first two quantum numbers are applied.

Quantum Numbers

- First quantum number (n)
  - This is also known as the Principle quantum number.
  - It is represented by n = 1, 2, 3, ...
  - It gives the energy of electrons.
  - It represents shells or orbits (k, l, m, ...).
  - It gives the distance of an electron from the nucleus.
  - It is the only quantum number related to Bohr's model.
- Second quantum number (l)
  - It is also known as an angular or azimuthal quantum number.
  - It gives the angular momentum of an electron.
  - It represents subshells (s = 0, p = 1, d = 2, f = 3...) and l = 0, 1, ... (n – 1).
- Third quantum number (m_l)
  - It gives energy states in subshells.
  - There are (2l +1) values of m_l ranging from –l to +l.

    s = 1 energy state

    p = 3 energy state

    d = 5 energy state

    f = 7 energy state
- Fourth quantum number (m_s)
  - It is also known as a spin quantum number.
  - Electrons can have either clockwise or anticlockwise spin; hence spin quantum number can be $m_s = +\frac{1}{2}$ or $-\frac{1}{2}$

Modern Concept of Atomic Model or Electron Cloud Model or Wave Mechanical Model

- The modern concept is based upon quantum mechanical theory or waves mechanical model. It takes into account Heisenberg's uncertainty principle, which states that $\Delta p.\ \Delta x \geq \frac{h}{4\pi}$

  Δp = uncertainty in momentum

  Δx = uncertainty in position

  h = Plank's constant
- In this model, the likelihood of an electron's presence at a certain point around the nucleus is taken into account instead of treating the electron as a particle travelling in a series of discrete orbits. Therefore, electrons cannot have fixed, distinct orbits. In other terms, a probability distribution, or electron cloud, describes where something is.
- Atomic electrons are constantly moving in orbit around the nucleus. Thus, the following Schrodinger wave equation may be used to describe the path of an electron through the nucleus's force field.

$$\nabla^2\Psi + \frac{2m}{\text{?}}(E - V)\Psi = 0$$

Where

h = Plank's constant

ψ = complex wave function

$\psi\ \psi^*$ = proportional to electron density

m = mass of an electron

E = Total energy

v = potential energy, which is a function of position

that is, v = v(x,y,z)

$$\nabla^2\Psi = \frac{\partial^2\Psi}{\partial x^2} + \frac{\partial^2\Psi}{\partial y^2} + \frac{\partial^2\Psi}{\partial z^2} \quad \text{... (i)}$$

- To determine the dimensional distribution of electron density around the nucleus, equation (i) can provide us with the answer. Many solutions exist for equation (i), but only those in which the electrons are allowed to take on very particular energy values are physically relevant and acceptable. These permissible

energy values, or the allowable atomic electron states, are given by the answers in terms of above mentioned four quantum numbers.

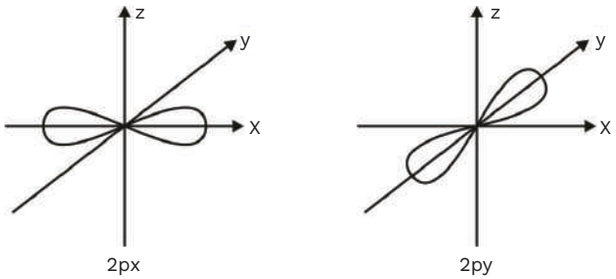


Fig. 2.6

- It is evident that the electrons in the three different 2p substates have directional features as they are scattered in directions that are perpendicular to one another. However, if the 2p state is completely saturated, the distribution will take on a spherical shape.

Molecular Orbital Theory

Case 1: Let us take the example of the H_2 molecule.

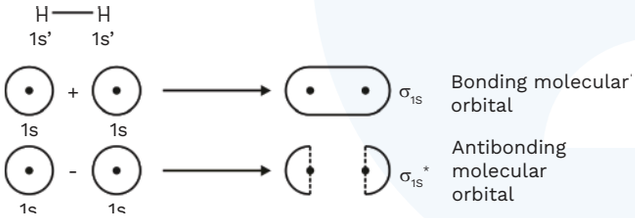


Fig. 2.7

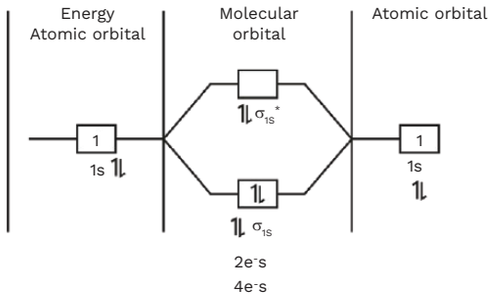


Fig. 2.8

Each electron is individually inserted into a molecular orbital, beginning with the lowest energy orbital. The stability of a molecule depends on whether there are more electrons in bonding molecular orbitals than the anti-bonding molecular orbitals.

Case 2: Consider the case of O_2 molecule.

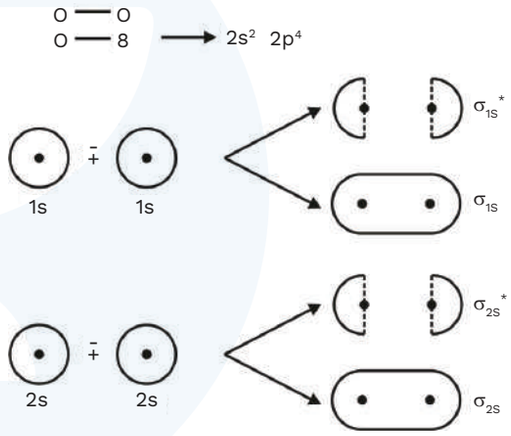


Fig. 2.9

End-to-End Interaction (Single Bond Formed)

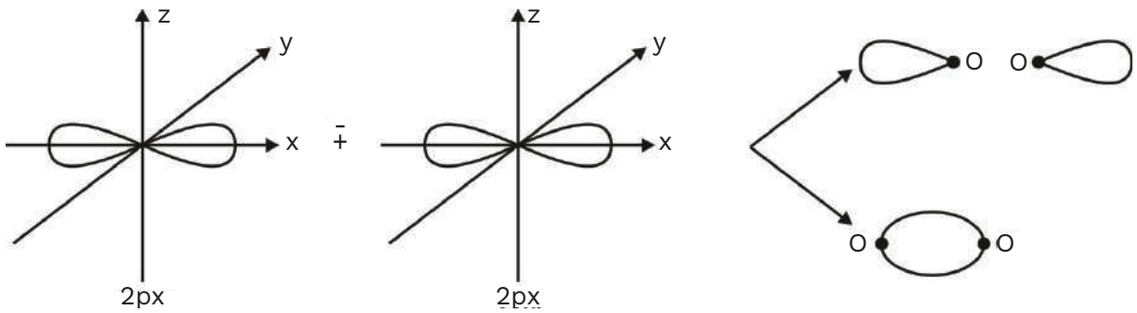


Fig. 2.10

Side-by-Side Interaction (Pie Bond Formed)

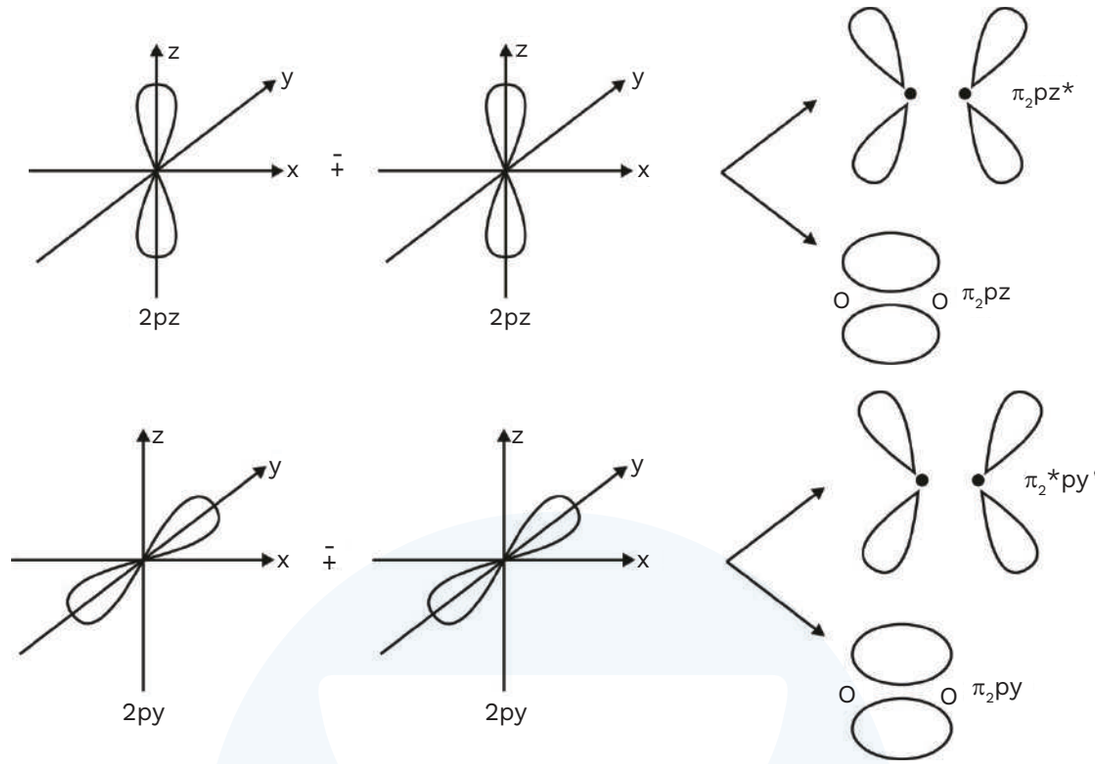


Fig. 2.11

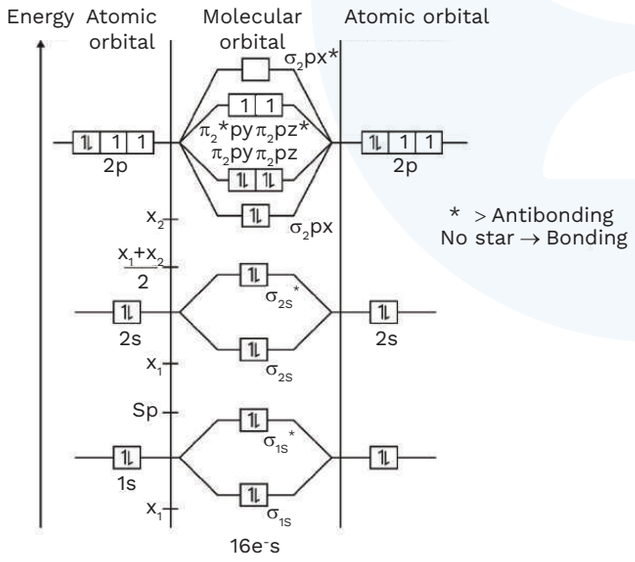


Fig. 2.12

Number of electrons in bonding molecular orbital = 10

Number of electrons in anti-bonding molecular orbital = 6

Bond order = $\frac{10-6}{2}$ = 2

When all electrons in a molecule's orbitals are coupled with one another, a phenomenon known as diamagnetism occurs. Molecular orbitals with unpaired electrons are associated with paramagnetism.

Fundamental Concepts of Chemical Bonding

- Atoms have tiny nuclei made of protons and neutrons at their centres. Moving electrons ring the nucleus.
- 'Elements' are compounds that cannot be broken down into simpler ones. The term 'atom' is often used to refer to the smallest component of an element in a chemical process.
- Atomic number (Z), or proton count, is a unique identifier for each chemical element.
- Each atom's atomic mass (A) may be written as the summation of the masses of its nucleus, i.e., protons and neutrons.
- An element's atomic weight is equal to the sum of the masses of its naturally occurring isotopes. Calculations of atomic weight may be done using the atomic mass unit (AMU).

- The mass of an electron is around 1/1836 that of a neutron, and it has a negative electric charge. The proton has a positive charge, whereas the neutron is neutral and has the same mass as the proton but no charge.
- Take an example, as both Chlorine-37 and Potassium-39 have 20 neutrons in their nuclei despite having different numbers of protons (17 and 19, respectively), they are isotones. Isotones vary from one another in terms of their chemical composition and atomic number. So, isotones are a set of atoms that share the same number of neutrons but differing numbers of protons.
- Isotopes are variations of an element that share an atomic number but a distinct atomic weight. Due to sharing the same chemical characteristics, isotopes cannot be physically separated. Similar elements may be distinguished by their isotopes, which all have the same atomic number and nuclear charge. Both 37Cl and 35Cl are stable isotopes of chlorine, whereas ^{1}H, ^{2}H, and ^{3}H are stable isotopes of hydrogen.
- If two atoms of distinct chemical elements have an equal mass number, they are isobars. Atoms that vary in atomic number but have a similar mass number, i.e., atomic weight or number of protons and neutrons, are termed isobars. In this sense, argon and calcium are isobars, as they both have a mass number of forty but differ in the number of protons and neutrons that make up their atoms.

Atomic Electrons

- Atoms of different elements have the same number of electrons, protons, and neutrons but otherwise differ in chemical properties.
- Electrons, protons, and neutrons are, thus, the essential building blocks of all matter. If this is the case, then why do different substances have varied properties? This is due to the fact that each atom is made up of a unique combination of protons, neutrons, and electrons in its own unique configuration.
- Not all electrons of an atom travel in the very same orbit. An atom's electrons have distinct homes or shells.
- The electrical configurations of the atoms of each element have been used to create a periodic table. The items in a single vertical column have a lot of characteristics.

Periodic Table

- In the periodic table, elements are arranged in groups based on the arrangement of their electrons. The elements here are laid down in seven horizontal rows, or periods, from highest to lowest atomic number.
- Group 0 elements are inert gases located at the far right of the periodic table. They have complete outer electron shells and stable inner electron configurations. Elements in groups VII A and VI A are unable to form solid crystal structures because they lack one or two electrons, respectively.
- Elements with partly full d electron states and, in certain circumstances, one or two electrons in the next higher energy shell are classified as transition metals and may be found in Groups III B through II B.
- In terms of their valence electron structures, the elements in Groups III A, IV A, and V A (Boron, Silicon, etc.) exhibit properties that fall somewhere in the middle of those of metals and non-metals.
- Halogens are a common name for the members of Group VII A (fluorine, chlorine, bromine, iodine, and astatine). Groups I A and II A are designated for the one or two electron above stable structure alkali and alkaline earth metals (lithium, sodium, potassium, magnesium, calcium, etc.).
- Elements in the same row or column have comparable chemical and physical characteristics due to their shared valence electron configuration. These characteristics evolve progressively as one goes across the various periods and down the various columns.
- Most elements in the periodic table are metals, as can be seen by inspecting the table itself. Electropositive elements are those that can donate one or more of their valence electrons to form a positively charged ion.
- In addition, elements to the right in the periodic table are more electronegative,

meaning they quickly take electrons to create negatively charged ions.

- When an atom's outer electron shell is almost full and it is less close to the nucleus, it is more likely to receive an electron from a higher orbit.

Chemical Bonding

- Chemical bonds are the interatomic forces that hold molecules together. Strong reciprocal forces of attraction bind the atoms and molecules together.
- These interactions between atoms are electrostatic in nature and are influenced by the element's electronic structure. Interatomic bonds are the strongest and most stable kind of bond.
- Intermolecular attraction forces, often known as Van der Waals forces, are responsible for the formation of secondary or molecular bonds. Molecules or atoms with entirely filled outer shells seldom absorb, lose, or trade valence electrons with one another. The bonding relies on Van der Waals's forces of attraction.
- The electrostatic repulsion felt by one atom's nucleus toward another's electrons is the source of the Van der Waals force. These ties are flimsy and easily broken.

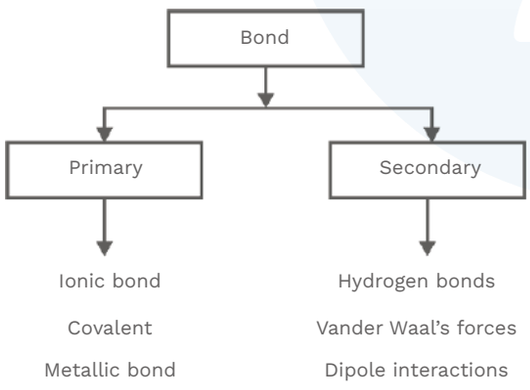


Fig. 2.13

Ionic Bond

- When positive and negative ions are brought close to one another, they create an ionic bond owing to their mutual attraction.
- Ions are created when atoms exchange or lose electrons in an effort to maintain a stable electronic configuration in their outer shells. In general, electrons are lost by electropositive components and gained by electronegative ones.
- Sodium contains one electron in its outer shell, and it has a strong tendency to lose it, turning it into a positively charged ion (Na^+). Non-metal chlorine contains seven electrons in its outer shell and tends to acquire one electron for stable configuration, making it a negative ion (Cl^-).

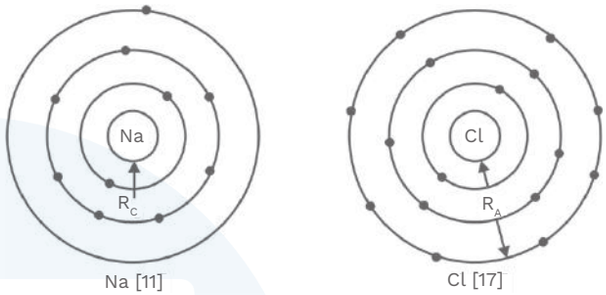


Fig. 2.14

- Glued together by electrostatic repulsion, positive and negative ions attract one another and form bonds. A possible chemical equation for the reaction is as follows.

  $Na \rightarrow Na^+ + e^-$

  $Cl + e^- \rightarrow Cl^-$

  $Na^+ + Cl^- \rightarrow NaCl$
- Ionic bonds are unidirectional, and they form compounds that have general characteristics as given below.
  - High strength
  - High melting and boiling points due to strong electrostatic forces between atoms
  - Have brittleness
  - Crystalline in nature
  - Electrically non-conducting
  - High hardness

Covalent Bond

- Covalent bonds are created when electrons are transferred between atoms. The atoms of many non-metallic elements create covalent connections because their outer electron orbits are often not entirely filled. Hydrogen, chlorine, carbon, and oxygen are all examples of these elements.

- Chlorine contains seven electrons in its outer shell. When you bring together two chlorine atoms, they each try to acquire an electron such that their electrical configurations cancel out. Two chlorine atoms share two electrons, creating a stable diatomic molecule and achieving this goal.

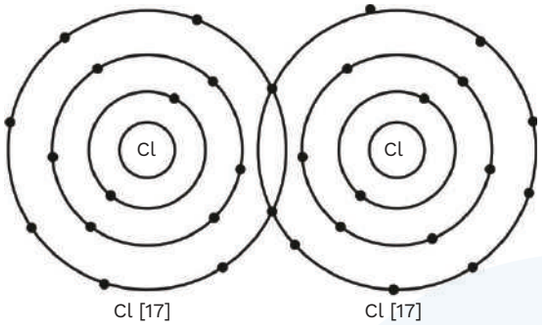


Fig. 2.15

- Covalent compounds may exist as solids, liquids, or gases, and their bonds are always directed. These general features apply to all compounds with covalent bonds.
  - They have a high boiling point and melting point.
  - Covalent bonds, like ionic bonds, often result in very sturdy materials.
  - Tin, with its weak covalent link, is electrically conductive, whereas silicon, germanium, and diamond are semi-conductive, and diamond is electrically insulating.
  - Due to the directed nature of covalent bonding, covalent solids do not form closed pack formations.
  - For the atoms inside the material to move (deform), a certain link must be broken, making the substance fragile.

Dispersion Bonds

- The electrons in a symmetrical molecule orbit the nucleus at a constant rate because they are all evenly spaced apart. Symmetrical molecules have identical positive and negative charge centres.
- However, this holds false in molecules when electron distribution is asymmetrical with respect to the nucleus. This results in a movement of the centres of positive and negative charges.

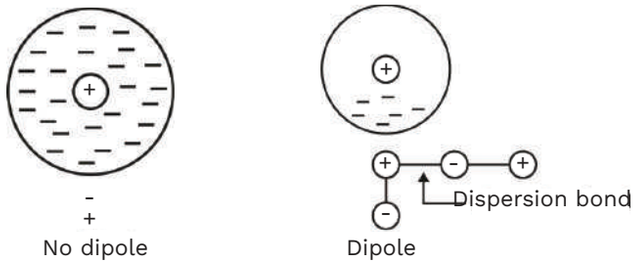


Fig. 2.16

- It causes polarisation, which is an imbalance in the electrical charge. This polarisation change is due to something called the dispersion effect, and it occurs when electrons are in motion.
- Dispersion bonds are a kind of weak attraction between molecules of the same element caused by the dispersion effect. Noble gases such as Ar, He, and Ne have this connection.

Metallic Bond

- Metals establish bonds by exchanging some of their valence electrons with one another. Those bonds are what keep the atoms in metals and alloys together. Elements with few loosely bound valence electrons may freely donate them to the shared pool.
- Each atom bonds with another when its valence electrons join those of other atoms to create an electron cloud, and the atoms become positive ions. As the valence electrons are no longer covalently bound to any specific atom, they are free to roam across the interatomic space.
- In other words, it indicates that the valence electrons are temporarily attached to a variety of distinct atoms.
- It is the electrostatic attraction forces between the delocalised electrons which is an electron cloud and the positively charged metal ions that produce metallic bonding.
- In a lattice of positively charged ions, bonding may be thought of as the sharing of free electrons (cations). When metals bind, electrons are shared across atoms, creating what is known as a sea of delocalised electrons around metal cations.

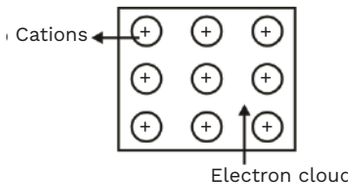


Fig. 2.17

Hydrogen Bonding

- Hydrogen bonds are a kind of dipole link formed when hydrogen atoms and oxygen atoms form covalent bonds.
- For the same reason that hydrogen atoms are less attracted to the shared electron pairs, oxygen atoms are more interested in them. One of the electrons in the shared pair moves to the oxygen atom. A dipole is created when an electron couple moves apart. Other water molecules also generate dipoles in a similar fashion. These dipoles are attracted to one other and create hydrogen bonds, which leads to the formation of water.

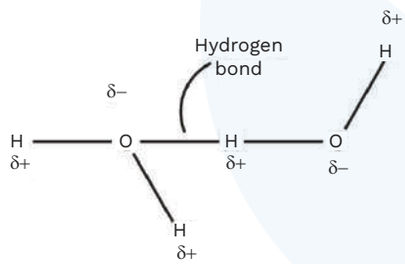


Fig. 2.18

Dipole Bonds

- Covalent bonds are formed when two atoms share electrons, and this sharing is not always equal. This occurs if one atom has a greater tendency to attract electrons than the others.
- As a result of an uneven distribution of electrons, molecules acquire diametrically opposed charges. The end outcome is the generation of persistent dipoles. These dipoles are attracted to one another and link with other dipoles in this way.

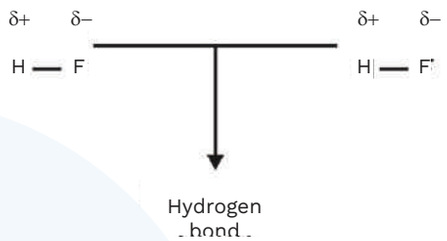


Fig. 2.19

- Dipole bonds are less strong than primary bonds but more powerful than dispersion bonds.
- When compared to hydrogen, fluorine's affinity is much greater. As a result, the covalent bond's shared electron pair moves closer to the fluorine atom. A dipole is formed when an electron pair moves, with a positive charge directed toward the hydrogen atom and a negative charge directed toward the fluorine atom. When molecules undergo dipole bonding, the resultant dipoles provide the attraction required to form the bonds.

Practice Exercise

1. In ionic crystals, electrical conductivity is:
(A) very high
(B) depends on the material
(C) depends on temperature
(D) practically zero

2. Crystals in which atoms are chemically highly inactive and do not form compounds with other atoms is:
(A) ionic crystals
(B) metal
(C) valence crystal
(D) Van Waals crystal

3. The bond between two identical non-metal atoms has a pair of electrons:
(A) with identical spins
(B) unequally shared between the two
(C) transferred fully from one to another
(D) equally shared between them

4. Which type of bonding would be expected between S and Cl?
(A) Ionic
(B) Non-polar covalent
(C) Polyionic
(D) Polar covalent

5. A strontium atom differs from a strontium ion in that the atom has a greater:
(A) umber of electrons
(B) number of protons
(C) mass number
(D) atomic number

6. Which combination of atoms can form a polar covalent bond?
(A) H and H
(B) Na and Br
(C) H and Br
(D) N and N

7. Which kind of bonding can be found in a sample of H_2O?
(A) Hydrogen bonds only
(B) Both polar covalent and hydrogen bonds
(C) Ionic and non-polar hydrogen bonds
(D) Non-polar covalent bonds only

8. Atoms bond primarily to:
(A) increase their potential energy and lose stability
(B) reduce their potential energy and gain stability
(C) increase their potential energy and gain stability
(D) make more atoms

9. Which one of the following compounds is most likely to be an ionic compound?
(A) CO_2
(B) CS_2
(C) KF
(D) CCl_4

10. The forces of attraction that exist between non-polar molecules are called:
(A) Van der Waals/dispersion forces
(B) ionic bonds
(C) electrovalent bonds
(D) covalent bonds

Answer Key

1. (D) **2.** (D) **3.** (D) **4.** (D) **5.** (A) **6.** (C) **7.** (B) **8.** (B) **9.** (C) **10.** (A)

3 Crystallography

Introduction

- Crystallography is the study of crystals, including both their internal structure and characteristics, as well as, their exterior and internal symmetries.

Crystalline and Non-Crystalline Solids

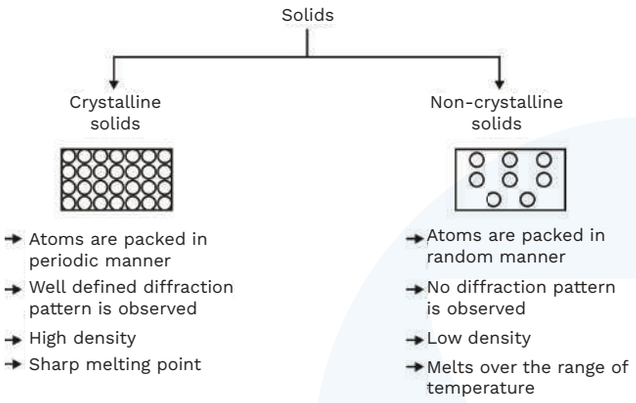


Fig. 3.1

- In crystalline solids, periodic packing of atoms is seen, a sharp diffraction pattern is produced as a result and extremely high density with a sharp melting point, whereas non-crystalline materials have the following characteristics: low density, melting across a wide temperature range, and no discernible atomic packing pattern.
- The atoms in non-crystalline materials are not arranged in a consistent and orderly fashion across relatively long distances. Because their atomic structure is so similar to that of a liquid, materials are sometimes referred to as amorphous materials, or supercooled liquids.
- The ease with which an initially disordered atomic structure in the molten state may be organised in an ordered state determines whether a crystalline or amorphous solid arises. Further, given the ordering process has so little time to complete, non-crystalline solids are more likely to develop when temperatures are quickly lowered below the freezing point.

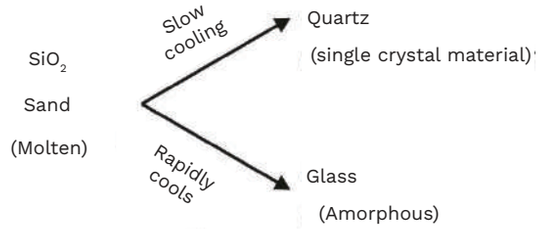


Fig. 3.2

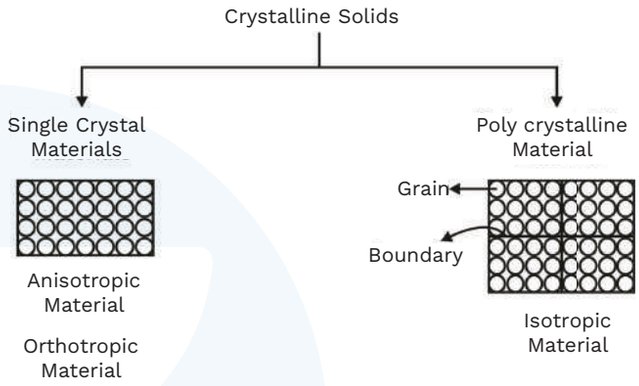


Fig. 3.3

If n = Grain size number

N = Number of grain per square inch at a magnification of 100×

Then, $N = 2^{n-1}$

Single Crystal Materials

For a crystalline solid, the term 'single crystal material' refers to a state in which the atomic arrangement is uniform and continuous over the whole specimen. It has been shown that if the ends of a single crystal are allowed to expand without being confined in any way, the crystal will take on a regular geometric form with flat faces, like those seen in some of the jump stones. The crystal structure may be inferred from the shape.

Polycrystalline Materials

- Many distinct sized crystals or grains with varying crystallographic orientations make up polycrystalline material. Within the area where two grains meet, there is an atomic mismatch. One can see a grain boundary.
- Grain boundaries prevent dislocation from moving, making polycrystalline materials

stronger than single crystal materials. Crystalline minerals are often found in polycrystalline form, however, some, like quartz, diamond, sodium chloride, sugar, etc., may be found in single crystal form.

Polymorphic or Allotropic Material

Some materials, called polymorphic or allotropic, may exhibit more than one crystal structure at different temperatures and pressures. Allotropic or polymorphic materials are the term for these kinds of substances.

Distinguishing between Anisotropy, Orthotropy, and Isotropy

- It is the crystallographic orientation that determines which physical characteristics may be measured in single crystal materials. Anisotropy refers to the directional nature of certain traits.
- To be classified as orthotropic, a material must exhibit directional anisotropy in three planes that are perpendicular to one another.
- Isotropic materials are those whose measured characteristics do not rely on the direction of measurement.
- For several polycrystalline materials, the crystallographic orientations of individual grains are fully random. Under these conditions, a specimen made up of a grain aggregate behaves isotopically, even though each individual grain is anisotropic.

Homogenous Material

Material is said to be homogeneous if its qualities are uniform in all directions across its volume.

Some More Points

- Before the development of X-ray technology, crystallography research was restricted to examining crystals based only on their outward appearance and symmetry. X-ray crystallography has made it feasible to determine the true existence of a crystal's shape pattern, the precise form and size of a unit cell, and the true periodic arrangement of that cell in three-dimensional (3D) space.
- Many useful substances are solid crystals, and X-ray crystallography might shed light on this fact. In crystallography, the most fundamental concepts are lattice, space lattice, unit cell, basic symmetry, Miller indices, and so on.
- Today, crystallography relies on analysing diffraction patterns from samples that have been hit by a beam of X-rays, neutrons, or electrons. Each of these three forms of radiation affects the specimen in a slightly different manner. To better determine where charges are located, an electron beam may be used.

Lattice Points, Space Lattice, and Crystal Structures

- Crystalline materials have their atoms organised in a regular, repeating pattern throughout their 3D structure. A crystal or space lattice is the 3D arrangement of atoms in a crystalline substance. Every single point on a space lattice is surrounded by the same space, and this is the defining feature of a space lattice.
- In describing the lattice structure of a crystal, a 3D coordinate system is used, with the axes of the system aligned with the crystal's faces. Every point in a space lattice, therefore, exists in a perfectly regular 3D grid.
- The atoms (or ions and molecules) of a substance are positioned at certain positions called lattice points, and a lattice is simply a network of lines constructed in space in such a manner that the space is split into equal volumes, and the sites of the intersection of these lines are the lattice points.
- As it turns out, there are only 14 unique methods to arrange points in 3D space that yet meet the requirements of being considered a space lattice. The name 'Bravvis lattice' is used to refer to this set of 14 different possible space lattices. These 14 Bravais lattices provide the basis for all possible crystal formations.
- A lattice is defined as a 3D translationally periodic arrangement of points, whereas a crystal is defined as a 3D translationally periodic arrangement of atoms. Accordingly, a crystal is equivalent to a lattice plus a basis, or motif.
- A motif or basis is an atom or a collection of atoms associated with each lattice point, the formation of crystal structure takes place from the combination of lattice and motif/basis.

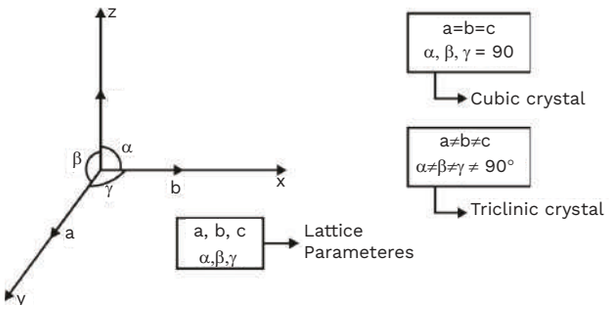


Fig. 3.4

Bravais Lattices

For reasons of perfect symmetry, there are only 14 unique ways to arrange points in a 3D space. A space lattice is considered to be a Bravais lattice if and only if all the atoms at the lattice points are the same. There are 14 different Bravais lattices, all of which fall under one of seven different crystal systems. The following is a list of crystal structures.

Crystal System	Axial Relationship	Interaxial Angles
Cubic	a = b = c	$\alpha = \beta = \gamma = 90°C$
Hexagonal	a = b ≠ c	$\alpha = \beta = 90°C, \gamma = 120°C$
Tetragonal	a = b ≠ c	$\alpha = \beta = \gamma = 90°C$
Rhombohedral (Trigonal)	a = b = c	$\alpha = \beta = \gamma \neq 90°C$
Orthorhombic	a ≠ b ≠ c	$\alpha = \beta = \gamma = 90°C$
Monoclinic	a ≠ b ≠ c	$\alpha = \gamma = 90°C \neq \beta$
Triclinic	a ≠ b ≠ c	$\alpha \neq \beta \neq \gamma \neq 90°C$

Table 3.1

Unit and Primitive Unit Cell

- The real crystal structure may be determined by repeating unit cells in three dimensions, which is the smallest geometrical form that can do so.

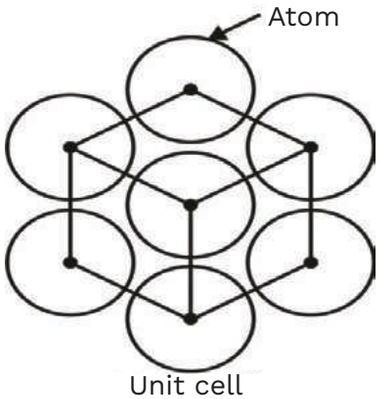


Fig. 3.5

- The concept of a unit cell provides a useful way to characterise a space lattice. The space lattice may be thought of as the result of an infinite number of repetitions of the smallest unit, the unit cell. Connecting any four adjacent lattice points creates a square, which is a unit cell in two dimension.
- The effective number of lattice points in a unit cell is one as each corner of the square is shared by a four unit cell meeting at that point.
- If one considers a unit cell with a single lattice point in the middle of a square and none at the corners, one can find that the effective number of lattice points in the cell is still single.
- A simple or basic unit cell is a unit cell with a single lattice point. Therefore, primitive unit cells have the fewest atoms overall and the

smallest atomic volume. Non-primitive cells are those that have more than one lattice point.

Cubic Crystal Structures

There are four cubic crystal structures as given below.

- Simple cubic (SC) crystal structure
- Body centered cubic (BCC) crystal structure
- Face centered cubic (FCC) crystal structure
- Diamond cubic (DC) crystal structure

Simple Cubic (SC) Crystal Structure

Every one of the eight corners of the unit cell in a basic cubic construction serves as a lattice point. The coordination number of a basic cubic lattice is six because the central atom is surrounded by six neighbours at equal distances from all sides. Each corner atom in this architecture serves as a part of an eight cell matrix.

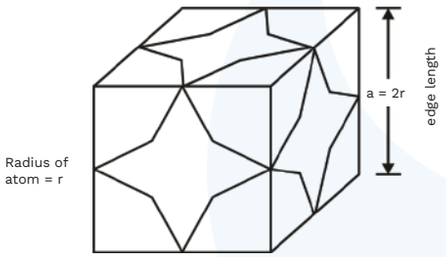


Fig. 3.6

Atomic Radius

- Atomic radius is defined as, 'half the distance between nearest neighbours in a crystal of pure element.'
  Let 'r' be the radius of the atom.
- In simple cubic crystal (SCC) lattice
  $a = 2r$
  $r = \frac{a}{2}$

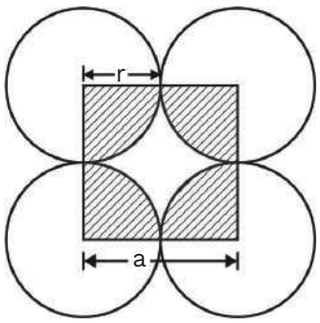


Fig. 3.7

Atomic Packing Factor

The packing of atoms in a unit cell of the crystal structure of a material is known as the atomic packing factor (APF). It is also known as packing efficiency (η).

$$APF = \frac{\text{Total volume of the atoms per unit cell}}{\text{volume of the unit cell}}$$

For Simple Cubic Structure

$$APF = \frac{N \times \text{volume of each sphere}}{\text{total volume of each cell}}$$

Where *N* is the total number of atoms in each unit cell.

For simple cubic, $N = 1$

Radius of each sphere = $r = \frac{a}{2}$

The volume of each sphere

$$= \frac{4\pi}{3} r^3 = \frac{4\pi}{3} \times \left(\frac{a^3}{8}\right) = \frac{\pi}{6} a^3$$

APF = 0.52

APF in per cent = 52.4 per cent filled

Body Centered Cubic (BCC) Crystal Structure

A unit cell in a molecule with a BCC structure has eight atoms in the unit cell's eight corners and one atom in the cell's centre. None of the corner atoms make contact with one another, but all four of them make contact with the atom in the centre of the body along the diagonal. Thereby, the BCC structure has eight coordinates.

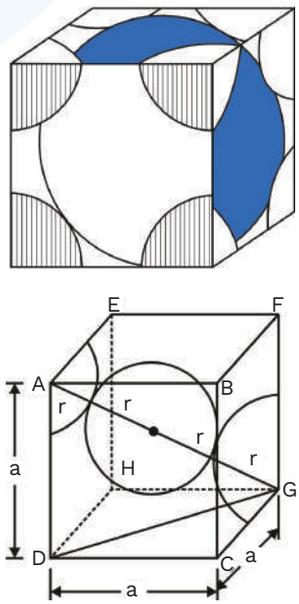


Fig. 3.8

Atomic radius of BCC, AG = r + 2r + r = 4r

In, $\Delta DGA\ DG^2 = DC^2 + CG^2$

$DG = \sqrt{a^2 + b^2}$

Diagonal of face = $a\sqrt{2}$

So, In ΔDGA, $AG^2 = a^2 + (a)^2$

AG = a. = diagonal of the cube

Also, $4r = a\sqrt{3}$

$r = a\frac{\sqrt{3}}{4}$

Atomic packing factor

For BCC, $N = 2$ and $r = a\frac{\sqrt{3}}{4}$

The volume of each sphere

$$= \frac{4\pi}{3}r^3 = \frac{4\pi}{3}\left(\frac{a\sqrt{3}}{4}\right)^3 = \frac{\pi\sqrt{3}}{16}a^3$$

$$APF = \frac{N \times \text{volume of each sphere}}{\text{total volume of each cell}}$$

$$= \frac{\frac{2\times\pi\sqrt{3}}{16}}{a^3} = \frac{\pi\sqrt{3}}{8} = 0.68$$

APF in per cent = 68 per cent filled

Face Centered Cubic (FCC) Crystal Structure

- One atom is located at each corner of the cube and one atom is located in the middle of each face in the FCC structure. In a corner atom, four neighbouring atoms are located at each of the four corners.
- It is worth noting that there is a literal atom in the core of every face. The coordination number of a face-centered cubic structure is 12, as there are four closest atoms on each of the four atomic planes below and above the atom in question.

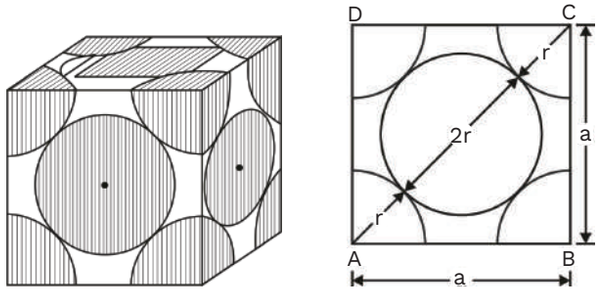


Fig. 3.9

Atomic radius of FCC

From figure, $AC^2 = AB^2 + BC^2$

$(r + 2r + r)^2 = a^2 + a^2$

$16r^2 = 2a^2$

$r = \frac{a\sqrt{2}}{4}$

Atomic Packing Factor

For FCC, $N = 4$ and $r = \frac{a\sqrt{2}}{4}$

The volume of each sphere

$$= \frac{4\pi}{3}r^2 = \frac{4\pi}{3}\left(\frac{a\sqrt{2}}{4}\right)^2 = \frac{\sqrt{2}\pi}{24}a^3$$

$$\text{Therefore, APF} = \frac{4\times\frac{\sqrt{2}\pi a^3}{24}}{a^3} = \frac{\sqrt{2}\pi}{6} = \frac{\pi}{3\sqrt{2}} = 0.74$$

%APF = 74% filled

Diamond Cubic (DC) Crystal Structure

Carbon exists in two forms, namely diamond and graphite. Both have quite different characteristics and properties.

Diamond

- For the (sp3) hybrid covalent bond, each of its atoms is linked to three others through a network of four connections. For diamonds, the bond angle in a particular direction is 109.5°. Diamonds have what is called a diamond cubic structure.
- Regarding DC crystal sturucture, it should be noted that:
  - It consists of two interpenetrating FCC arrangements of carbon atoms.
  - A unit cell of DC contains 18 atoms. Of these eight are placed on corners of the cube, one on each of the six faces and four inside it i.e., (8 + 6 + 4 = 18) atoms.
  - The space lattice of DC is face-centered cubic (FCC).
  - There is tetrahedral coordination of carbon atoms.
  - Number of atoms in the DC unit cell = $\frac{1}{8}$ x 8 (corner atom) + $\frac{1}{2}$ × 6 (face-centered atoms) + 1 × 4 (atom inside the cell) = 1 + 3 + 4 = 8.
  - The distance of separation between the two atoms of the basis (nearest neighbours) is $a\sqrt{\left(\frac{1}{4}-0\right)^2 + \left(\frac{1}{4}-0\right)^2 + \left(\frac{1}{4}-0\right)^2} = \frac{a\sqrt{3}}{4}$

(where a is the lattice parameter 3 of DC unit cell.)

- Atom packing factor of DC crystal structure

For DC, $N = 8$ and $r = \frac{a\sqrt{3}}{8}$

$$APF = \frac{8 \times \frac{4\pi}{3}\left(\frac{a\sqrt{3}}{8}\right)^3}{a^3} = \frac{\pi\sqrt{3}}{16} = 0.34$$

%APF = 34% packed

Hexagonal Closed Packing

- Crystal structures of metals often feature hexagonal unit cells rather than cubic ones. This arrangement, known as hexagonal close packed, is shown in the image with a reduced sphere unit cell (HCP).

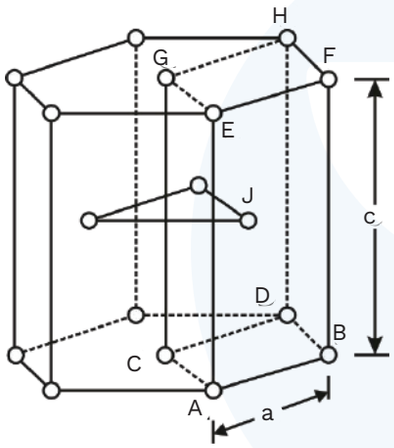


Fig. 3.10

- This arrangement, known as hexagonal close packed, is shown in the image with a reduced-sphere unit cell (HCP). Six atoms make form a regular hexagon on the top and bottom sides of the unit cell, surrounding a single atom in the middle. Between the two, there is a third plane that adds three more atoms to the unit cell. In this intermediate plane, atoms have the closest neighbours in both of the two planes on either side.
  Consequently,

$$\text{Atom/cell} = 12 \times \frac{1}{6} + 2 \times \frac{1}{2} + 3 = 6$$

- As there are 12 atoms in the 12 corners of the hexagonal, two atoms at the top and bottom, and three atoms inside. The c/a ratio, where a and c are the short and long unit cell measurements of a figure, should be 1.633, although, it is not the case for particular HCP metals. The HCP crystal structure has the same coordination number (12) and atomic packing factor (0.74) as the FCC structure.
- Some examples of HCP metals include cadmium, magnesium, titanium, and zinc.

Miller Indices

- Miller indices are used to label the orientations and planes of crystals and their constituent units. To study distinct characteristics of different elements, such labels become essential. Miller indices are represented in the most generic form as (h k l), with no comma between the three terms. In this case, the unknown numbers are denoted by the symbols h, k, and l.
- Different kinds of brackets have special meanings, which are elaborated as follows.
  - (h k l) denotes a plane
  - [h k l] denotes a direction
  - {h k l} denotes a family of planes
  - <h k l> denotes a family of directions
- The Miller indices (h, k, l) are written as the reciprocal of the intercepts (p, q, r) the plane makes along the three rectangular axes (x, y, z). Intercepts are expressed as a fraction of a unit cell's length. These are the three axis distances from the origin expressed as units. Hence,
  $h = 1/p$, $k = 1/q$, $l = 1/r$
  where p is the intercept of the plane on the x-axis, q is the intercept of the plane on the y-axis, and r is the intercept of the plane on the z-axis.
- The inverse of these intercepts is then turned into a whole integer using the reciprocal function. To do this, the LCM of the denominator may be used as a multiplier for each reciprocal. The Miller indices for the specified plane are obtained in this way. To represent the Miller indices, only the three smallest integers are used.

Determining the Miller Indices of a Given Plane

- Refer to the figure below, showing a plane ABC in a cubical unit cell. One has to determine

Miller indices of plane ABC. Origin O and coordinate axes x, y, and z are shown in the unit cell in which OP = OQ = OR = a. The plane in question intercepts at A, B, and C on x, y, and z axes in such a way that,

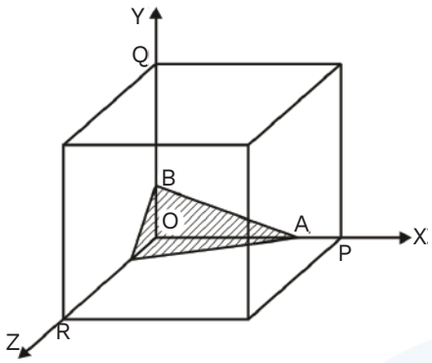


Fig. 3.11

$OA = \left(\frac{2a}{3}\right)$ along x-axis

$OB = \left(\frac{2a}{5}\right)$ along y-axis

$OC = \left(\frac{a}{3}\right)$ along z-axis

- To obtain the required Miller indices, one has to proceed as below.
  The intercepts are C_1 = OA, C_2 = OB, and C_3 = OC; therefore,

$$p \text{ (along x-axis)} = \frac{OA}{OP} = \frac{\left(\frac{2}{3}\right)a}{a} = \frac{2}{3}$$

$$q \text{ (along y-axis)} = \frac{OB}{OQ} = \frac{\left(\frac{2}{5}\right)a}{a} = \frac{2}{5}$$

$$r \text{ (along z-axis)} = \frac{OC}{OR} = \frac{\left(\frac{1}{3}\right)a}{a} = \frac{1}{3}$$

as h = 1/p, k = 1/q and l = 1/r; therefore,

$$h = \frac{1}{\frac{2}{3}} = \frac{3}{2}\ k = \frac{1}{\frac{2}{5}} = \frac{5}{2}, \text{ and } l = \frac{1}{\frac{1}{3}} = \frac{3}{1}$$

$$(h\ k\ l) = \left(\frac{3}{2}\frac{5}{2}\frac{3}{1}\right)$$

- As (h k l) should have the smallest possible integers; hence, write like this:

$$(h\ k\ l) = \frac{1}{2}\ (3\ 5\ 6)$$

- Miller indices of planes in non-cubical unit cells may be determined in the same manner.

Planar Density

- The number of atoms per unit area of a crystal plane is known as planar density. This density expresses the packing of atoms on a plane. The planar density p can be expressed as

$$\rho_p = \frac{N_e}{A}$$

  where *N* is the effective number of atoms on the plane whose area is A
- Planar density in SCC on (111) plane

$$\rho_p = \frac{0.58}{a^2}$$

- Planar density in BCC on (110) plane

$$\rho_p = \frac{1.414}{a^2}$$

- Planar density in FCC on (100) plane

$$\rho_p = \frac{2}{a^2}$$

Inter Planar Spacing

Inter planar spacing is defined as the distance between two parallel planes, one of which passes through the origin and the other of which does not. The symbol for this is 'd' and is measured perpendicular to the planes. Cubic end tetragonal crystals have the following relations for interplanar spacing.
For cubic unit cells,

$$d_{(h\ k\ l)} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

and for tetragonal unit cell,

$$d_{(h\ k\ l)} = \frac{a}{\sqrt{h^2 + k^2 + l^2\left(a^2 / c^2\right)}}$$

Members of a family of planes (h k l) have the same inter planar spacing.

Bragg's Law of X-Ray Diffraction

$n\lambda = 2d_{h\,k\,l}.\sin\theta$

$$d_{h\,k\,l} = \frac{a}{\sqrt{h^2 + k^2 + l^2}}$$

a = Lattice parameter

$d_{h\,k\,l}$ = Inter planar spacing

n = Order of reflection

λ = Wavelength of *x*-Ray

Crystal Imperfections

- In contrast to ideal crystals, which have a completely regular crystal structure, actual crystals always contain certain faults or imperfections. As natural crystals grow in an unregulated environment, they always have a wide variety of faults.
- Certain material qualities may be altered upwards or downwards depending on the nature of the defect and the constituents have chosen.
- When chromium replaces some of the aluminium oxides in a ruby crystal, the resulting change in colour makes the crystal more valuable. For the same reason, tiny amounts of impurities like phosphorus and arsenic are essential for silicon to be used in electronics. With a few exceptions, most flaws in crystals are not wanted.

Types of Imperfections Present in Solids

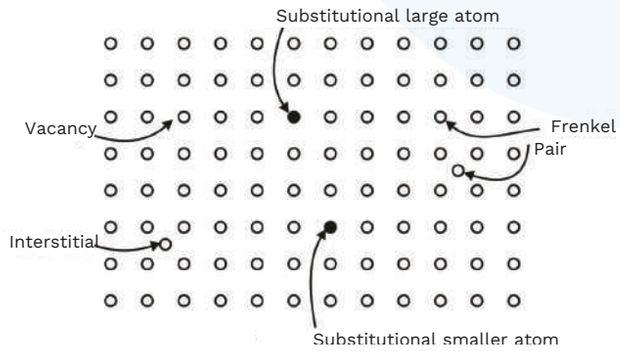


Fig. 3.12

Point Imperfections

- The point flaws are sometimes referred to as zero dimensional flaws. They form in crystal like regions as imperfect points. The normal size of a point defect is just one or two atomic diameters.
- An imperfection at the spot might be caused by thermal fluctuations, or when fast cooling (quenching) is used to bring materials down from high temperatures.
- Both forging including hammering and rolling and irradiation (displacing atoms in a material with the bombardment of high energy particles) are methods of deformation.
- The point imperfections are discussed below.

Interstitial Defect

- The illustration displays the formation of an interstitial defect, which occurs when an atom occupies a location in the lattice that would not typically be occupied in the perfect crystal. This flaw manifests itself when atoms move into the empty spaces between those of a perfect crystal.

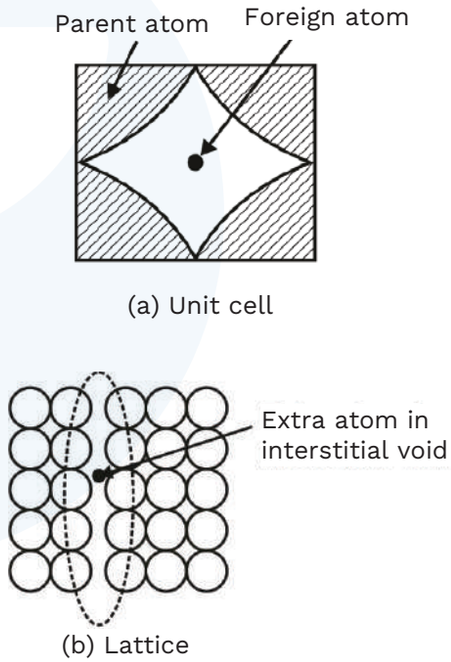


Fig. 3.13

- An interstitial atom may be a native crystal atom or an outside atom. In a crystal with a low atomic packing factor, the interstitial atoms are more likely to become permanently ensconced in the crystal lattice.
- Interstitial defects cause atomic distortion because, depending on the size of the interstitial atom and the space between the normal atoms, the interstitial atom tries to push the surrounding atoms further apart.

- In a crystal structure, an interstitial defect is an additional atom that has found its way into the structure due to a poor atomic packing factor. An alloying agent or an impurity might be the source of the alien atom.
- It turns out that vacancies and interstitial flaws are opposite phenomena. An example of this kind of flaw is the presence of carbon as an interstitial solute in FCC iron.

Vacancies

- It is the simplest kind of point fault there is. It happens whenever there is a free spot (an atomic position) in a crystal.
- A point defect is just an atomic spot that is either absent or unoccupied. One source of this flaw is improper packing during the initial crystallisation process, while the other is atomic vibrations caused by elevated temperatures. A crystal in thermal equilibrium will have a certain percentage of vacancies, and these flaws will enhance the unpredictability of the structure.
- Due to the clustering of atoms around a vacancy, the defect causes the lattice planes to be distorted, as seen in the image. The number of available slots might be one, two, or even more.
- Vacancies aid in the transit or movement of atoms across the lattice during the manufacturing operations of annealing, surface hardening, and normalising.

Frenkel's Defect

- An interstitial defect arises when an atom occupies a definite position in the lattice, which is not normally occupied in the ideal crystal by anyone as shown in figure. In this defect, atoms occupy positions between the atoms of an ideal crystal.

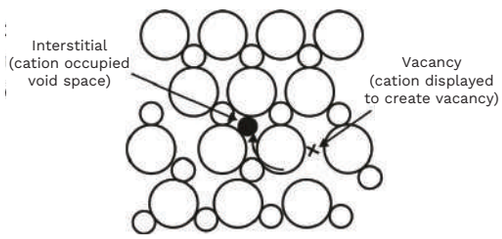


Fig. 3.14

- Ionic crystals often exhibit Frenkel's defect, which is an interstitial imperfection. Ionic crystals include both cations and anions. Cations fill a void of smaller ions, whereas anions are bigger.
- As cations are the smallest kind of ion, they may readily fill the gaps left behind by other ions in an ionic solid. However, unlike cations, anions cannot be moved into empty space because of their bigger size.
- The existence of this defect is referred to as Frankel's defect, and it is caused by cations filling interstitial spaces rather than lattice points. The crystal's overall neutrality is unaffected by the existence of this imperfection.
- Frankel's defect combines vacancy and interstitial flaws. The crystals of CaF_2 and silver halides are affected by Frankel's defect.

Schottky's Defect

- This defect develops in ionic crystals when a cation and anion pair are absent.

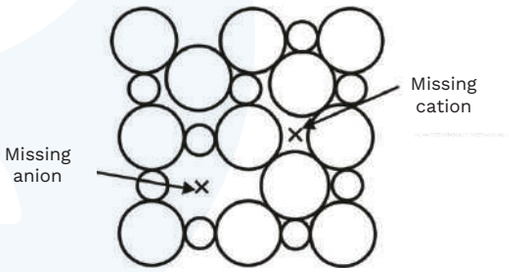


Fig. 3.15

- In alkali halides like LiCI, LiBr, and Lil, the electrical neutrality of the ionic crystal is maintained due to the vacancy of the missing pair of ions.

Frenkel's Defect	Schottky's Defect
It occurs in ionic solids.	It also occurs in ionic solids.
It occurs when cations are displaced into the interstitial space formed between anions.	It occurs when a pair of cation and anion are missing from the ionic crystal.

Frenkel's Defect	Schottky's Defect
The overall electrical neutrality of the ionic crystal is maintained despite this imperfection.	The overall electrical neutrality of the ionic crystal is maintained despite this imperfection.
It is mainly an interstitial type point defect.	It is mainly a vacancy type of defect.
It occurs mainly in CaF_2 and silver halides.	It occurs mainly in alkali halides.

Table 3.2

Substitutional Imperfections

- Substitutional imperfections are created in crystals when an outside atom displaces, or 'substitutes' a parent atom at a lattice point.

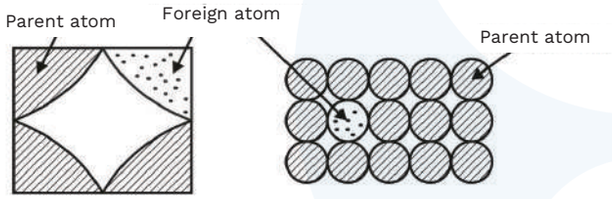


Fig. 3.16

- Solutes are the atoms with which their parents are being replaced, whereas solvents are the parent atoms. Elements from both the solute and the solvent combine at random to generate a new element.
- Alpha bras crystals are formed when copper and zinc atoms combine. Boron and antimony atoms are also doped into germanium crystals to create a substitutional impurity.

Atomic Vibrations

- Each atom in a solid substance is in a constant state of motion relative to its lattice location; these motions may be thought of as flaws. The vibrations of neighbouring atoms are connected by the chemical bond/atomic connection, rather than being completely autonomous.
- These vibrations are timed in such a manner that elastic waves, called lattice waves, propagate across the medium.
- A material's vibrational thermal energy is a sum of elastic waves with different amplitudes and wavelengths. There is a limit on the energy that may be used. One unit of vibrational energy is called a 'phonon' in the language of quantum mechanics. Nonetheless, phonons may also be vibrational waves.
- Note, any kind of crystal defect available in the material function as a scattering point for the movement of electrons.

Practice Exercise

1. The difference between the number of atoms in a unit cell of a BCC crystal and an FCC crystal is
(A) 1
(B) 2
(C) 4
(D) 6

2. When BCC iron is heated, it changes to FCC iron resulting in
(A) contraction in volume
(B) increase in volume
(C) no change in volume
(D) crack in the material

3. In a silicon crystal, the arrangement of atoms repeats periodically. This type of material is classified as
(A) amorphous and non-crystalline
(B) non-crystalline and epitaxial
(C) epitaxial and single crystal
(D) amorphous and single crystal

4. Consider the following statements concerning FCC structure.
I. Number of nearest neighbour atoms is 12
II. Packing efficiency is 0.74.

III. There is an atom at the body centre of the unit cell.

Which of the statements given above is/are correct?

(A) I, II, and III
(B) I and II
(C) I and III
(D) I and III

5. Which one of the following exhibits the most well-defined X-ray diffraction pattern?

(A) A polycrystalline aggregate
(B) An amorphous material
(C) A single crystalline material
(D) A plastically deformed crystal

6. In a crystal lattice, what are the vacancies created by the absence of certain atoms known as?

(A) Hertz defects
(B) Schottky defects
(C) Pauli defects
(D) Crystal defects

7. Match List I with List II and select the correct answer using the code given below the lists.

List I

A. Hexagonal crystal
B. Rhombohedral crystal
C. Triclinic crystal
D. Monoclinic crystal

List II

I. $a = b = c$; $\gamma = \alpha = \beta \neq 90°$
II. $a = b \neq c$; $\alpha = \beta = 90°$; $\gamma = 120°$
III. $a \neq b \neq c$; $\alpha = \beta \neq \gamma \neq 90°$
IV. $a \neq b \neq c$; $\alpha \neq \gamma \neq 90° \neq \beta$

Codes

	A	B	C	D
(A)	I	II	III	IV
(B)	II	I	IV	III
(C)	I	II	IV	III
(D)	II	I	III	IV

8. The atomic packing factor for face centered cubic (FCC) crystal structure is:

(A) 0.63
(B) 0.74
(C) 7.4
(D) 6.3

Answer Key

1. (B) **2.** (A) **3.** (C) **4.** (B) **5.** (C) **6.** (B) **7.** (B) **8.** (B)

4 Magnetic Properties of Materials

Introduction

- Magnetic materials are those materials in which a state of magnetisation can be induced. On magnetisation, a material can create a magnetic field in the surrounding area. The magnetic properties of materials originate from the motion of their electrons and their magnetic moments (dipoles). Though the electron orbits are associated with magnetic moments, the orbital arrangements may be such that they may prevent a net permanent magnetic moment resulting from the atoms. Magnetic moments can interact in different ways, and these moments may align in a parallel or anti-parallel way with equal or unequal magnitude.
- Magnetic effects exist outside the material when it has a net permanent magnetic moment. Magnetism is a form of energy. The magnetic material can perform work when it lifts iron fillings against the force of gravity.

Magnetic Parameters

Magnetic Dipole Moment

- The magnetic moment of a magnet is a quantity that determines the force that the magnet can exert on electric currents and the torque that a magnetic field will exert on it.
- A loop of electric current, a bar magnet, an electron, and a molecule possess magnetic moments.
- The magnetic field produced by a small current loop is similar to the electric field produced from a small electric dipole. For this reason, a small current loop is called a magnetic dipole. The magnetic dipole moment of a current loop is defined as the product of the loop area and the current flowing through the loop.
- It is a vector quantity, and the vector direction of the dipole moment is perpendicular to the plane of the loop and is along the direction of a right-hand screw when moved in the direction of current in the loop. Hence, the magnetic dipole moment is given as

$$\vec{p}_m = iA\hat{n}$$

where $\hat{n}$ = unit vector normal to the plane consisting of a current loop
I = current in the loop
A = cross sectional area of the loop

Magnetisation

- All materials are affected by the presence of a magnetic field, thereby they acquire magnetic moments.
- The magnitude of this magnetic moment per unit volume is called magnetisation of the medium and it is described by the vector M. Magnetisation (M) of a material can be expressed in terms of a magnetic dipole (p_m) as given below

$$M = N \times p_m$$

where N = number of magnetic dipoles per unit volume

- Now consider a solenoid of length l, area of cross section (A), and the number of turns N with current flowing through it as I. If the solenoid is placed in a vacuum, then the flux density is

$$B_0 = \mu_0 H$$

where $H = \frac{NI}{l}$ = magnetising force of magnetic field strength
0 = permeability of free space = $4\pi \times 10^{-7}$ henry/meter

- The flux density can be increased if the vacuum is replaced by a homogeneous magnetic medium with a relative permeability as μ_r

$$B = \mu_0 \times \mu_r H$$

- Therefore, the increase in flux density is given by

$$B - B_0 = \mu_0 \times \mu_r \times H - \mu_0 H$$
$$= \mu_0 H(\mu_r - 1)$$
$$= \mu_0 H'$$

- This increase in flux density can be considered to be resulting due to an increase of magnetising force from H to H' of the solenoid rather than due to the change of the medium. If I- is the corresponding increase in current, then

$$H' = \frac{NI}{l} = \frac{NI'A}{Al}$$

where A = cross sectional area, A.l = volume, NI'A = dipole moment, H' = dipole moment per unit volume

- Hence, the above relation represents the total magnetic dipole moment per unit volume, which is nothing but the magnetisation of the medium, which can be denoted by the letter M.

$$H' = (\mu_r - l) \times H = M$$

or $$\frac{M}{H} = \mu_r - 1 = \chi$$

or $$M = \chi H$$

where χ is a dimensionless quantity called the magnetic susceptibility of the medium.

Magnetic Susceptibility

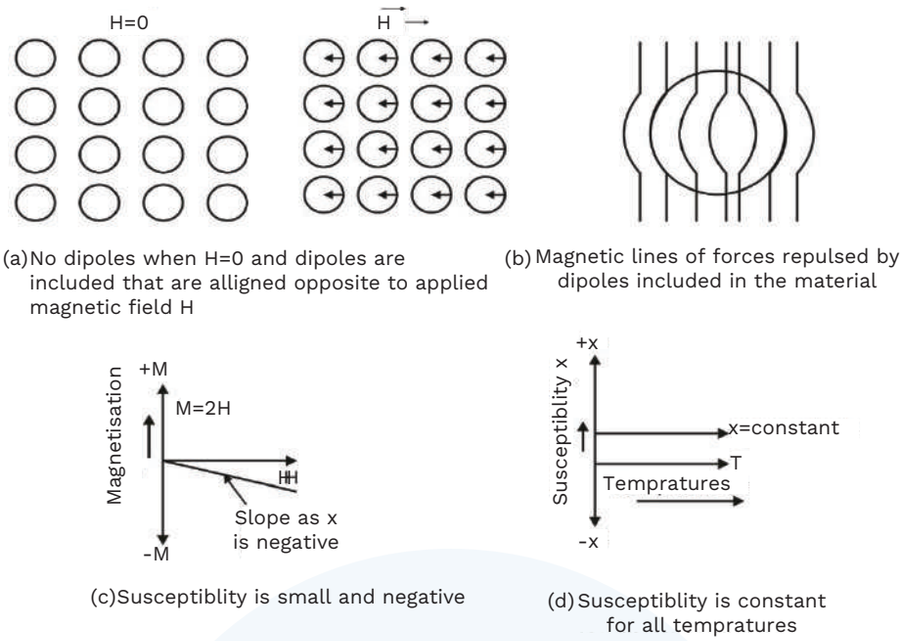
- The magnetic susceptibility χ is dependent on the relative permeability μ_r, of the medium and it is given as
  $x = (\mu_r - 1)$
- The magnetic susceptibility depends on the nature of the magnetic material and its state (temperature). The susceptibility of the material may change on cold working done on the material. The susceptibility of copper changes from a negative to positive value, for example, on cold working. It again changes to a negative on annealing.
- The susceptibility may be determined by measuring the force exerted on a magnetic material when it is placed in a magnetic field.

Classification of Magnetic Materials

- The origin of magnetism lies in the orbital and spin motion of electrons and how the electrons interact with one another. All materials are magnetic, but some materials are more magnetic than others. In some materials, there is a very strong interaction between atomic moments, whereas there are no or small interactions between atomic magnetic moments in other materials.
- There are five classes into which all the magnetic materials can be grouped according to their magnetic behaviour, which include the following.
  - Diamagnetic substances
  - Paramagnetic substances
  - Ferromagnetic substances
  - Ferrimagnetic substances
  - Anti-ferromagnetic substances

Diamagnetic Substance

- Diamagnetic substances show a very weak form of magnetism when placed in a magnetic field. They become weakly magnetised in a direction opposite to that of the magnetisation of the external field. Therefore, purely diamagnetised substances are repelled by a magnetic field.
- The reason is that in the presence of a magnetic field, dipoles are induced in the substance that is aligned opposite to the magnetic field, as shown in Figure (a). The substances have a negative magnetic susceptibility, as shown in Figure (c), and magnetic permeability of less than one. The lines of force of an applied magnetic field are slightly repulsed, as shown in Figure (b).



(a) No dipoles when H=0 and dipoles are included that are alligned opposite to applied magnetic field H

(b) Magnetic lines of forces repulsed by dipoles included in the material

(c) Susceptiblity is small and negative

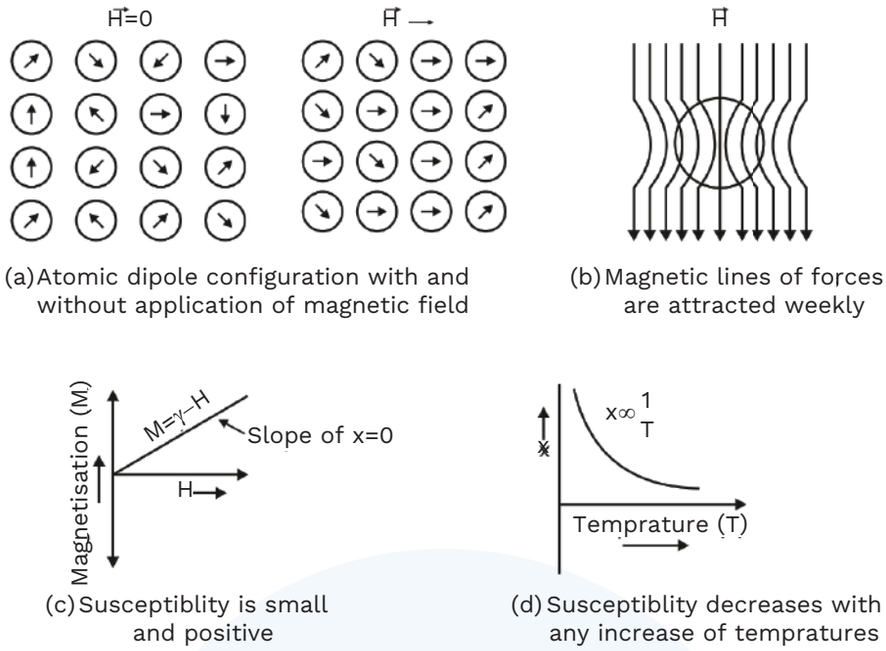
(d) Susceptiblity is constant for all tempratures

Fig. 4.1

- The diamagnetic substances include organic solids like naphthalene, benzene, etc., metals like gold, silver, and copper, and rare gases like helium, neon, and argon.
- A very small fraction of the applied magnetic field is shielded or excluded from the interior of diamagnetic substances by the induced diamagnetic currents.

Paramagnetic Substances

- These are the substances that exhibit an induced magnetic field, which is parallel and proportional to the applied magnetic field. Paramagnetic materials always have a permeability greater than one.
- In paramagnetic substances, each atom possesses a permanent dipole moment, which results by virtue of incomplete cancellation of electron spin and orbital magnetic moments.
- In the absence of any applied magnetic field, these magnetic moments of atoms are randomly orientated, thereby resulting in zero overall magnetic moments or magnetisation of the substance.
- These magnetic dipoles are free to rotate. Therefore, these magnetic dipoles tend to preferentially align by rotation when subjected to an external magnetic field, as shown in Figure (a).
- There is no mutual interaction between adjacent dipoles. Due to preferentially alignment of dipole moments, they enhance the magnetic field, thereby giving rise to a relative permeability that is greater than one. Hence, these materials have a relatively small and positive susceptibility as shown in Figure (c).
- The susceptibility is temperature dependent and it reduces with temperature as shown in Figure (d).
- Magnetic lines of forces are attracted weakly by the paramagnetic materials as shown in Figure (b).



(a) Atomic dipole configuration with and without application of magnetic field

(b) Magnetic lines of forces are attracted weekly

(c) Susceptiblity is small and positive

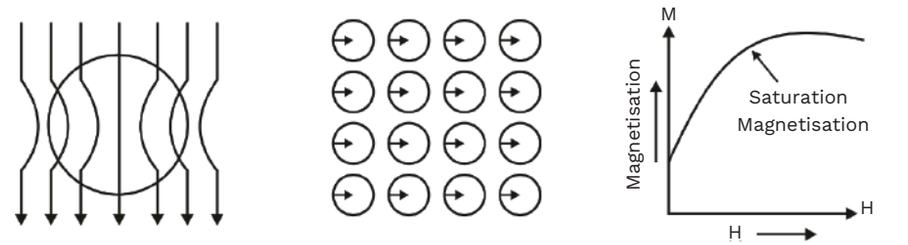
(d) Susceptiblity decreases with any increase of tempratures

Fig. 4.2

- Both paramagnetic and diamagnetic substances are considered to be non-magnetic because they exhibit magnetisation only in the presence of an external field. Both substances on magnetisation can have a small flux density B within, which is almost the same as that is possible in free space or vacuum.

Ferromagnetic Substances

- The outstanding characteristic of a ferromagnetic substance is that it is very strongly attracted by a magnet. The importance of iron as a magnetic substance has led to the name of these substances as ferromagnetic substances.
- The lines of magnetic forces of the external field are attracted very strongly by ferromagnetic substances, as shown in Figure (a).
- Permanent magnetic moments in ferromagnetic substances result from atomic dipole moments due to uncancelled electron spins. Each spinning electron can be considered to be equivalent to a moving charge (current), thereby forming a magnetic dipole moment (a small magnet).
- A substance having an even number of electrons has an equal number of electrons spinning in one direction as in the other direction. In ferromagnetic substances, more electrons spin in one direction than in the other, thereby resulting in a net magnetic moment in their atoms.
- In materials such as iron, cobalt, and nickel, the magnetic dipole moments are quite strong, which helps the atomic dipoles to align themselves in parallel directions as atoms are sufficiently close to interact with each other, as shown in Figure (b). Hence, ferromagnetic substances show spontaneous magnetisation even in the absence of an external magnetic field.
- Two distinct characteristics of the ferromagnetic substance are spontaneous magnetisation and the existence of spontaneous magnetisation occurs only below the Curie temperature.
- Above the Curie temperature, ferromagnetic substances have the same behaviour as that of paramagnetic materials, i.e., small magnetisation and susceptibility.
- Below the Curie temperature, the magnetisation of ferromagnetic materials increases rapidly to a high value called saturation magnetisation which is the maximum induced magnetic moment that can be obtained in a magnetic field. Beyond this magnetisation no further increase is possible, as shown in Figure (c).



(a) Lines of force are attracted very strongly

(b) Dipole moments are aligned even when H=0 which is called spontaneous magnetism

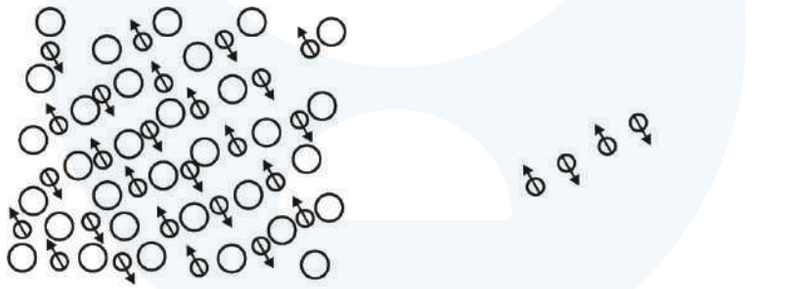
(c) Saturation magnetisation is the maximum induced magnetic moment

Fig. 4.3

- Saturation magnetisation is an intrinsic property, and it is independent of particle size but dependent on temperature.
- In addition to the curie temperature and saturation magnetisation, ferromagnetic materials can retain a memory of an applied field once it is removed, i.e., a magnetic flux density (remanence) that persists even when the magnetisation field is removed. This property is called retentivity.

Anti-ferromagnetic Substances

- In ferromagnetic substances, the atomic dipole moments align parallel to each other, but in anti-ferromagnetic substances, there is a tendency for anti-parallel alignment of electron spins (dipole moments) of neighbouring atoms, as shown in the figure below. Consequently, there is no net magnetic moment in the anti-ferromagnetic materials.



(a) Alignment of magnetic moments

(b) Antiparellel alignment of electron spins which are equal

Fig. 4.4

- The atoms in the first transition metals, such as copper and zinc, are anti-ferromagnetically coupled. The oxides of iron, cobalt, and manganese, as well as, the compounds of nickel and manganese, exhibit antiferro-magnetisation.
- The susceptibility of anti-ferromagnetic sub-stances depends upon their temperature. The variation of susceptibility of anti-ferromagnetic substances with their temperature is shown in the figure. There is a temperature called the Neel temperature (T_N) where maximum susceptibility occurs in the material.

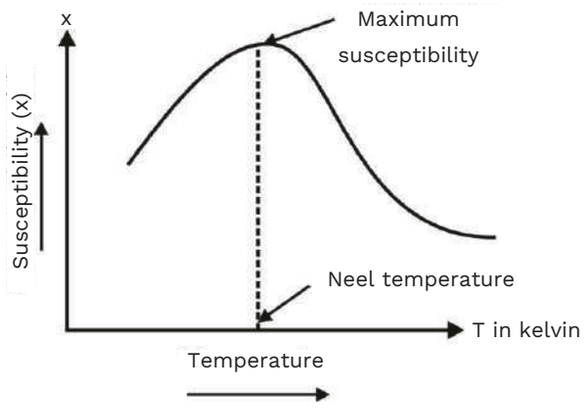


Fig. 4.5

- The Neel temperature generally occurs at a much lower level than the room temperature. The Neel temperature for MnF_2 is at 100 K (–173°C). Below the Neel temperature, the susceptibility increases with an increase of temperature while the susceptibility decreases with an increase in temperature above the Neel temperature.

Ferrimagnetic Substances

- Ferrimagnetic substances have constituent atoms which are anti-ferromagnetically coupled (dipole moments of two neighbouring atoms are aligned in opposite directions), but these neighbouring atoms have a different magnitude of magnetic moments. Hence, the total cancelling of dipole moments does not take place as it occurs in anti-ferromagnetic substances.
- There is a net magnetic moment in each coupling, and the sum of the moments of all couplings results in total magnetisation, which is quite comparable to the magnetisation occurring in ferromagnetic substances. The coupling of atomic dipole moments is shown in the figure.



(a) Alignment of magnetic moments of sub-lattices A and B

(b) Magnetic moment of sub-lattice A is more than sub-lattice B

Fig. 4.6

- This type of magnetism occurs in materials such as ferrites, which are the oxides of various metals.
- These ionic materials can be represented by the chemical formula such as MFe_2O, where M represents metals such as magnesium, barium, and manganese.
- The most versatile ferrimagnetism can be obtained from the 'rare earth iron garnets.' With a suitable choice of rare earth ions, it is possible to design a ferrimagnetic material with a prescribed magnetisation and temperature behaviour.
- Ferrites are preferred over ferromagnetic materials as they have high resistivity (good electrical isolator) to prevent eddy current losses in high frequency transformers.
- Ferrites are used in the core of coils in the transformer. Ferrites are brittle and have poor machinability. Barium ferrites are used for the fabrication of permanent magnets.
- The magnetic hysteresis loop of ferrites varies from very narrow to very wide. Ferrites having narrow hysteresis loops are suitable for soft magnetic applications such as audio transformers, television transformers, gyrators, cores of induction coils, and memory cores of computers (Mn–Mg ferrite).

Curie Temperature

- At a critical temperature, the alignment of magnetic moments vanishes in magnetic materials, and this temperature is called Curie temperature or Curie point. The magnetic moments in magnetic materials remain aligned in one direction below the Curie temperature, but further heating above this temperature in these magnetic materials leads to the random orientation of magnetic moments due to more thermal energy in the materials. This is the reason why ferromagnetic materials behave as paramagnetic materials at temperatures above their Curie temperatures.
- Below the Curie temperature, the susceptibility of a ferromagnetic material decreases with an increase in the Curie temperature. The Curie temperature is different for ferromagnetic and paramagnetic materials.

- These are called ferromagnetic Curie temperature (T_{FC}) and the paramagnetic Curie temperature (T_{PC}).
- The saturation magnetisation in a material is a function of the critical temperature. The saturation magnetisation vanishes or becomes zero at Curie temperature, as shown in the figure, and the material behaves as non-magnetic.

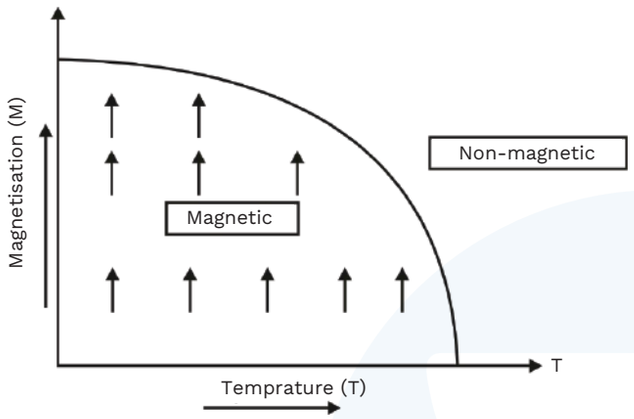


Fig. 4.7

- It is possible to modify the Curie temperature by proper alloying to meet the requirements for different applications.
- For example, to maintain magnetic properties, the materials must have a Curie temperature higher than the temperatures to which they are subjected.
- The magnetic materials used in transformers, motors, alternators, relays, and permanent magnets must have higher Curie temperatures, so that magnetic properties can be maintained in these devices during their normal working temperatures.
- To ensure non-magnetisation in the materials, the Curie temperatures must be lower than the operating temperatures. Certain devices require non-magnetic quality, which necessitates that operating temperatures must be higher than Curie temperatures so that materials must not be magnified as non-magnetic quality is desired in these devices such as spectrometers, cathode ray tubes, and compass bearings.
- Curie temperature of material has to be varied to get the desired magnetic properties by suitably alloying the material.

Laws of Magnetic Materials

Curie Law

- Susceptibility is independent of the magnetising field, but it varies inversely with absolute temperature.

$$\chi \alpha \frac{1}{T}$$

or $\chi = \frac{Ce}{T}$

where C_e = Curie constant
- The Curie law is applicable to the elements in their pure state, the oxides, and salts of paramagnetic nature.

Curie Weiss Law

- Several paramagnetic materials in their bulk state obey the Curie–Weiss law. The law is expressed as

$$\chi = \frac{Ce}{T - T_{pc}}$$

where C_e = Curie constant T_{pc} = paramagnetic Curie temperature
- This law is also applicable for ferromagnetic materials when the temperature (T) is greater than ferromagnetic Curie temperature (T_{tc}), i.e., $T > T_{tc}$, as then ferromagnetic materials behave as paramagnetic materials.

Neel Law

- The susceptibility of anti-ferromagnetic materials is a function of temperature. The maximum susceptibility (χ_{max}) occurs at a particular temperature known as the Neel temperature (T_N). The susceptibility is given by

$$\chi = \frac{C_e}{T + T_{pc}} \text{ for } T > T_{pc}$$

where C_e = Curie constant
T_{pc} = paramagnetic Curie temperature
- The variation of reciprocal susceptibility with temperature as per these three laws for different magnetic materials is shown in the figure.

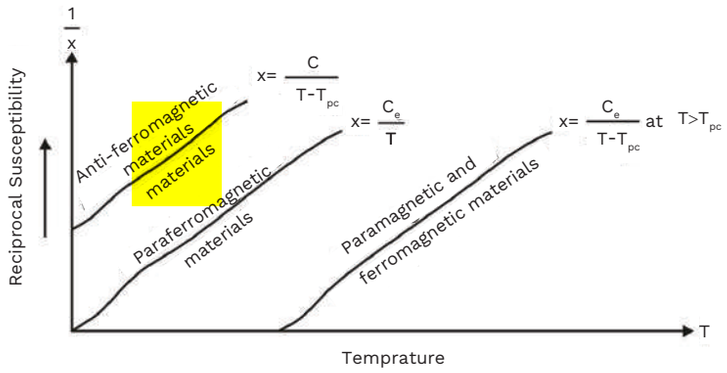
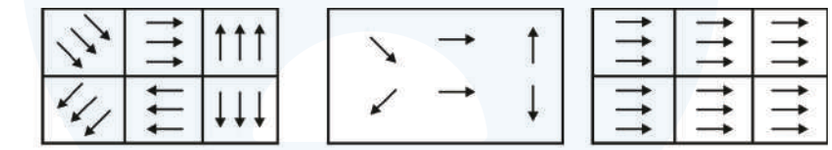


Fig. 4.8

Domain Theory

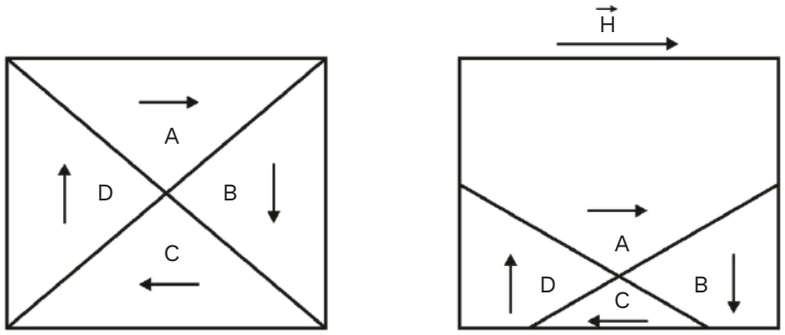
- According to Weiss's domain theory, a ferromagnetic material consists of a large number of regions or domains, which are permanently magnetised.
- The magnetic moments in the individual domains are all aligned parallel to one another when the temperature is lower than the Curie temperature. Each domain is magnetically saturated and has a net magnetic moment.
- The direction of the net magnetic moment in each domain varies, thereby resulting in the overall magnetic moment or magnetisation of the material being negligible or zero.
- Above the Curie temperature, the one directional alignment of magnetic moments in different domains may disrupt, and the material may lose its ferromagnetisation on disruption of orderly aligned domains.
- A ferromagnetic material with magnetic moments aligned in different domains is shown in the figure.



(a) Domains with difference directions of magnetic moments | (b) Random magnetic moments orientation | (c) Aligned magnetic moments

Fig. 4.9

- The domains are separated by domain walls in which magnetic moments (spin direction) change from the preferred direction of one domain to the preferred directions of the neighbouring domains, as shown in Figure (c).
- To understand how the saturation of the magnetisation takes place, consider a ferromagnetic specimen having four domains with a preferred direction of orientation of magnetic moments, as shown in Figure (a), which gives zero resultant magnetisation. This is the reason why steel is ferromagnetic, but it is not a magnet in the absence of a magnetic field.
- When an external magnetic field is applied to the material, the domains having an alignment of magnetic moments parallel to the applied magnetic field grow in size at the expense of other domains having magnetic moments aligned in other directions.
- The sizes of domains change by virtue of the motion of domain walls. As shown in Figure (a) domain A grows at the expense of a reduction in the sizes of domains B, C, and D. The motion of domain walls can be considered to take place to reduce the size of domains from which certain magnetic moments have been removed and realigning them in the direction parallel to the applied magnetic field, thereby domains having a parallel orientation to the applied magnetic field grows in size.



(a) Domains configuration giving zero resultant magnetisation

(b) Magnetisation by preferred domain growth

Fig. 4.10

- Due to an increase in the orientation of magnetic moments in the direction of the magnetic field, the magnetisation of material increases. A stage is reached when the whole material consists of one enlarged domain, thereby, giving rise to the saturation of the magnetisation inside the material.
- The saturation of magnetisation occurs due to the motion of domain walls and the rotation of existing magnetic moments in the direction of the magnetic field.
- It can be seen that as follows.
  - The spontaneous magnetisation takes place in the material in certain preferred directions, thereby forming domains. When an external field is applied, some domains which are easy to rotate, align themselves in the directions of the field.
  - The coercive force needed to neutralise the spontaneous magnetisation in ferromagnetic material can be understood in terms of the mobility of domain walls. The mobility of domain walls depends on the impurities and lattice imperfections. Hence, it is possible to design a material by adding or subtracting impurities in the material so as to obtain the desired value of coercive force.

Magnetisation Curve and Magnetic Hysteresis Loop

- Magnetic hysteresis means that the magnetisation of a ferromagnetic material depends not only on the present magnetic field but also on its past subjection to magnetic fields.
- When an external magnetic field is applied to a ferromagnetic material such as iron, the magnetic moments in different domains rotate and align themselves with the field. When the field is removed, part of the alignment of magnetic moments in different domains is retained, and the material behaves as a magnet.
- Once magnetised, the material requires a magnetic field in the opposite direction to achieve its demagnetisation. This is the retentive magnetic effect that provides the element of memory to a hard disk drive.

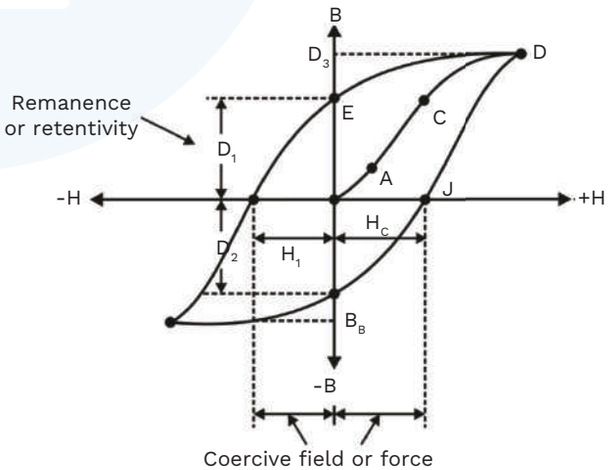


Fig. 4.11

- The relationship between magnetic field strength (H) and magnetisation (M) in the material is non-linear, and this is plotted as the BH curve, which is called the

magnetisation curve. In a virgin material, the curve rises in the first region OA. The second region, AC, has almost a constant slope. The third region, CD, has a decreasing slope, and the curve almost becomes horizontal at D. At this condition, the material has attained the saturation of magnetisation (Bs).

- The amount of magnetisation does not increase further from here even when the magnetic field is increased, as shown in the figure. If the magnetic field is reversed, the magnetisation decreases slowly at first, and its residual value remains B_r (shown by OE) even when the magnetic field (H) becomes zero.
- It means that magnetic moments in certain domains continue to remain aligned in the direction of the erstwhile applied magnetic field even when H = 0. This residual in a ferromagnetic material is called retentivity or remanence in a ferromagnetic material.
- In case the magnetic field is applied in the opposite direction, the magnetic moments in domains now tend to reverse their alignment, thereby, losing their magnetisation. At a certain value of reverse magnetic field (-H), the complete magnetisation is lost (M = 0), and this reverse magnetic field is called the coercive magnetic field or H_C.
- The process of reversal of alignment of magnetic moments in domains continues to give a net magnetisation in the opposite direction as the applied magnetic field is increased in the opposite direction and the magnetisation curve E-G is obtained.
- The saturation of magnetisation is obtained at point G, similar to that obtained at point D but in the opposite direction.
- When the magnetic field in the opposite direction is removed, the B–H curve follows path Gl, and there is magnetic retentivity of OE (B_r). On further application of magnetic field, the B–H curve follows path/J–D. The curve DEEGIJD is called a magnetic hysteresis loop.
- In the earlier stages of magnetisation below the saturation regions of the hysteresis curve, domain growth is dominant, and this domain growth is almost complete as the saturation region is approached. After this, the fully grown domain has to rotate to align completely with the magnetic field directions. The energy needed to rotate the fully grown domain is much greater than that required to move the domain walls during the domain growth. This is why the slope of the B–H curve decreases as the curve approaches saturation regions.
- When an alternating magnetic field is applied and the B–H curve makes a hysteresis loop, energy equal to the area of the loop is dissipated as heat. The power loss due to hysteresis in devices such as transformers is dependent on the number of times the full hysteresis loop is made per second.

Soft Magnetic Materials

- Soft magnetic materials are materials having high permeability, low hysteresis losses, and low coercive force.
- These materials are characterised by a steeply ascending magnetisation curve and a narrow B–H curve.
- These materials are easy to magnetise but lose their magnetisation quickly and almost completely when the magnetisation force is removed.
- The process of magnetisation is rapid when it involves moving the domain wails so that favourably oriented domains can grow, while unfavourable domains can shrink. As the domain walls can move easily, the soft magnetic material is easily magnetised. This is the reason why such a material is called a soft magnetic material.
- The hysteresis loop of soft magnetic material is narrow, as shown in the figure, which has a small area under the B–H curve. Thereby, the power losses are low, and it can be used as a magnetic material at high frequency.

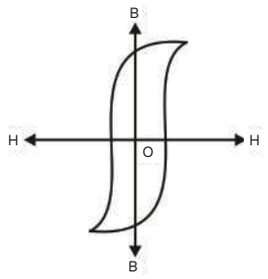


Fig. 4.12

- Soft magnetic materials must have easily movable domain walls, and domain walls tend to become immovable when they have dislocations, voids, and imperfections. The cold worked material has a high number of dislocations and has to be annealed to reduce the amount of dislocation to make it suitable as soft magnetic material. Non-metallic soft magnets made of ferrite and garnet must be free of voids.
- These materials are prone to inheriting voids when they are manufactured through the powder compacting process.
- These materials cannot be used as soft magnetic materials with voids, as voids prevent easy movement of domain walls.
- Besides, hysteresis loss the other source of energy loss in soft magnetic materials is eddy current loss. When the magnetic flux in a material is changing, an emf is induced, which sets up an eddy current in the material, thereby resulting in $i^2 R$ resistance losses. As the resistance of soft materials is small, the resistance losses are considerable due to the high value of current.
- The transformer core tends to have high eddy current losses when the core is not laminated or made of sheets, as shown in the figure. The laminated core has high resistivity, thereby lower eddy current losses.

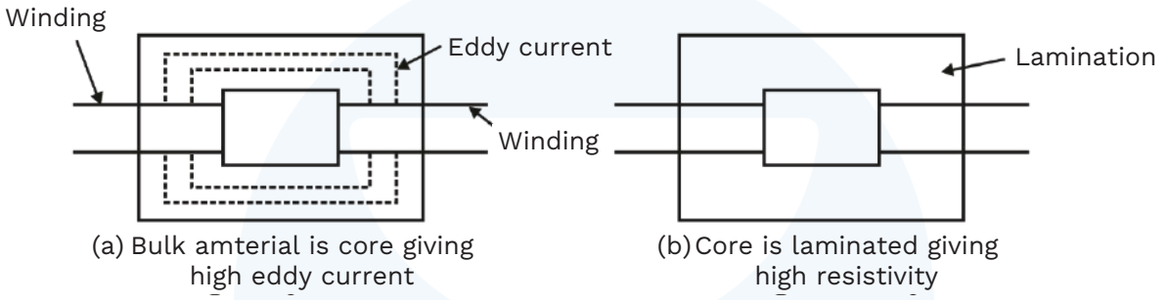


Fig. 4.13

- There are two ways to modify the properties of a soft material, which includes controlling the movement of domain walls by inclusions and grain boundaries and alloying the soft magnetic material to increase the electrical resistivity for minimising eddy current losses.
- The soft magnetic materials are given below.
  - **Iron silicon (0.8 to 4.8 per cent) alloys:** they are widely used soft magnetic materials. The addition of silicon to iron increases the electrical resistivity, thereby reducing (i) eddy current and (ii) hysteresis losses. Silicon also improves magnetic permeability, hardness, tensile strength, and brittleness. Suitable rolling and annealing of iron silicon sheets induce preferred crystal orientation. The direction of easy magnetisation has to lie in the rolling direction. Therefore, it is easier to magnetise the sheet of iron silicon alloy.
  - **Iron nickel alloy:** soft iron nickel alloys are preferred over iron silicon alloys in the applications of communication equipment for high sensitivity and high fidelity. Materials such as 45 Permalloy, Mumetal, and Supermalloy are commonly used soft iron nickel alloys.
  - **Ferrites:** they are soft magnetic materials that have a composition of MeO, Fe_2O_3, where Me stands for a bivalent metal such as nickel, manganese, zinc, copper, magnesium, and cadmium. The important characteristics of ferrites are their fairly constant permeability and a narrow hysteresis loop, or losses in an alternating magnetic field. They are used mainly for making cores for coils operating at elevated temperatures and high frequencies, as used in telephony and radio. The ferrites are made by a powder metallurgy technique, also called a sintering process. The sintering

process in blocks is shown in the figure. As ferrites are ceramics with covalent bonding (no free electron), the resistivity of ferrites is very high. They are mechanically hard, brittle, and difficult to be machine.

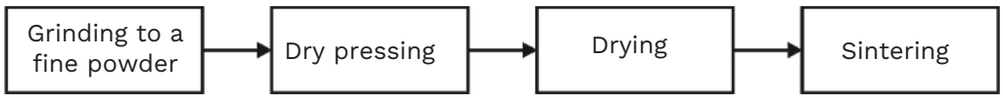


Fig. 4.14

- **Metallic glasses:** These are iron base alloys containing 15 to 25 per cent of Si, B, and C. These materials do not crystallise but solidify into a metallic glass. Grain boundaries are absent in the glassy matrix, so domain walls can move without any interference from grain boundaries. This reduces the hysteresis losses, owing to the larger concentrations of the impurity atoms, the electrical resistivity is higher, thereby reducing the eddy current losses.

- The types of applications for soft magnetic materials fall into two categories of AC and DC.
- In DC applications, the material is magnetised to operate and then demagnetised at the end of the operation. An electromagnet, for example, on a crane being used at a scrap yard will be magnetised to attract the scrap iron items and then demagnetised to drop the iron items at the desired place.
- In AC application, the soft magnetic material is continuously cycled from being magnetised in one direction to the other during the period of operation, as it happens in a power supply transformer.
- For DC applications, the important consideration is the permeability of material, while for AC applications, the important consideration is the energy loss in the hysteresis loop.
- Hysteresis loss can be reduced by reducing the coercive force of the material. Another loss by eddy current can be reduced by increasing the resistivity of the material.

Hard Magnetic Materials

- Hard magnetic materials can retain a magnetic field. They have high coercive force and sufficient permeability. As they have high remanence (retentivity), they are commonly used for making permanent magnets. As they have a large coercive force, they have a large area in the B–H curve or hysteresis loop, as shown in the figure, in comparison to soft magnetic materials.

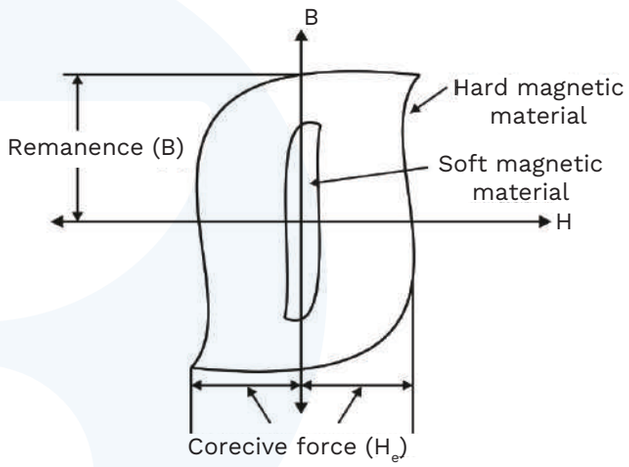


Fig. 4.15

- The hysteresis behaviour of hard magnetic materials is related to the difficulty with which the domain walls can move. By impending domain wall motion, the coercivity and susceptibility are enhanced to such an extent that a large external field is needed for the demagnetisation of the material.
- The hard magnetic materials must have a high residual induction or retentivity (B_r), a large coercive field or force (H_c), and a high energy product ($B_r \times H_c$) needed to demagnetise the material.
- Hard magnetic materials can be classified as conventional hard materials or high energy hard materials.
- The conventional hard magnetic materials have BH products ranging between 2 and

80 J/m³. The materials such as magnet steels, Cunife alloys (Cu–Ni–Fe), and Alnico (AL–Ni–Co) alloys are conventional hard materials.

- The hard 'magnet steels' are generally alloyed with tungsten, or chromium. These two elements can easily combine with carbon in the steel to form tungsten and chromium carbide precipitate particles, which help prevent domain wall motion, thereby making it difficult to magnetise and demagnetise these materials.
- High energy hard magnetic materials have a high value of BH product of over 80 J/m³. The examples of high energy hard magnetic materials are Samarium cobalt permanent magnets (SmCo) and Neodymium-iron-boron permanent magnets (Nd, Fe, B). The material SmCo is an alloy having a combination of cobalt, or iron with a light rare earth element.
- It has a BH value of 120 to 240 kJ/m³ and relatively large coercivity. It is manufactured using the powder metallurgy technique. Samarium (Sm) is relatively expensive, and the supply of Co (cobalt) is uncertain. Hence, sintered neodymium iron boron alloys are commonly used for hard magnets. These alloys are also manufactured by a powder metallurgy technique.
- Hard magnetic materials are used in electric motors, loudspeakers, electric meters, and holding devices. Permanent magnets are far superior to electromagnets as their magnetic field is continuously maintained without any expense of electric power. Permanent magnets are much smaller in size.
- The applications of hard magnetic material are in devices such as cordless drills and screwdrivers, fan motors, wipers and washers of automobiles, clocks, audio recorders, video recorders, lightweight earphones, hearing aids, computer peripherals, and speakers with audio systems.

Magnetic Storage

- Magnetic storage is nothing but the storage of data on a ferromagnetic medium that can be magnetised.
- A magnetic storage device works on the principle of magnetisation of magnetic medium using a magnetic field varying as per the applied electric signals.
- The basic principle is that a magnetisable medium can be magnetised by an electric magnet. It can then be 'read' magnetically, thereby recording data can be played back.
- Magnetic storage was first used for audio recording using a magnetisable type and an electric magnet.
- The modern audio recording now uses a digital magnetic storage device. A digital magnetic storage pattern has a digital stream of data in the form of a binary signal containing a sequence of 1's and 0's, i.e., a digital signal consists of only two main states.
- Each head of the magnetic storage device has a tiny electromagnet, which consists of an iron core wrapped with wire, as shown in the figure.

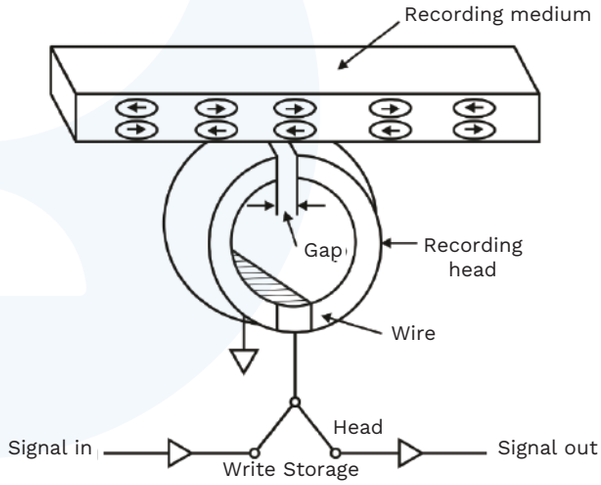


Fig. 4.16

- The electromagnet applies varying magnetic flux on the iron-oxide coated tape, and signals are recorded as magnetisation patterns on the tape.
- There are many types of magnetic storage devices, which include audio tape, consisting of plastic tape coated with iron oxide (ferromagnetic material), which is permanently magnetised, when exposed to a varying magnetic field, a computer hard disk, a floppy disk, videotape, and ATM credit cards.

Practice Exercise

1. Basic source of magnetism is:
(A) charged particles alone
(B) movement of charged particles
(C) magnetic dipoles
(D) magnetic domains

2. The unit for magnetic flux density is:
(A) Wb/m^2
(B) Wb/A.m
(C) A/m
(D) Tesla/m

3. Magnetic permeability has units as:
(A) Wb/m^2
(B) Wb/A.m
(C) A/m
(D) Tesla/m

4. Magnetic field strength units are:
(A) Wb/m^2
(B) Wb/A.m
(C) A/m
(D) Tesla/m

5. Examples of diamagnetic materials:
(A) superconductors
(B) alkali metals
(C) transition metals
(D) ferrites

6. Examples of paramagnetic materials:
(A) superconductors
(B) alkali metals
(C) transition metals
(D) ferrites

7. Examples of ferromagnetic materials:
(A) superconductors
(B) alkali metals
(C) transition metals
(D) ferrites

8. Examples of anti-ferromagnetic materials:
(A) salts of transition elements
(B) rare earth elements
(C) transition metals
(D) ferrites

9. Example for ferrimagnetic materials:
(A) salts of transition elements
(B) rare earth elements
(C) transition metals
(D) ferrites

Answer Key

1. (B) **2.** (A) **3.** (B) **4.** (C) **5.** (A) **6.** (B) **7.** (C) **8.** (A) **9.** (D)

5 Electric Properties of Materials

Introduction

- The degree to which a material conducts electricity, or fails to do so, is a defining feature. This characteristic is used to categorise materials into three broad categories: conductors, semiconductors, and non-conductors.
- More than 25 orders of magnitude separate the conductivity (σ) of various materials at ambient temperature. This range increases to 40 orders of magnitude when low-temperature superconductivity is included (with an estimated conductivity for superconductors of roughly 10^{20} $(\text{-cm})^{-1}$.
- The proportion between the diameter of the universe (10^{26} m) and the radius of an electron (10^{-14} m) is equivalent to this magnitude of variation in a physical attribute. Conductors, semiconductors, insulators, and superconductors are all types of electronic materials. The world has been radically altered by this class of materials.
- Electronic materials are used in a wide variety of contexts, including spark plugs made of alumina and copper wires for electric power transmission, components for wireless communications, insulators, capacitors, active matrix displays, solar cells, active matrix displays, and Si and Ga arsenide based computer chips.
- In the field of electronics, some significant gains have been made thanks to recent developments in material science. Recently, advances in technology have allowed us to create ceramics that serve not only as superb insulators but also as semiconductors and superconductors. In a similar vein, there are polymers that both conduct electricity and show superconductivity. The table below provides a quick and generalised breakdown of the many types of electrical components.

Ohm's Law and Electrical Conductivity

- Current, measured in amperes (A), or the rate at which charge is transferred over a circuit, is related to voltage, in volts (V), through Ohm's law.

  V = IR

  where R is the resistance (Ohm, Ω)
- Current, measured in amperes (A), or the rate at which charge is transferred over a circuit, is related to voltage, in volts (V), through Ohm's law.
- The resistance (in Ohms), R is the variable in the formula V = IR.
- V is represented by joule per coulomb (J/C), I by ampere (C/s), and R by ohm.
- R's value is affected by the shape of the specimen and, for certain materials, is independent of the current. Size, shape, and material attributes all contribute to a resistor's resistance (R), which is defined as

$$R = \rho \frac{1}{A} = \frac{1}{\sigma A}$$

  where l is the resistance of the resistor (in m), or the distance between the two voltage measurement locations.

  'A' is the cross sectional area (m^2) perpendicular to the direction of the current.

  ρ' is the electrical resistivity (Ω – m)

  σ which is the reciprocal of 'ρ' is the electrical conductivity [$(\Omega - m)^{-1}$].
- Electrical conductivity (σ) is used to specify the electrical character of a material. It is reciprocal of the resistivity, i.e.,

$$\sigma = \frac{1}{\rho}$$

- Thus, resistivity, or conductivity allows one to compare different materials: resistivity is a microstructure sensitive property.

  In addition, Ohm's law may be expressed as

  J = σ E

  'J' (A/m^2) is the current density (I/A), i.e., the current per unit area of the specimen.

  'E' is the electric field intensity or the voltage difference between the two points divided by the distance (l) separating them, i.e.,

$$E = \frac{V}{l}$$

One can also determine the current density (J) as

$J = nq\bar{v}$

'n' is the number of charge carriers (carriers/m³)
'q' is the charge on each carrier (1.6×10^{-19}C)
$\bar{v}$ is the average drift velocity (m/s) at which charge carriers move.
Thus,
$\sigma E = nq\bar{v}$
Or

$$\sigma = nq\frac{\bar{v}}{E}$$

- The term $\frac{\bar{v}}{E}$ is the mobility of carriers and is denoted by ji (m²/V-sec). In case of metals, μ is the mobility of electrons.

  Therefore, $\mu = \frac{\bar{v}}{E}$

  Thus, $\sigma = nq\mu$
- Charge carriers in metals are electrons, whereas charge carriers in semiconductors are both electrons and holes. Expression for conductivity gets modified for a semiconductor as below.

  $\sigma = nqu_n + pq\mu_p$

  Where μ_n and μ_p are the mobilities of electrons and holes respectively. The terms n and p represent the concentrations of free electrons and holes in a semiconductor.
- It is possible to divide the universe of electronic substances into four types based on their electrical conductivity: superconductors, regular conductors, semiconductors, and insulators or dielectrics.
- To be classified as a dielectric or insulator, a material must have a resistance of more than 10^{12} Ω-cm or a conductivity less than 10^{-12} Ω^{-1} cm^{-1}. Those materials classified as conductors have a conductivity of more than 10^3 $\Omega^{-1}cm^{-1}$ and a resistivity of less than 10^{-3} Ω cm. Materials with a conductivity of less than 10^3 Ω^{-1} cm^{-1} but more than 10^{-12} Ω^{-1} cm^{-1} are considered semiconductors.

Energy Band Structure in Solids

- The amount of electrical conductivity is discovered to be substantially reliant on the number of electrons accessible to engage in the conduction process in all conductors, semiconductors, and some insulating materials.
- Conduction does not include every single electron in every atom. The organisation of electron states, or energy levels with regard to energy, and subsequently the way in which these states are filled by electrons, are directly connected to the number of electrons accessible for conduction in a given material.

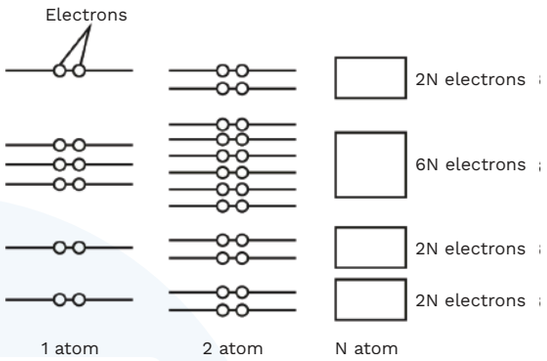


Fig. 5.1

- The energy levels occupied by electrons in a single atom are quantised. Pauli's exclusion principle states that there can be no more than two electrons in any given energy level, hence the 2s level of a single atom consists of a single energy level and two electrons.
- In all, there are three energy levels and six electrons in the 2p level. For a solid to be produced from a collection of atoms, the Pauli principle stipulates that no more than two of those electrons may share the same energy.
- When a solid is produced, the electrons' discrete energy levels merge to form continuous energy bands. This means that there is a total of N energy levels in the 2s band of a solid and that each of these levels is occupied by 2N electrons.
- In addition, there are N energy levels and 2N electrons spread throughout each of the 2p levels. As there is some overlap between the 3 2p bands, but they may also be described as a single, wide 2p band with 3N energy levels, and 6N electrons.
- Take the example of sodium. Sodium's electrical configuration is $1s^2$ $2s^2$ $2p^6$ $8s^1$ and it has 11 electrons. The equilibrium interatomic spacing of atoms in solid sodium determines

the energy inside the bands. The empty 3p energy levels, which are separated from the 3s band by an energy gap, make up the conduction band, whereas the 3s band is the valence band. The amount of splitting is related to the distance between atoms, and it starts at the outermost electron shells as they are the first to be disrupted when atoms collide. Although, there are distinct energy levels inside each band, the gaps between them are quite narrow.

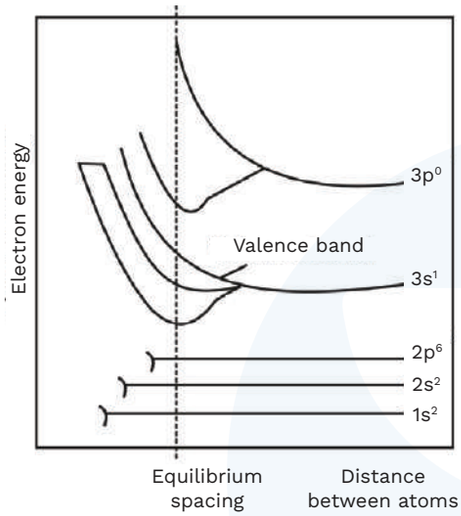


Fig. 5.2

- For the electron subshells closest to the nucleus, band formation may not happen at the equilibrium spacing. Furthermore, there may be gaps between neighbouring bands, and often the energies that fall inside these band gaps are not accessible to electron occupancy.

Classification of Materials Based Upon Energy Band Diagram

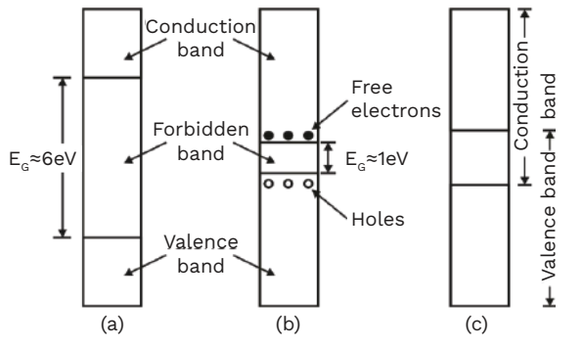


Fig. 5.3

Metals

- No prohibited energy gap exists between the valence and conduction bands. As can be seen, the two bands truly overlap. (Figure c)
- A huge number of free electrons are already present in metal, making it an excellent conductor even without the input of other energy sources like heat or light. Metals such as aluminium, and copper are some examples.

Semiconductors

- As shown in Figure (b), it is the energy band gap present in a semiconductor. At absolute zero, semiconductors and insulators are structurally identical, except for the band gap EG, which is significantly lower in semiconductors (EG = 1 eV) than in insulators.
- The comparatively tiny band gaps of semiconductors make it possible for a fair amount of thermal or optical energy to excite electrons from the lower (valence) band to the higher (conduction) band.
- Substantial increases in conductivity due to thermal, or optical energy set semiconductors apart from insulators. Ge and Si are two such elements.

Insulators

- As shown in Figure (a), it is the energy band diagram for an insulator. The gap between the full valence band and the unoccupied conduction band is relatively large (6 eV). This makes it very difficult, if not impossible, for an electron in the valence band to cross across to the conduction band.
- An insulator does not conduct electricity at normal temperatures. At high temperatures, or when a high voltage is put across it, however, it may conduct. This is referred to as an insulator failure. Diamond is a good example.

Metals' Electrical Resistivity

- Nearly all metals are excellent conductors of electricity. The high conductivities of metals may be attributed to the abundance of free electrons in the material's conduction band.
- The resistivity of metal rises as the number of crystallographic flaws rises because these

imperfections act as scattering centres for the electrons that carry the metal's conduction current (or lower the conductivity). The severity of these flaws is affected by the metal's temperature, composition, and level of cold work.

- Experiments have shown that the overall resistivity of a metal is equal to the sum of the contributions from thermal vibrations, impurities, and plastic deformation. For those who prefer numbers over words, consider the following representation.

$$\rho_{total} = \rho_t + \rho_i + \rho_d$$

Where t, i, and d each stand for the resistivity due to temperature, impurities, and deformation, respectively. The aforementioned formula is sometimes referred to as Matthiessen's rule.

- Cold working is one example of a mechanical treatment that causes localised strain in the material, which in turn increases the material's resistivity. In comparison to annealed copper, the conductivity of a hard drawn copper wire is much lower.
- When a material ages, its hardness increases, and its resistance rises, while its conductivity falls.
- Outside of this, temperature and alloying are the most important factors in determining metals' resistivity, the effects of which are explored further down.

Temperature

- As the temperature of a conducting material rises, the average collision time of its electrons decreases. As conductivity is related to average collision duration, conductivity decreases as temperature rises.
- Given that resistivity is inversely proportional to conductivity, the resistivity of a conducting material will grow as its temperature rises. What follows is a formula showing how the resistivity of conducting material changes with temperature.

$$\rho_{T2} = \rho_{T1} (1 + \alpha\Delta T)$$

- The temperature difference, denoted by T, is defined as T2 minus T1. A constant proportional to the temperature, denoted by the 'α' symbol.

Alloying

- At low temperatures, the resistivity's thermal component drops to zero, and at very low temperatures, it almost disappears altogether.
- The residual component of resistivity arises from material impurity and lattice imperfection, and it is the third component of resistivity.
- Even at 0 Kelvin, the residual component will continue to exist. As a result, the resistivity may be induced by alloying. The electrical conductivity of a solid solution alloy quickly decreases with increasing alloy percentage because alloys tend to have a less uniform structure than pure metals.

$$\rho_{alloy} = \rho_{thermal} + \rho_{residual}$$

- Once the temperature reaches a particular threshold, known as the Debye temperature (D), the substance begins to undergo a phase transition. The alloy's resistance rises linearly with temperature.

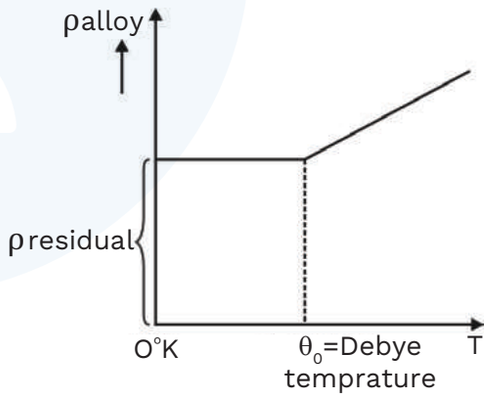


Fig. 5.4

Wiedemann-Franz Law for Thermal Conductivity of Metals

- All solids are heat conductors. Metals, in general, are excellent thermal conductors. Some metals are excellent thermal, as well as electrical conductors. Similar to electrical conductivity, heat is conducted mostly through valence (free) electrons in such conductors.

- The word thermal conductivity is used to describe the rate at which heat is transferred through a substance and is defined as the product of two quantities. Electrical lattice phrase, when the electrons in a material are held securely by individual atoms, or molecules, the material is said to be insulating because thermal energy is absorbed, or transported only by the lattice vibrations, and the electrons do not play a role.
- However, metal electrons are not strongly attracted to the cores of individual ions, thus they can absorb and transfer heat energy by means other than thermal vibrations.
- In the positive x-direction of the bar, the temperature gradient dT/dx is directly proportional to the flux of thermal energy (heat), J_Q, across a unit area per unit time (heat energy divided by time and area), and the resulting formula is

  $$J_Q = -K\left(\frac{dT}{dx}\right)$$

  where J_Q is expressed in Watts/m^2, in degrees C per metre
- The thermal conductivity (k) is the proportionality constant and has the unit of (J/msK) or (W/mK). The negative sign indicates that heat is transferred from hotter to colder areas, the reverse of what would happen in a temperature gradient. Wiedemann-Franz law is a link between two characteristics of metals that reflects their proximity.

  $$K = \sigma LT = \frac{LT}{Á}$$

- The Lorentz number, denoted by the symbol L, whose value is theoretically anticipated to be $2.443 \times 10^{-8} W\text{-}\Omega/K^2$.
- Thermal conductivity at 300 K varies from 15 W/m-K to 430 W/m-K for the most popular metals.

Thermoelectric Phenomenon

- An emf is created across the junction when two wires of different metals are linked at one end and the junction ends are kept at different temperatures. Up to a particular temperature, this emf induces a current to flow from the hot junction to the cool junction. Thermoelectric currents are those generated by the heating of metal connections. This kind of emf is known as a thermal emf.

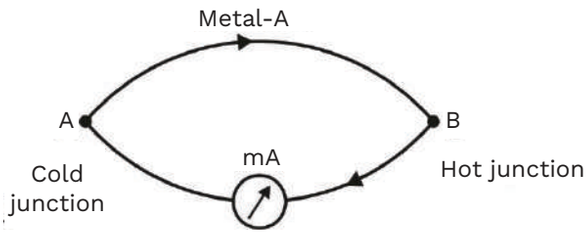


Fig. 5.5

- In other words, the contact potential changes as the temperature does. Let's pretend a loop is formed by joining the ends of two wires of different materials (say, a copper wire and an iron wire). A single iron/copper interface. After that, a voltage difference, denoted as

  $$\Delta V = a\left(T - T_0\right) + b\left(T - T_0\right)^2$$

  It is seen between two 'thermocouples.' Where T and T_0 are high and low temperatures, respectively, and a and b are constants.
- Often referred to as thermoelectric power, a thermocouple's output voltage varies with temperature as shown by the equation.

  $$\frac{\Delta V}{\Delta T} = a + b\left(T - T_0\right)$$

- Thermoelectric power may be thought of as a linear function of temperature for particular combinations where the constant b is close to zero.
- Thermocouple materials must be stable over long periods, in terms of the emf they create; they must also be resistant to corrosion and oxidation; the emf generated per degree Celsius must be high, and the devices must be suitable for usage over a wide temperature range. At times, the price may be a major factor. Even in out-of-the-way places, thermocouples are utilised as sturdy, cheap, and quick probes for monitoring temperatures.

Superconductivity

- Materials in a superconductive state have a resistance of zero. In the superconducting condition, diamagnetism is seen in the material. In the first superconductor to be found,

mercury (Hg), the resistance dropped entirely below 4.2 K, with the transition from normal to superconductivity happening across a short range of temperature in the order of 0.05 K.

- The critical temperature or transition temperature is the temperature at which superconductivity is observed (Tc). Materials that exhibit superconductivity at a certain temperature and magnetic field are called superconductors. The normal state of a superconductor is a condition in which the material does not exhibit any superconducting behaviour.

Meissner's Law

The Meissner effect is 'the outflow of magnetic flux from the inside of a piece of superconducting material when the material transitions to the superconducting phase.' It is not the case that zero resistance and Lenz's law always lead to the Meissner effect.

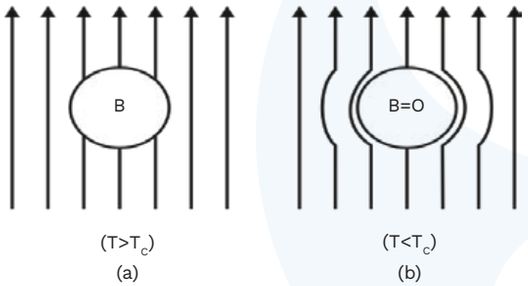


Fig. 5.6

Effect of Magnetic Field (H)

- The superconducting state may be removed not only by increasing the temperature, but also by applying a magnetic field to the material (H). At the transition, the lines of induction are pushed outward inside the specimen when a superconductor is cooled in a magnetic field below its critical temperature T (transition temperature). This is represented by the equation below.

B = 0 in superconducting state,

$$B = \mu_0 \left(\vec{H} + \vec{M}\right)$$

$$0 = \mu_0 H + \mu_0 M$$

$$\frac{M}{H} = \chi_m = -1$$

Since $\chi_m = \mu_r - 1$

Therefore, $\mu_r = 0$

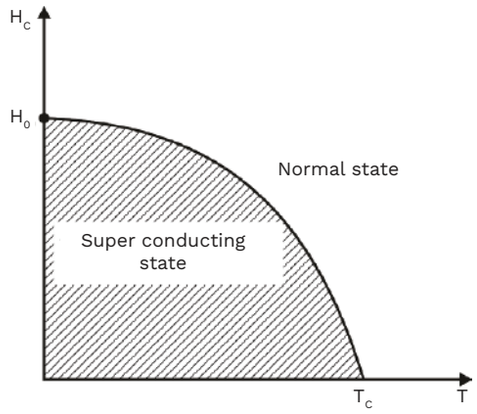


Fig. 5.7

- Perfect diamagnetism is an attribute of a superconductor. This is because super-conductors are diamagnetic and hence, strongly oppose magnets. Critical field (Hc) is defined as 'the minimal applied magnetic field required to eliminate superconductivity and restore the normal resistivity.' Superconductivity may be destroyed by applying a magnetic field greater than its critical value, (Hc), and this value depends on the temperature in the range T = T (c).

$$H_c = H_0 \left[1 - \left(\frac{T}{T_c}\right)^2\right]$$

Where, H_0 is the critical field at absolute zero and T_c is the transition temperature.

Silsbee's Rule

- The magnetic field which causes a super-conductor to become normal from a superconducting state is not necessarily externally applied, it may arise as a result of electric current flow in the conductor. In general,

$$\int_C \vec{H} \cdot d\vec{l} = I_{enclosed}$$

$$H.2\pi r = 1$$

$$H = \frac{1}{2\pi r}$$

As H increases, I also increase.

When $I = I_c$ then H becomes equal to H_c

$I_c = 2\pi r\, H_c$

- This is called Silsbee's rule. When ($I < I_c$), refers to superconductivity. When ($I > I_c$), superconductivity is destroyed. It limits the use of superconductor wire in strong electromagnets.

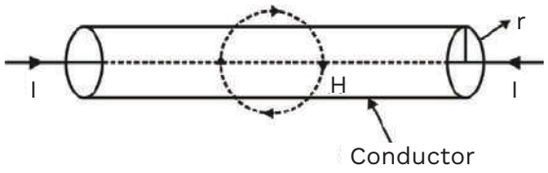


Fig. 5.8

More Properties of Superconductivity

- **Frequency effect:** superconductivity is mainly observed for DC and up to radio frequencies. It is not observed for high frequency. As frequency increases, the resistance of material increases due to the skin effect, and the superconductivity is observed only from DC to RF. At infrared frequencies (10^{13} Hz), superconductivity is destroyed and it becomes a normal conductor.
- **Entropy:** entropy increases on going from a superconducting state to a normal state.
- **Thermal conductivity:** in an ideal super-conductor, there is a marked drop in thermal conductivity when superconductivity sets in.
- **Isotope effect:** here critical temperature (T_c) of the superconductor varies with isotopic mass (M) as the relation,

$$T_c = \frac{1}{\sqrt{M}}$$

Where 'M' is the mass of the isotope.

Superconductor Subclasses

- There are two distinct classes of super-conductors, distinguished by their magnetic characteristics.

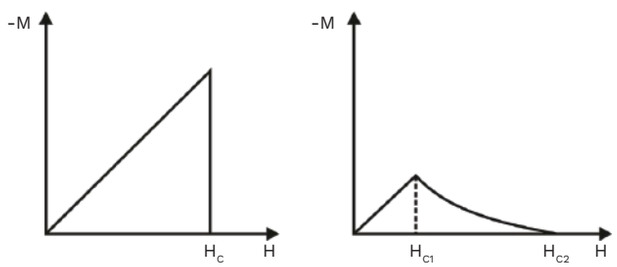


Fig. 5.9

Type I Superconductors

- Soft superconductors are type l, or perfect superconductors that are entirely diamagnetic in the superconducting state up to the critical field H_c and then return to the normal state (or ideal) as shown in the figure above. This class of metals includes the more malleable ones, such as lead and indium.
- These superconductors have a relatively weak critical field. Their melting point is rather low. These materials exhibit the Meissner effect and follow Silsbee's rule. There is a sudden change in behaviour from the normal to the superconducting state. The Hc and Tc values are rather low.

Type II Superconductors

- Hard metals and alloys with various magnetisation properties are referred to as type ll superconductors.
- High transition temperature, high critical field, incomplete Meissner effect, breakdown of Silsbee's rule, and/or a wide transition area are all features of these phases.
- It is also known as a hard superconductor and exhibits perfect behaviour up to a lower critical field H_{c1}, after which the magnetisation progressively decreases and becomes zero at an upper critical field H_{c2}.
- The zone between H_{c1} and H_{c2} where the Meissner effect is incomplete is called the vortex region. This is a mixed state, another term for it. The superconductor is in a partially diamagnetic condition here due to the presence of magnetic flux that penetrates the material.

Uses of Superconductors

- In electric power production and transmission, superconducting materials are of interest because of the promise of reduced joule heating losses. It also has use in electronics. Spacecraft design and performance may see significant improvements from the use of superconducting technology.
- Strong magnetic fields may be produced with its help. There is potential for its use in the construction of modulators and rectifiers for

the detection of modulated high-frequency waves. Cryotrons, electronic switching devices, utilise these components. It has been found that superconducting magnets may be used in situations such as superconducting magnets for energy storage, Magnetic Resonance Imaging (MRI) and other uses, magnets for nuclear fusion, and generators and motors, magnetically levitated transportation, and magnets for high-energy physics.

Insulators

- According to the literature, a perfect insulator has no free electrons and uses all of its valence electrons to create bonds.
- An electrical insulator, therefore, is a non-metal with a full valence band at zero degrees Kelvin and a sizeable energy gap. As a result, the electrical conductivity at ambient temperature is very small, being less than around 1010 $(\Omega-m)^{-1}$. The resistivity of a good insulator may be as high as 1014 $(\Omega-m)^{-1}$.
- Polyethylene, bakelite, lucite, mica, PVC, rubber, porcelain, and so on are only some of the common electrical insulating materials. At normal temperatures, insulating materials like polymers and ionic ceramics predominate.
- Insulating qualities are degraded by thermal agitation and defects, but like with actual materials, some electrons are always present to carry electricity. Higher temperatures improve the conductivity of insulators because more and more conducting electrons are made available.
- Materials that are insulators at 0 K but exhibit large conductivities at normal temperatures are transformed into semiconductors. Good insulating materials must meet electrical, mechanical, thermal, and chemical standards.
- To limit electrical leakage, an insulating material should have high resistance and strong dielectric strength to prevent breakdown or puncturing at a higher voltage and little dielectric loss. As insulators are often selected based on their volume rather than their weight, a low density is generally favoured.
- The thermal and electrical characteristics of liquid insulators may be guaranteed to be constant, if their viscosity is kept constant. Transform oil, hydrogen, and helium are all examples of liquid and gaseous insulators that double as coolants. Good thermal conductivity is a desired attribute for such materials.
- The insulator's thermal expansion has to be low to avoid mechanical damage. And, if it can catch fire, it ought to be able to put itself out. Insulating materials should have chemical resistance to oils, liquid gas fumes, acids, and alkalis. Water decreases insulation resistance and dielectric strength; thus the insulator should not soak it up either.
- Different types of insulation have different mechanical requirements. Insulators used in electric machinery must be robust enough to tolerate vibration. In such a scenario, it would be ideal, if the material could also transmit heat well. Insulating materials having high permittivity, such as those rich in electronic and ionic polarizabilities, are utilised to create dielectric capacitors. One such substance is titanium oxide, which has a permittivity of 100.

Dielectrics

- There are not enough electrons in electrical insulators for them to conduct electricity normally. When an electric field is applied to one of these materials, electric dipoles are formed, and the substance subsequently becomes polarised, resulting in some fascinating electrical characteristics.
- Dielectrics are a class of materials distinguished by the prominence of polarisation effects in their operation. If a material's primary purpose is to prevent electrical current from passing through it, then it is referred as an insulator. A substance is considered dielectric, if its primary use is as a charge storage medium and it does not carry electricity. The most prevalent characteristics of dielectric materials are as follows:
  - Six parameters may be used to characterise dielectrics, which are dielectric strength, dielectric constant, surface resistivity, insulation resistance, loss factor, the

tangent of loss factor in terms of a capacitor or phase difference, and materials that are polar or non-polar.

- When moisture and impurities are eliminated, materials like silicon, oils, and liquid dielectrics, which are used in electrical conductor separators and circuit breakers, exhibit high dielectric constant, high resistance, and high dielectric strength. Because of their low dielectric losses and strong ability to dissipate heat, they can be used in very high temperatures without overheating.

Dielectric Constant

- Consider a parallel plate capacitor (a capacitor is a device that stores electric energy) with metal plates of area A separated by a distance d in a vacuum, as shown in the figure.

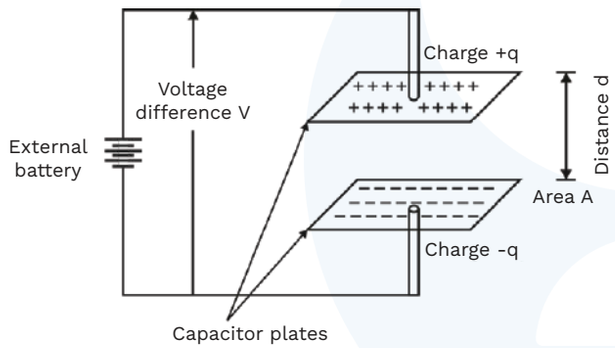


Fig. 5.10

- If a voltage V is applied across the plates, one plate will acquire a net charge of +q and the other a net charge of –q. The charge q is found to be directly proportional to the applied voltage V as given below.

$$q = CV \text{ or } C = \frac{q}{V}$$

where C is a proportionality constant called the capacitance of the capacitor.

- The SI unit of capacitance is coulombs per volt (C/V), or the farad (F). Thus

$$1 \text{ Farad} = \frac{1 \text{ coulomb}}{\text{volt}}$$

- The capacitance of a capacitor is a measure of its ability to store electric charge. The more charge stored at the upper and lower plates of a capacitor, the higher its capacitance.
- The capacitance C for a parallel plate capacitor, whose area dimensions are much greater than the separation distance of the plates is given by

$$C_o = \frac{\epsilon_0 A}{d}$$

ϵ_0 = permittivity of free space
= 8.854×10^{-12}F/m

- If the space between the plates is filled with a dielectric material, the capacitance of the capacitor increases and is given by

$$C = \frac{\epsilon_0 \epsilon_r A}{d}$$

ϵ_r is the relative dielectric constant of the dielectric material and is the property of the material as compared to vacuum.

- The value of a ϵ_r of a material can be determined by measuring the capacitance of a capacitor with vacuum as dielectric and then with material as the dielectric. ϵ_r will be given by

$$\epsilon_r = \frac{C}{C_0}$$

- Different materials have different values of ϵ_r. Its value is unity for vacuum, for air 1.0006, for helium (STP) 1.0000684, for silicon 12, and for germanium 16.
- The capacitance of a capacitor is directly proportional to the relative dielectric constant.
- The energy stored in a capacitor of a given volume at a given voltage is increased by the factor of the dielectric constant when the dielectric material is present.
- By using a material with a very high dielectric constant, very small capacitors with high capacitances can be produced.

Dielectric Strength

- Dielectric strength is a measure of the ability of the material to hold energy at high voltage. It is defined as the voltage per unit length (electric field, or voltage gradient) at which failure occurs and thus is the maximum

electric field that the dielectric can maintain without electrical breakdown.

- If the dielectric is subjected to a voltage gradient that is too intense, the strain of the electrons, or ions in trying to pass through the dielectric may exceed its dielectric strength. If the dielectric strength is exceeded, the dielectric material begins to break down and the passage of current (electrons) occurs.
- Dielectric strength is structure sensitive property of the material. Note, the actual thickness of the material affects the breakdown potential per unit thickness i.e., dielectric strength. Usually thicker materials possess lower dielectric strength than thinner materials.
- The fundamental breakdown mechanisms in a solid are intrinsic, thermal, discharge, and electrochemical breakdown.

Electric Dipole Moment and Polarisation

- Perhaps the greatest approach to understanding the phenomena of capacitance is with the assistance of field vectors. To begin, for every electric dipole, there is a gap between a positive and a negative electric charge as depicted in the figure.

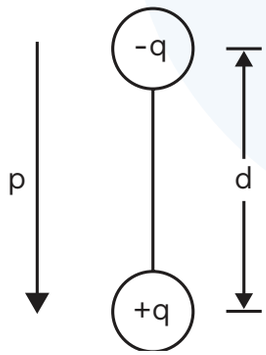


Fig. 5.11

- An electric dipole moment p is associated with each dipole as follows:

  $p = q \cdot d$

  where q is the magnitude of each dipole charge and d is the distance of separation between them.
- In actuality, a dipole moment is a vector that is oriented from the negative to the positive charge, as depicted in the diagram above. As seen in the image below, an electric dipole experiences a force (or torque) that causes it to align with an external electric field E, which is itself a vector quantity.

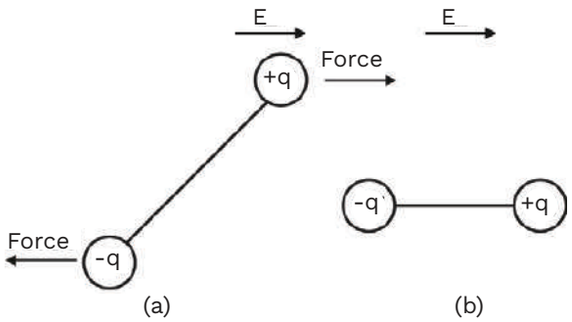


Fig. 5.12

- Polarisation refers to the process of bringing dipoles into a more symmetrical arrangement. Again, back to the capacitor, the surface charge density D, or the amount of charge per unit area of the capacitor plate (C/m^2), is proportional to the electric field.
  When a vacuum is present, then

  $D_0 = \epsilon_0 E$

  The constant of proportionality being ϵ_0. Furthermore, an analogous expression exists for the dielectric case; that is,

  $D = \epsilon E$
- Sometimes, D is also called the dielectric displacement or electric flux density.
- Total electric flux density inside dielectric material under the influence of electric field is due to two components: external electric field and polarisation inside the dielectric material

  $D = \epsilon_0 E + P$

  where P is the polarisation, or the increase in charge density above that for a vacuum because of the presence of the dielectric.
- The polarisation P may alternatively be thought of as the total dipole moment per unit volume of the dielectric material, r as a polarisation electric field inside the dielectric that comes from the mutual alignment of the numerous atomic or molecular dipoles with the externally applied field E.
- For any dielectric material

  $D = \epsilon_0 \epsilon_r E$ ... (i)

and

$$D = \epsilon_0 E + P \quad \text{... (ii)}$$

From equations (i) and (ii)

$$\epsilon_0 \epsilon_r E = \epsilon_0 E + P$$

$$P = \epsilon_0 (\epsilon_r - 1) E$$

or

$$P = \epsilon_0 \chi_e E$$

where $\chi_e = \epsilon_r - 1$

- The χ_e is known as dielectric susceptibility. The quantity χ_e is a measure of the ability of the material to become polarised and differs from one dielectric to another.

Types of Polarisation

- Molecular, or atomic dipole moments may be permanently aligned with an externally applied electric field, and this is what it means when one talks about polarisation. Polarisation may be explained by four main factors or processes. There are four types of polarisation: electrical, ionic, orientation, and space charge.
- Like the mechanical behaviour of materials relies on the strain rate, their occurrence is frequency dependent when an electrical field is applied. A material's ability to deform through processes like plastic deformations is inhibited when strain is applied at a very high pace. Similarly, certain polarisation processes may not be able to create polarisation in the material, if one applies a fast oscillating electric field.

Orientation Polarisation

The only compounds that exhibit orientation polarisation are those with permanent dipole moments. When an external field is introduced, the permanent moments will rotate to face that direction, causing polarisation. Because atomic thermal vibrations work against this alignment tendency, polarisation diminishes as temperature rises.

Electronic Polarisation

It is possible to produce electronic polarisation in all atoms to some extent. Caused by the electric field, it occurs when the centre of the electron cloud moves away from the atomic nucleus. All dielectrics exhibit this polarisation form in the presence of an electric field.

Ionic Polarisation

Only ionic materials experience ionic polarisation. By moving cations in one direction and anions in the other, and applied field creates a net dipole moment. The dipole moment, shown by the symbol p_i, is proportional to the product of the ion charges and the relative ion displacements, written as $q.d_i$.

$$p_i = q \cdot d_i$$

Space Charge Polarisation

- This polarisation arises from the presence of impurity centres and lattice vacancies.
- Some dielectrics have free charge carriers that can travel between their ends. Lattice vacancies and impurity centres may act as barriers to the free movement of charge carriers across the dielectric in the presence of an electric field, potentially trapping the charge carriers. This will cause a localised build-up of charges inside dielectrics. This mirror charge, which is diametrically opposed to the original, will accumulate at the electrodes. This leads to the development of the dipole and the introduction of polarisation, the space charge polarisation.
- The accumulation of charge carriers at interfaces and electrodes may result in space charges and a tiny field, as seen in figure.

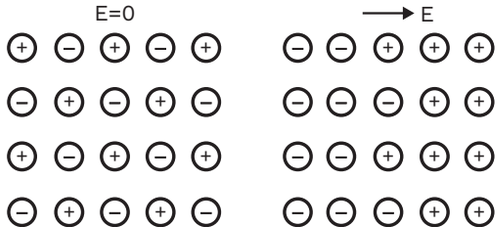


Fig. 5.13

Phase Difference and Dielectric Loss

- A dipole tends to align itself along the direction of the applied field and ac fields tend to follow the field and be in a phase with it. However, the interaction of this dipole with

other dipoles in the medium prevents this and this leads to dielectric loss. This loss appears as heat. This dielectric loss is connected with $\in$, the imaginary part of the dielectric constant. The rate of loss of energy in a unit volume of the material is obtained as:

$$w = \frac{1}{2} \in_0 \in_r'' \omega E_0^2$$

- Usually, the dielectric loss is expressed in terms of a quantity called 'the loss tangent,' tan δ. This is defined as $\tan \delta = \in_r'' / \in_r'$. The angle δ is the compliment of angle φ, i.e., $90° - \varphi = \delta$, where φ is the angle between the applied field and the resultant current vector.
- The energy losses in a dielectric material are due to ionisation, leakage current, polarisation, and structural inhomogeneity.

Power Factor

Power factor is defined as the ratio of a material's power loss to the sum of the voltage and current being applied. Loss angle and power factor are both affected by environmental factors such as humidity, temperature, and frequency.

Polar Molecules

- Some molecules have permanent dipole moments due to an asymmetry in the distribution of their positive and negative charges. Substances that exhibit this property are called 'polar.' In case of HCl, the hydrogen electron spends more time orbiting the chlorine atom than the hydrogen atom itself.
- Thus, the molecule possesses a dipole moment that points from the Cl atom to the H atom, and the centre of negative charges does not correspond with the centre of positive charges. The HCI molecule has an electric dipole of $p = 3.43 \times 10^{-30}$ C-m.

Non-polar Materials

- Hydrocarbons are mostly non-polar. In non-polar materials, molecules are often diatomic, meaning they are made up of two atoms of the same kind.
- These molecules may be seen as a positively charged nucleus surrounded by a negatively charged electron cloud, with the charges q and –q being symmetrically distributed.
- With no external electric field, the positive and negative charge distributions have the same centre of mass. In the presence of an external electric field, the positive and negative charges on the molecules experience electric forces that tend to pull them apart along the direction of the field.
- As the displacement is restricted by restoring forces, which increase with increasing displacement, the distance travelled is relatively tiny (10^{-10} m). When molecules are polarised, the centres of their positive and negative charges no longer coincide.

Some More Electrical Characteristics of Materials

Ferroelectricity and piezoelectricity are two more unique electrical properties identified in certain materials that are worth mentioning briefly.

Ferroelectricity

- A class of dielectrics known as ferroelectrics displays polarisation in the absence of an applied electric field.
- Similar to how ferromagnets, which may exhibit persistent magnetic behaviour, are the electric counterpart of insulators, ferroelectrics serve as a model for describing this phenomenon.
- For ferroelectrics, an applied electric field may alter or even invert the polarisation state.
- As a ferroelectric crystal's structure may be generated from a non-polarised structure by minor displacement of ions, spontaneous polarisation is reversible.
- Above a specific temperature, the ferro-electric Curie temperature, this non-polarised arrangement becomes stable in most ferroelectric crystals (T_c). That is to say, the crystal passes through a phase transition, changing from its polarised phase (ferroelectric phase) to its unpolarised phase (paraelectric phase).
- Applying an electric field to a piece of a ferroelectric crystal causes the polarisation to increase fast up to a certain point, beyond which the dependency becomes linear. By

linearly extrapolating from zero fields, one may calculate saturation or spontaneous polarisation.

- There is always some degree of polarisation that persists even after the applied field is eliminated, a phenomenon known as residual polarisation. The term 'coercive field' is used to describe the negative field necessary to eliminate polarisation. Applying an electric field may toggle a ferroelectric crystal, and this process is accompanied by a hysteresis loop.
- The presence of a dielectric hysteresis loop in a dielectric material indicates the presence of spontaneous polarisation, Ps. The value of spontaneous polarisation (depending on the shape of the hysteresis loop) is affected by a wide range of variables, including specimen size, temperature, crystal texture, and thermal and electrical properties.
- The loop of hysteresis seen in the figure is representative of the norm. To attain such high levels of polarisation, P values of 105 to 108 V/cm are often required of ferroelectrics. A ferroelectric material's hysteresis loop distorts as its temperature rises. Hysteresis loop height and breadth both shrink as temperature rises.

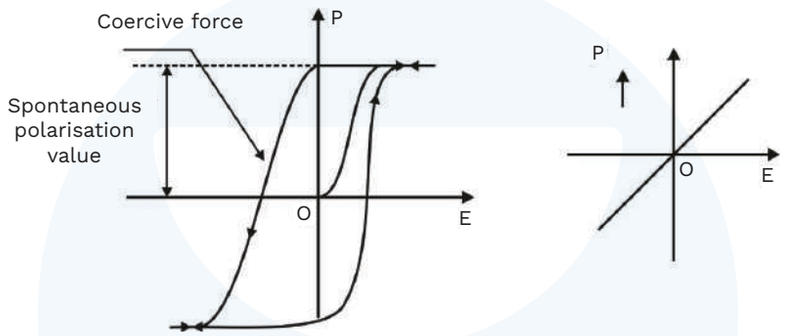


Fig. 5.14

- The Curie temperature (T_c) is the critical temperature over which ferroelectric behaviour ceases to exist in a material.
- For temperatures above the Curie temperature (Tc), the paraelectric phase, the electric susceptibility, χ_e, is temperature dependent according to the Curie-Weiss law.

$$\chi_e = \frac{C}{T - Tc}$$

where, C = Curie constant and T_c = Curie temperature

Piezoelectricity

- Piezoelectricity, often known as pressure electricity, occurs when mechanical pressure is applied to a dielectric crystal, producing an electric current.
- The application of mechanical stress in some dielectric crystals causes an electric polarisation (electric dipoles) which is proportional to this stress. This polarisation shows itself as a voltage across the crystal under load, if the crystal is isolated, and as a current, through the crystal under load, if the crystal is shorted.
- In contrast, mechanical deformation of the material is brought about by applying a voltage between certain faces of the crystal. Piezoelectricity describes this mutually beneficial connection. The direct piezoelectric effect refers to the phenomenon where a voltage is generated when a crystal is subjected to mechanical stress, while the converse piezoelectric effect refers to the strain that is induced in the crystal when an electric field is applied.
- To illustrate the piezoelectric effect, observe the figure below. Piezoelectric effect Figure (a) dipoles within a piezoelectric material

and Figure (b) a voltage is generated when the material is subjected to a compressive stress.

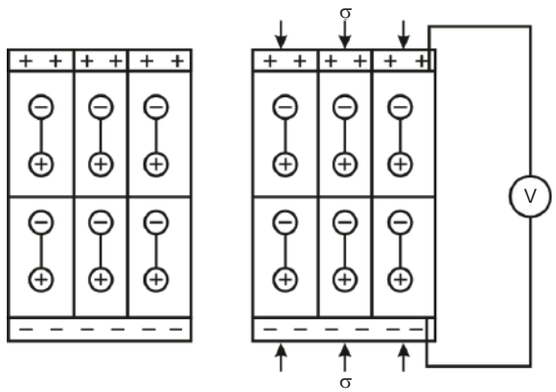


Fig. 5.15

- Titanates of barium and lead, zirconate (PbZrO3), ammonium dihydrogen phosphate (NH_4 H_2PO_4), natural quartz, etc., are all examples of piezoelectric materials.
- Piezoelectric materials have the following major applications: frequency resonators, gramophone pickups, filters, ultrasonic flow detectors, underwater sonar transducers, and air transducers.

Applications of Dielectrics

- The most common applications for dielectrics are capacitors, resistors, strain gauges, piezoelectric and electro optical gadgets, thermionic valves, radiation detectors, and electric gadgets.
- As insulators, dielectrics are often found in electrical components such as power cables, signal cables, electric motors, etc.
- In order to eliminate the heat generated by transformers, switchgear, and generators, dielectrics are often insulated using transformer oil, also known as mineral oil.

Semiconductor Materials

- Crystalline semiconductors predominate; both elemental (such as silicon and germanium) and compound (GaAs, GaP, etc.) semiconductors exist. All semiconductors (whether elemental or compounded) have a few key characteristics.
  - The range of electrical resistivity is around 10^{-5} to 10^{7} Ω-m.
  - Semiconductors have greater conductivity in highly illuminated conditions (optical effects).
  - As the temperature increases, semi-conductors become more conductive.
  - Semiconductors, like dielectrics, can tolerate electrical fields inside themselves.
  - Most devices can only function with single crystal materials.
  - To continue with the theme of hardness and fragility, these materials are often quite stable.
  - Semiconductors are non-linear devices because the current in semiconductors does not follow Ohm's Law and will grow faster than voltage.

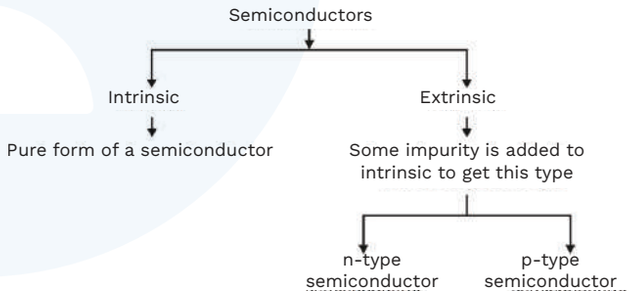


Fig. 5.16

Electrons and Holes in an Intrinsic Semiconductor (Pure Semiconductor)

- Take, for example, a Ge crystal structure at 0 K (first figure) and one at normal temperature (second figure).

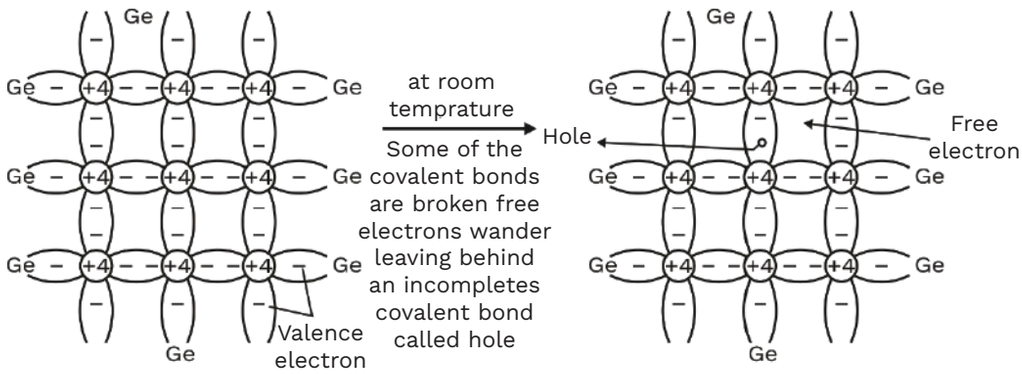


Fig. 5.17

- The total number of holes and free electrons are balanced in a perfect semiconductor. Thermal agitation continues to form new electron hole pairs, but other hole electron pairs perish as a consequence of recombination. The ratio of electrons to holes determines the concentration of each.

  $n = p = n_i$

  where, n_i = intrinsic concentration

Extrinsic Materials

A semiconductor material doped with impurity atoms is called extrinsic material. Two extrinsic materials exist for device fabrication: n-type and p-type.

n-type Material

- It is created by introducing pentavalent impurity elements such as antimony, arsenic, and phosphorous.
- In n-type semiconductors, electrons are the majority charge carriers and holes are the minority charge carriers.

p-type Material

- It is created by introducing trivalent impurity elements such as boron, gallium, indium, etc.
- In a p-types semiconductor, holes are the majority charge carriers and electrons are the minority charge carriers.

Charge Densities in Semiconductor

- Let N_D be the concentration of donor atoms and N_A be the concentration of acceptor ions. As the semiconductors are electrically neutral, the total magnitude of positive-charge density must be equal to negative concentrations.

  $N_D + p = N_A + n$

  For n – type materials $N_A = 0$ and $n >> p$

  $n_n \approx N_D$

- So, one can say in n-type material free electron concentration is approximately equal to the density of donor atoms.

  For p-type Material $N_D = 0$ and $n >> p$

  $P_p \approx N_A$

- So, one can say in p-type material hole concentration is approximately equal to the density of acceptor atoms.

Electrical Properties of Semiconductors

Conductivity

Current density (J) = $eE(n\mu_n + p\mu_p) = \sigma E$

Conductivity (σ) = $e(n\mu_n + p\mu_p)$

As, for intrinsic semiconductor, n = p = ni

$\sigma = n_i e(\mu_e + \mu_p)$

Intrinsic Concentration

Variation of ni with temperature T is given as

$n_i^2 = A_0 T^3 e^{-3E_{G_o}/kT}$

where E_{G_o} is the energy gap at 0^0K in electron volts, k – Boltzmann's constant. In eV/k and A_o is a constant independent of T.

Hall Effect

The Hall effect has played a decisive role in revealing the mechanism of conduction in semiconductors If a specimen (metal or semiconductor) carrying a current 'T' is placed in a transverse magnetic field 'B,' an electric field 'E' is induced in the direction

perpendicular to both I and B. This phenomenon is known as the 'Hall effect.' In the figure, *I* is in (+) ve X-direction and Bis in (+) ve Z-direction; a force will be exerted in the (–) ve Y-direction on the charge carriers.

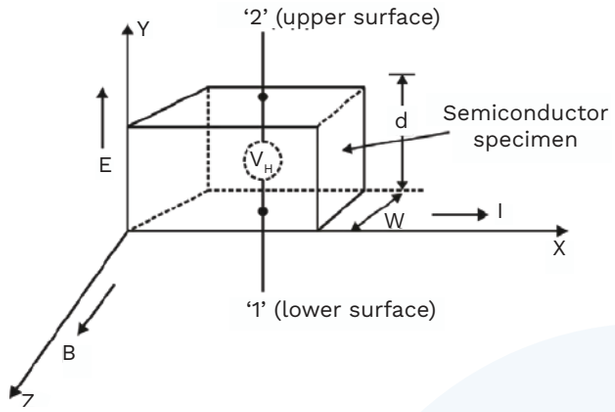


Fig. 5.18

- The current (i) may be due to holes (if the specimen is p-type) moving from left to right or to free electrons (if the specimen is n-type) moving from right to left in the specimen. Hence, independently of whether the carriers are holes or electrons, they will be forced downward toward side '1.'
- If the semiconductor is n-type material, then all free electrons will accumulate on side '1' and so the lower surface becomes (–) vely charged with respect to side '2.' Hence a potential, called the Hall voltage (V_H) appears between surfaces '1' and '2.' If the polarity of 'V_H' is (+) ve at terminal '2,' then as explained above, the carrier must be electrons and so the specimen is n-type.
- On the other hand, if terminal '1' becomes charged positively with respect to terminal '2,' the semiconductor bar must be p-type

Applications of Hall Effect

- It is used to determine whether a semi-conductor is n-type or p-type.
- It is also used to determine the carrier concentration, mobility, drift velocity, the conductivity of the specimen, and field 'B.'
- It is also used in the manufacturing of magnetic flux metres and Hall effect multipliers.

Thermistors

- These are semiconductors with a negative coefficient of resistance. Note that metals have positive coefficient of resistance, as with the increase in temperature, the mean path of free electrons becomes less, and mobility is decreased. By including a thermistor, it is possible to compensate for temperature change over a wide range of up to 100ºC.
- In addition, thermistors are used in microwave frequency metres, as a thermal relay, and in control devices actuated by a change in temperature. Si and Ge are not used as thermistors because their properties are too sensitive to impurities. Examples of thermistors are NiO, Mn_2O, and $Co_2 O_3$. Heavily doped semiconductors exhibit a positive temperature coefficient of resistance, called sensors having a temperature coefficient of resistance of + 0.7 per cent/C.

Photoconductors

- As shown in the figure, it is the energy diagram of semiconductors having both acceptor and donor impurities. Two types of transition are possible, if photons of sufficient energies illuminate the specimen.

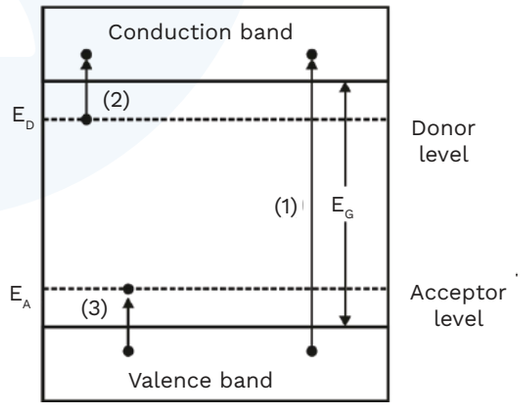


Fig. 5.19

- Intrinsic excitation: in this excitation, an electron hole pair can be created by a high-energy photon.
- Extrinsic excitation: where a photon may excite a donor electron into the conduction band or a valence electron may go into an acceptor state.

Spectral Response

- $\lambda_c = \frac{1.24}{E_G}$

  Where λ_c is in microns and E_G is in eV
- If the wavelength (λ) of the radiation exceeds λ_C then the energy of the photon is less than E_G and such a photon cannot cause a valence electron to enter the conduction band hence λ_C is called the cut-off wavelength or critical wavelength or long wavelength threshold of the material.
  - Note that with varying wavelengths, the number of free carriers generated would be different, i.e., spectral response changes. Thereby it means that a photoconductor is a frequency selective device.

Practice Exercise

1. In a sodium atom the valence band and conduction band are represented by:
(A) 3s, 3p energy levels
(B) 3s, 4s energy levels
(C) 3p, 4s energy levels
(D) 4s, 4p energy levels

2. The Fermi energy (E_p) is the energy level at which half of the possible energy levels in the band are occupied by:
(A) protons
(B) neutrons
(C) electrons
(D) none of these

3. With the increase in temperature, the conductivity of semiconductors:
(A) increases
(B) decreases
(C) remains constant
(D) sometimes increases and sometimes decreases

4. Imperfections in the crystal structure of a metal scatter electrons, reducing its:
(A) mobility and conductivity
(B) mobility only
(C) conductivity only
(D) none of these

5. The motion of magnetic flux lines limits the current transport in a superconductor within the:
(A) grains
(B) surface
(C) vortices
(D) none of these

6. The temperature dependence conductivity in intrinsic semiconductors is nearly:
(A) linear
(B) exponential
(C) parabolic
(D) none of these

7. When trivalent impurity such as gallium or boron is added in Si and Ge, the semiconductor is called:
(A) intrinsic
(B) p-type extrinsic
(C) n-type extrinsic
(D) none of these

8. A measure of how much electrical energy is lost due to the motion of charged entities that respond to the electric field via different polarisation mechanisms is called:
(A) heat loss
(B) dielectric loss
(C) heat, as well as, dielectric loss
(D) none of these

9. If a small amount of Cu is added to a Ni conductor, then:
(A) the resistivity of Ni will increase at all temperatures because Cu is a better conductor than Ni.
(B) residual resistivity of Ni at low temperature will increase as Cu atoms act as defect centres.

(C) the resistivity of Ni will increase at all temperatures as Cu destroys the periodicity of Ni and acts as defects.
(D) the resistivity of Ni remains unaltered as Cu atoms give the same number of free electrons as Ni atoms.

10. Which one of the following is the best definition of a superconductor?

(A) It is a material showing perfect conductivity and the Meissner effect below a critical temperature.
(B) It is a conductor having zero resistance.
(C) It is a perfect conductor with the highest diamagnetic susceptibility.
(D) It is a perfect conductor but becomes resistive when the current density through it exceeds a critical value.

Answer Key

1. (A) **2.** (C) **3.** (A) **4.** (A) **5.** (A) **6.** (B) **7.** (B) **8.** (B) **9.** (C) **10.** (A)

6 Mechanical Properties of Materials

Introduction

- The mechanical properties of materials such as strength, rigidity, and ductility are important in selecting materials for fabrication and other engineering applications. Materials exhibit a wide range of mechanical properties ranging from the ductility of pure copper and the hardness of diamond to the high elastic behaviour of rubber.
- Several materials are found to behave differently when they are stressed in different ways. The knowledge of mechanical properties is very essential to a material scientist and design engineer.

Normal Stress

- The fundamental concept of stress can be understood by considering a prismatic bar that is loaded by axial force P at the ends as shown in the figure.

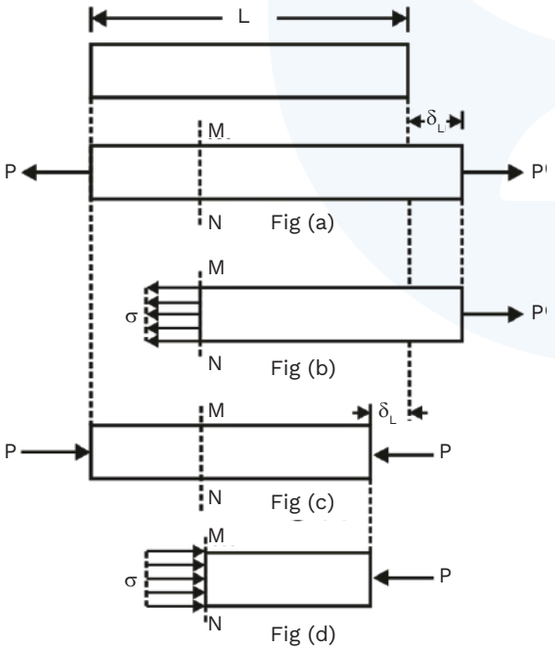


Fig. 6.1

- A prismatic bar is a straight structural member having a constant cross sectional area throughout its length. In Figure (a), axial force produces a uniform stretching of the bar; hence, the bar is said to be in tension, and in Figure (c) force produces uniform compression of the bar; hence, the bar is said to be in compression.
- To investigate the internal stresses produced in the bar by axial forces, make an imaginary cut at section MN Figure (b and d). This section is taken perpendicular to the longitudinal axis of the bar; hence, it is known as a cross section. Now isolate the part of the bar to the right of the cut and consider the right of the cut as a free body. The force P tends to move the free body in the direction of load, so to restrict the motion of the bar an internal force is induced which is uniformly distributed over a cross sectional area. The intensity of force that is force per unit area is called stress.
- Thus, stress can be defined as 'stress is the internal resistance of material offered against deformation, which is force per unit area.'
- Stress induced in a material depends upon the nature of force, point of application, and cross sectional area of a material. Stress can be tensile or compressive depending on the nature of the load. Generally, stress is represented by the Greek letter. One can calculate stress mathematically as

$$\sigma = \frac{P}{A} \text{ N/mm}^2 \text{ or MPa}$$

Tensile stresses = positive, compressive stresses = negative

Strain

- An axially loaded bar changes length, becoming longer when in tension and shorter when in compression. Thus, the elongation or shortening in axially loaded members per unit length is known as strain. Strain is represented by $\in$. Mathematically strain can be calculated as follows.

$$\in = \frac{\delta}{L}$$

- Unit: strain is a dimensionless quantity. It is always expressed in the form of a number. If the member is in tension then strain is called tensile strain and if the member is in compression then strain is called compressive strain.

Tension Test for Mild Steel

- The mechanical properties of materials used in engineering are determined by experiments performed on small specimen. These experiments are conducted in laboratories equipped with testing machines that are capable of loading in tension or compression.
- The American Society for Testing and Materials (ASTM) has published guidelines for conducting the test. A tension test is conducted on a Universal Testing Machine (UTM).
- The specifications of the specimen are the specimen is a solid cylindrical rod, Gauge length of 2.0" (inches), diameter of middle section 0.5", and LID ratio = 4.0.

Stress–Strain Curve for Tension

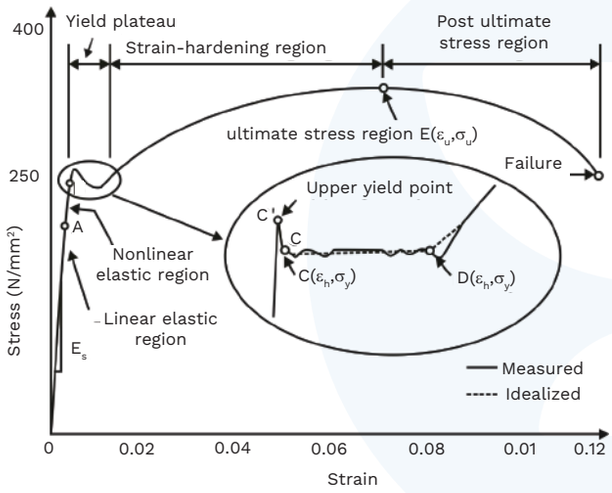


Fig. 6.2

- Beyond point A, linear variation (percentage) stops. The highest stress at which a specimen regains its original length upon release of the applied load is known as B's elastic limit. For mild steel, B is close to A. On the other hand, B may be larger than A for certain other materials.
- The maximum yield point is denoted by C. The stress equivalent to C varies in magnitude based on the test setup, the geometry of the specimen, and the size of the cross section. It is completely irrelevant in the real world.
- This yield point is also known as the true yield point, and it is denoted by the letter C. The strain at C represents the yield stress, often valued at (for mild steel). When stressed to this level, yielding starts to occur. When stressed to this level, yielding starts to occur.
- The area represented by CD is perfectly plastic; it is the stress free strain that develops beyond the yield point C. For mild steel, the strain at point D is roughly 1.4 per cent, whereas at point C, it is about 0.12 per cent. This means that plastic strain is around 10–15 times more than elastic strain.
- To illustrate strain hardening, look at DE: in this range, more stress results in more strain. Nonetheless, tension increases more rapidly in this area.
- As the material's atomic and crystalline structure changes this temperature range, its resistance to additional deformation increases. This section is not included in the structural planning.
- The ultimate stress at point E is called the ultimate stress, and the ultimate strain at point E is around 20 per cent for mild steel.
- F is the fracture point, and the stress and strain that correspond to it are, respectively, known as the breaking stress and the fracture strain. For mild steel, that figure is about 25 per cent.
- Necking occurs when the area of the cross section is dramatically reduced between points E and F.

Actual Curve vs Engineering Curve in Tension

- Note: the fracture strain depends upon per cent the carbon present in the steel.
- With an increase in percentage carbon, fracture strain reduces.
- With increased carbon content, steel has higher yield stress and higher ultimate stress.

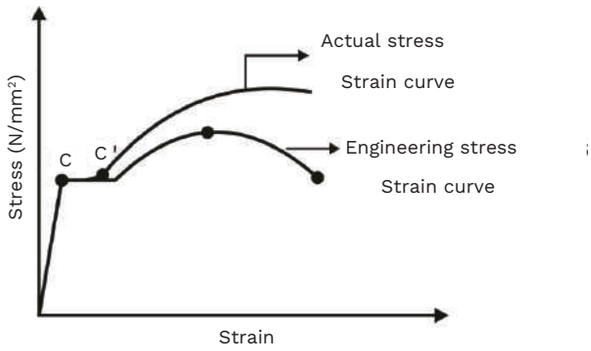


Fig. 6.3

Compression Curve for Mild Steel

- In compression, the engineering stress–strain curve lies above the actual stress–strain curve. In compression, mild steel yields stress σ_y = 263 N/mm^2 slightly greater than tension.
- Mild steel has the same Young's modulus of elasticity in compression and tension E = 2.1 × 10^3 N/mm^2

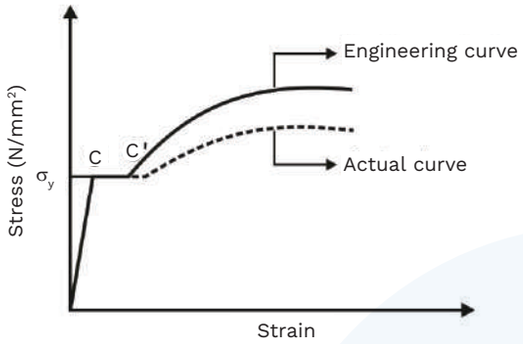


Fig. 6.4

Relation between engineering and the actual stress

$$\sigma_a = \sigma_0 \left(1 \pm \epsilon_0\right)$$

σ_a = Actual stress
σ_0 = Engineering stress
ϵ_0 = Engineering strain

- For tension take the positive (+ve) sign and take the negative (–ve) sign for compression.

Basic Terms of Mechanical Properties

- Isotropy: a material is said to be isotropic in case its physical properties are the same in all directions. Isotropic materials have properties that are independent of direction. Polycrystalline materials have grains, which randomly oriented and these materials behave isotopically. Isotropic materials have the same value of strength, refractive index, dielectric constant, resistivity, and hardness in all directions.
- Anisotropy: a material is said to be anisotropic in case its physical properties are not the same in all directions. Anisotropic materials have properties that are dependent on direction. The resistivity of certain materials is higher along a particular crystallographic direction. Wood, laminated plastics, and other composite materials have better strength in a certain direction.
- Strength: the mechanical strength of the material may be defined as the ability of the material to sustain load without distortion or failure. The material must have adequate strength when subjected to tension (tensile strength), compression (compressive strength), shear (shear strength), bending (bending strength), and torsion (torsion strength) loading during the intended use in the span of life.
- Stiffness: it is defined as the property of a material due to which it resists its deflection or any deformation. It is also rigidity. The degree of stiffness of a material is specified by its modulus of elasticity. A material that deforms less under a given load is stiffer than a material that deforms more. The modulus of elasticity is the ratio of stress to the strain.
- When a material lacks ductility, also known as brittleness, it becomes very difficult to bend or shape it. Brittle materials shatter soon after the elastic limit, at a lesser distortion. For brittle materials, the strain at the fracture and ultimate sites is identical, and there is a proportionate limit beyond which the strain becomes negligible. In general, materials with a post elastic strain of less than 5 per cent are considered brittle. Cast iron, concrete, and glass are all materials that may be used as examples.
- To stretch anything is to have the attribute known as ductility. Large deformations are, therefore, feasible in ductile materials before the total failure or rupture takes place. The plastic strain in these materials is more than 5 per cent after elastic deformation. Materials like mild steel, aluminium, copper, manganese, lead, nickel, brass, bronze, etc., are all examples of such alloys.
- A metal may be pressed into a thin sheet is a result of its malleability. A malleable substance exhibits a high degree of plasticity. Forging, hot rolling, drop (stamping), and similar processes all benefit greatly from this quality.
- Resistance to scratching or abrasion is measured by a material's hardness. It may be determined how hard something is using one of two methods.

- Mohr's test is widely used to determine scratch resistance.
- Hardness (abrasion resistance) is determined by the Brinell technique, Rockwell hardness scale, Vickers hardness scale, and Knoop hardness scale.

- Toughness is the ability of a substance to take a blow without breaking. Under cyclic or shock loads, this is a very useful trait to have. Metals with high resistance to deformation can accumulate significant amounts of strain energy before cracking. Toughness may be measured in terms of its modulus, which is the total strain energy per unit volume up to breakage. The area under the stress–strain curve up to the point of failure is equal to this value.

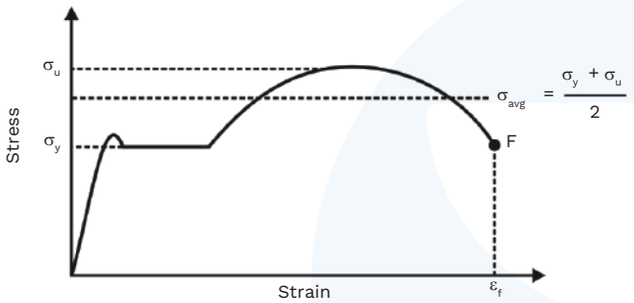


Fig. 6.5

$$\text{Modulus of toughness} = \left(\frac{\sigma_y + \sigma_u}{2}\right) \times \epsilon_f$$

σ_y = yield tensile strength
σ_u = ultimate tensile strength
ϵ_f = strain at the fracture point

- The modulus of toughness depends upon ultimate tensile strength and strain at failure (fracture strain).
- Hence, the very ductile material will exhibit a higher modulus of toughness than the case with mild steel.

Hardness	Toughness
It is the property to resist deformation.	It is the property to absorb maximum energy before rupture. More deformation before rupture increases energy absorption.

Hardness	Toughness
Materials with a small yield are hard as they resist penetration.	Materials with large yields are tough as they can absorb more energy. More ductility and more strength increase toughness.
Hardness can be improved by certain heat treatment (quenching), case hardening, and work hardening processes.	Toughness can be improved by increasing ductility such as tempering, annealing, and normalising.
More brittle	More ductile
The hardness of the iron can be improved by increasing carbon content.	The toughness of iron can be improved by reducing carbon content.

Table 6.1

- Elasticity: assume, for instance, that one applies a tensile load to a specimen so that strain and stress follow a path from O to B on the stress–strain curve shown in the figure. Further, when the load is removed, the material follows the same curve back to the origin O. So, the property by which original dimensions (i.e., length and cross section) can be recovered after unloading is known as elasticity.

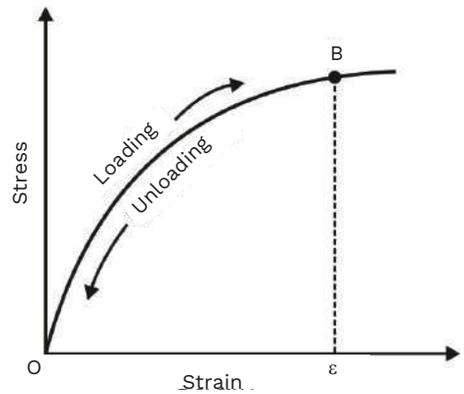


Fig. 6.6

- Within the elastic limit, the curve may be linear or non-linear. In the figure above, during

loading material stores elastic strain energy. The total strain energy that can be stored in the given volume of the metal and can be released after unloading is called resilience. It is also equal to the area under the load deflection curve within elastic limit (B). When elastic limit coincides with a yield stress, the maximum, elastic energy per unit volume is known as the modulus of resilience.

- Modulus of resilience is equal to the area under the stress–strain curve within the elastic limit for mild steel.

$$U_f = \frac{1}{2} \times \sigma_y \times \epsilon_y = \frac{1}{2} \times \sigma_y \times \frac{\sigma_y}{E} = \frac{\sigma_y^2}{2E} \quad \left[\because \epsilon_y = \frac{\sigma_y}{E} \right]$$

- The modulus of resilience depends upon yield strength and hence, a material with higher yield strength will have a higher modulus of resilience.
- Higher resilience is desirable in suspension, spring, and where load absorption is required.
- Proof Stress: some of the ductile metals like aluminium (Al), copper (Cu), and silver (Ag) do not show a clear yield point in the tension test; therefore, yield stress (σ_y) is not known. For such metals, design stress is calculated by the offset method. An offset of permanent plastic strain equal to 0.2 per cent is marked on the x-axis and a straight line is drawn, which is parallel to the initial portion of the stress–strain curve. The point of intersection of the stress–strain curve with a straight line is the proof point and corresponding stress at that point is proof stress.
- Fatigue: it has been found that materials behave differently under static and dynamic loading. In cyclic, or reverse cyclic loading, if total accumulated strain energy exceeds the toughness then fracture failure may occur.

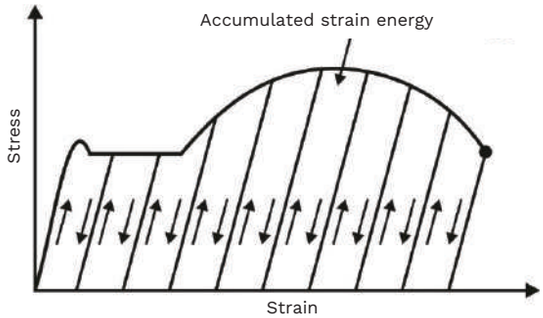


Fig. 6.7

- Factors affecting fatigue are loading condition, frequency of loading, corrosion, temperature, and stress concentration.
- The number of load cycles required to initiate surface crack is called fatigue initiation life and the additional number of load cycles required to propagate surface crack is called fatigue propagation life.
- To prevent fatigue failure the developed stress should be kept below the endurance limit. Endurance limit is that stress below which a material has a high probability of no failure even at an infinite number of load cycles.
- Examples of fatigue failure are the crashing of aircraft due to a crack in the turbine blade, failure of flywheels, and breaking of the wire due to cyclic bending.

Creep

- Creep is permanent deformation that is recorded with time at constant loading. Total creep deformation continues to increase with time asymptotically.

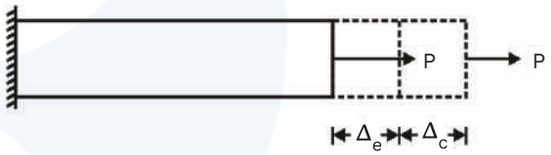


Fig. 6.8

Δ_e = elastic deflection = $\frac{PL}{AE}$

P = static load

Δ_c = deformation due to creep

- Factors affecting creep are the magnitude of load, type of loading (static or dynamic), time or age of loading, and temperature.
- Most materials weaken and have lower elastic constants as temperature increases, as a result of the increased atomic mobility.
- Therefore, even under steady stress, higher deformation occurs at increased temperatures. Because of this, engine and furnace designers need to take creep into account when working with materials at high temperatures.
- At temperatures equal to or higher than the homologous temperature, the creep becomes perceptible. This value is half the absolute melting point.

Fatigue	Creep
It is the degradation of mechanical properties due to cyclic loading, i.e., loading is not constant, or stresses are fluctuating.	It is the progressive deformation of material at a slow rate with time at constant loading, i.e., steady loading keeps on increasing deformation.
It is expressed in several cycles for a particular loading for the failure of the material. The endurance limit is specified below and fatigue does not take place.	It is expressed as the material's creep resistance up to the highest temperature below which material can be used without creep fracture.
It is a function of stress (loading) and many cycles of loading and unloading.	It is a function of time, temperature, and stress.
It occurs at lower stresses than the yield strength.	It occurs at lower stresses than rupture strength.
Fatigue failure starts on the surface at surface imperfections. Crack propagates in the interior of the material to rupture the material.	Creep failure takes place due to dislocation climb, vacancy diffusion, and grain boundary sliding.
Fatigue takes place without much deformation of a material.	Creep takes place with considerable deformation.
It has a typical curve of failure.	It has a typical curve with creep occurring in three stages. The creep rate varies in these three stages.
Surface finish and fine grains improve the fatigue of material.	Coarse grains and alloying to prevent boundary sliding improve the creep of a material.
Fatigue is an increase of stress due to crack propagation.	Creep is a stress relaxation and material deforms plastically under stress that is less than yield strength. Creep does not mean necessarily the failure of the material, but it is bad for the performance of the material, i.e., it weakens the material.

Table 6.2

Impact Strength

The resistance of a material to fracture under dynamic loading is called impact strength. It can also be defined as the specific work required to fracture a test specimen with a stress concentration notch at its midpoint so that it is broken with a single blow of a striker in a pendulum type impact testing machine.

Failure of Materials in Tension and Compression

Ductile Metals in Tension Test

Ductile metals are weak in shear and failure is due to shear strain along the plane forming 45° with the axis of the specimen. Inductile material, cup, and cone fracture take place. In ductile metals, necking is formed before fracture.

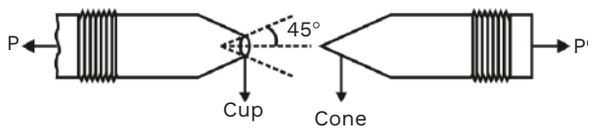


Fig. 6.9

Brittle Metals in Tension Test

Brittle metals are very weak in tension. Brittle metals fail due to the separation of particles along the surface, which is at 90° to the direction of load.



Fig. 6.10

Ductile Metals in Compression Test

Short compression members fail in compression yielding. The failure plane is at 90° to the compressive load. In compression yielding, bulging of material occurs.

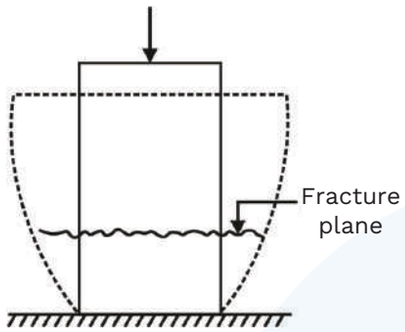


Fig. 6.11

Brittle Metals in Compression Test

In compression, brittle metals fail in shear, failure plane is at 45° to the direction of loading.

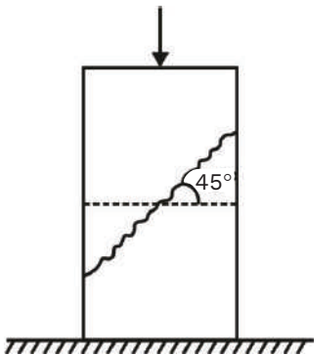


Fig. 6.12

Fracture

- Fracture is the separation of a specimen into two, or more parts by applied stress. Fractures are classified as brittle and ductile fractures.
- A brittle fracture occurs when a specimen breaks after little, or no plastic deformation. The ductile fracture occurs when the specimen breaks after undergoing a considerable amount of plastic deformation. The differences between ductile and brittle fractures are as given in table.

Ductile	Brittle Fracture
Material fractures after undergoing considerable plastic deformation.	Material fractures after undergoing little or no plastic deformation as it happens in materials like glass and china clay.
The appearance of the fractured surface is dull and fibrous	The appearance of a fractured surface is crystalline.
A fracture occurs after elongation of about 30 per cent.	The fracture occurs with about 0–0.5 per cent elongation.
There is a reduction in the cross sectional area of the specimen before the fracture.	There is an almost negligible reduction in the cross sectional area of the specimen before fracture.
Fracture takes place after necking and is accompanied by little sound.	Fracture takes place suddenly and rapidly, often accompanied by a loud sound.

Table 6.3

Hooke's Law

- It states that under direct loading, within proportionality limit, stress is directly proportional to strain stress prop to strain.

  $\sigma = E\varepsilon$

  E is a constant of proportionality and known as Young's modulus of elasticity.
- Hooke's law is valid up to the limit of proportionality. However, for mild steel proportionality limit and elastic limit are almost equal. But for other metals, the elastic

limit may be higher than the proportionality limit, for example, rubber.

- The slope of the stress–strain curve is called the modulus of elasticity (E). The modulus of elasticity (E) is the constant of proportionality, which is defined as the intensity of stress that causes unit strain. Thus, modulus of elasticity has unit same as that of stress.

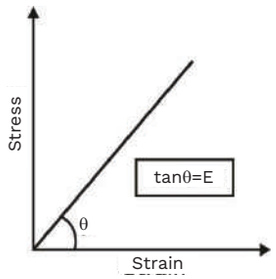


Fig. 6.13

- The assumptions in Hooke's law are material is homogenous (properties are equal at all points), isotropic (properties are equal in all directions) and elastic.
- If properties are different in three mutually perpendicular directions, then the material is called orthotropic material.
- If properties are different in all directions, then the material is called non isotropic or anisotropic, for example, crystal.

Elastic Constants

- **Young's Modulus (E):** Young's modulus is the slope of the stress–strain curve under direct loading.

$$E = \frac{\text{Direct stress}}{\text{Linear strain}} = \frac{\sigma}{\epsilon}$$

- **Shear Modulus (G):** when a body is subjected to shearing stresses, the shape of the body gets distorted. The measurement of this distortion is done by the angle of distortion. Under pure shear, the shape of the body gets distorted but the volume remains the same.

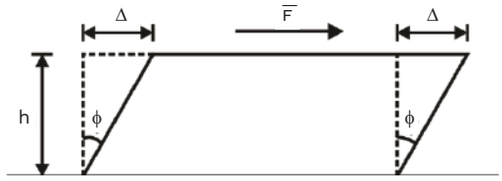


Fig. 6.14

- If under the shear, the shear strain is ϕ then the linear strain in the diagonal of the specimen given by $\epsilon = \frac{\phi}{2}$
- The linear strain of the diagonal is half of the shear strain in the body. Shear modulus is defined as the ratio of shear stress to shear strain. It is also called the modulus of rigidity.

$$G = \frac{\text{Shear stress}}{\text{Shear strain}} = \frac{\sigma}{\phi}$$

- **Bulk Modulus (K):** Bulk modulus is defined as the ratio of direct stress to the volumetric strain. It is represented by K.

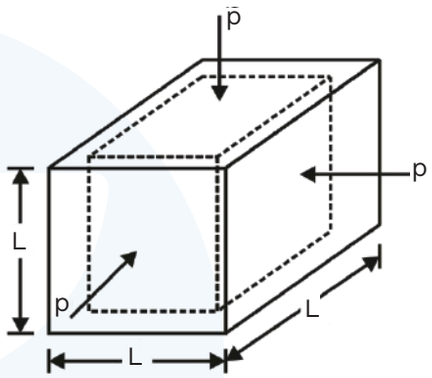


Fig. 6.15

$$K = \frac{\text{Direct stress}}{\text{Volumetric strain}} = \frac{\sigma}{\epsilon_v}$$

- The significance of Bulk modulus is concerning compressibility. In 3D hydrostatic loading,

$$\sigma_x = \sigma_y = \sigma_z = p$$

and volumetric strain $\epsilon_V = \frac{\Delta V}{V}$

then bulk modulus $K = \frac{p}{\left(\frac{\Delta V}{V}\right)}$

- Bulk modulus is inversely proportional to compressibility.
- **Poisson's Ratio:** Poisson's ratio is defined as the ratio of lateral strain to longitudinal strain within the elastic zone under direct loading. It is represented by μ.

$$\mu^* = \left|\frac{\text{Lateral strain}}{\text{Longitudinal strain}}\right| \text{ or } -\frac{\epsilon_L}{\epsilon}$$

- In general, for engineering purposes, Poisson's ratio varies from 0 to 0.5, but the actual range is –1 to 0.5.
- Poisson's ratio is a constant defined within the elastic zone. In the non elastic zone, the ratio of lateral strain to longitudinal strain is not constant, which is called the contraction ratio.

Relationship between Elastic Constants

- $E = 3K(1 - 2\mu)$
- $E = 3G(1 + \mu)$
- $E = \frac{9KG}{3K + G}$
- $\mu = \frac{3K - 2G}{6K + 2G}$
- Where E is the Young's modulus of elasticity, G is the modulus of rigidity, K is the bulk modulus, and μ is the Poisson's ratio.
- For orthotropic material, there are nine independent elastic constants and for non isotropic, there are 21 independent elastic constants.

Linearly	Non-linearly Elastic
It obeys Hooke's law.	It does not obey Hooke's law.
It has a specific and constant value of Young's modulus.	It does not have a constant value of Young's modulus.
It does not possess a secant modulus.	It possesses a secant modulus.
It does not possess a tangent modulus.	It possesses a tangent modulus.
The stress–strain curve is linear.	The stress–strain curve is non-linear.
It possesses a yield point.	It possesses a proof strain.
The strain energy is given by $U = \frac{\sigma^2}{2E} \times$ Vol	The strain energy is given $\propto \frac{1}{\text{Displacement}}$ per unit volume

Table 6.4 Difference between Linearly and Non-linearly Elastic Materials

Practice Exercise

1. Match List I (type of material) with List II (characteristics) and select the correct answer using the codes given below the lists.

List I

A. Elastic material
B. Rigid material
C. Plastic material
D. Resilient material

List II

I. Does not store energy
II. Has no plastic region in a stress–strain curve
III. Behave as a spring
IV. Offers resistance to deformation
V. Does not offer resistance to deformation

Codes:

	A	B	C	D
(A)	IV	I	V	III
(B)	IV	I	V	II
(C)	II	III	IV	V
(D)	I	III	IV	V

2. A structure is said to be linearly elastic if:
(A) load $\propto$ displacement
(B) load $\propto \frac{1}{\text{displacement}}$
(C) energy $\propto$ displacement
(D) energy $\propto$ load

3. Which one is not the characteristic of fatigue fracture?
(A) Rough fracture surface
(B) Rough and smooth areas on the fracture surface
(C) No plastic deformation
(D) Conchoidal markings on the fracture surface

4. A fatigue crack in a sound and smooth specimen takes:
(A) longer time in initiation than the propagation
(B) longer time in propagation than initiation
(C) equal time in initiation and propagation
(D) no time in propagation

5. Stress curve is always a straight line for:
(A) elastic material
(B) materials obeying Hooke's law
(C) elastoplastic materials
(D) none of the above

6. The term nominal stress in the stress strain curve for mild steel implies:
(A) average stress
(B) actual stress
(C) yield stress
(D) stress at necking

7. Consider the following statements.
The principle of superposition is applied to
I. linear elastic bodies
II. bodies subjected to small deformations
Which of these statements is/are correct?
(A) I only
(B) I and II
(C) II only
(D) Neither I nor II

8. The strain at a point is a:
(A) scalar
(B) vector
(C) tensor
(D) none of these

9. If the value of Poisson's ratio is zero, then it means that:
(A) the material is rigid.
(B) the material is perfectly plastic.
(C) there is no longitudinal strain in the material.
(D) the longitudinal strain in the material is infinite.

10. Materials having an elongation of less than 5 per cent are considered brittle. In such cases, the factor of safety is based on:
(A) yield stress
(B) endurance limit
(C) limit of proportionality
(D) ultimate stress

11. Assertion (A): strain is a fundamental behaviour of the material, while stress is a derived concept.
Reason (R): strain does not have a unit while stress has a unit.
(A) Both Assertion (A) and Reason (R) are individually true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are individually true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

12. Assertion (A): the amount of elastic deformation at a certain point, that an elastic body undergoes, under given stress is the same irrespective of the stresses being tensile or compressive.
Reason (R): the modulus of elasticity and Poisson's ratio are assumed to be the same in tension, as well as, compression.

(A) Both Assertion (A) and Reason (R) are individually true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are individually true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

13. Assertion (A): in a tensile test on a specimen, true stress in the specimen is more than nominal stress.
Reason (R): the grip of the universal testing machine introduces stress concentrations.
(A) Both Assertion (A) and Reason (R) are individually true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are individually true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

14. Assertion (A): the failure surface of a mild steel specimen subject to a torque about its axis is along a surface perpendicular to its axis.
Reason (R): mild street is relatively weaker in shear than in tension and the plane of maximum shear is perpendicular to its axis.
(A) Both Assertion (A) and Reason (R) are individually true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are individually true but Reason (R) is not the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

15. Statement I: fatigue failure is a stress which is lower than the yield strength of a metal.
Statement II: repeated straining lowers the yield strength.
(A) Both Statement I and Statement II are individually true and Statement II is the correct explanation of Statement I.
(B) Both Statement I and Statement II are individually true but Statement II is not the correct explanation of Statement I.
(C) Statement I is true but Statement II is false.
(D) Statement I is false but Statement II is true.

16. Statement I: physical properties of composite materials are generally isotropic.
Statement II: the stiffness of the composite panel will often depend upon the orientation of the applied forces and/or moments.
(A) Both Statement I and Statement II are individually true and Statement II is the correct explanation of Statement I.
(B) Both Statement I and Statement II are individually true but Statement II is not the correct explanation of Statement I.
(C) Statement I is true but Statement II is false.
(D) Statement I is false but Statement II is true.

17. Fatigue resistance of materials can be improved by:
I. shot peening
II. polishing the surface
Which of the above is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

Answer Key

1. (A)	**2.** (A)	**3.** (A)	**4.** (A)	**5.** (B)	**6.** (A)	**7.** (B)	**8.** (C)	**9.** (D)	**10.** (D)
11. (B)	**12.** (A)	**13.** (B)	**14.** (A)	**15.** (A)	**16.** (D)	**17.** (C)			

7 Ceramics

Introduction

- The interatomic bonds in ceramics are either entirely, or mostly ionic with some covalent character, making them inorganic, and non metallic materials.
- Ceramics derives from the Greek word Keramik, which means 'burnt substance,' suggesting that the desired qualities of these materials are often attained by a high temperature heat treatment procedure known as firing.
- Due to significant variances in bonding, the characteristics of these materials may also vary widely. Hard and brittle, with poor toughness and ductility, this describes the majority of these materials.
- In many cases, advanced ceramics are chosen over other materials because of their superior hardness, wear resistance, chemical stability, high temperature strength, and low coefficient of thermal expansion. Due to the lack of conduction electrons, these materials are often excellent electrical and thermal insulators.
- Ceramic materials are essential to many engineering designs due to their high melting points and chemical durability in a variety of adverse situations.
- In general, there are two categories of ceramics used in engineering, which are conventional ceramics and industrial ceramics.
- Clay, silica (flint), and feldspar are the three main ingredients of traditional pottery. Standard ceramics include things like bricks and tiles.
- Contrarily, engineering ceramics are made from chemical compounds that are either pure, or very close to pure, such as aluminium oxide, silicon carbide, and silicon nitride.
- High-tech applications of engineering ceramics include the experimental AGT-100 automobile gas turbine engine's high temperature high areas and the support foundation for integrated circuit chips in a thermal conduction module.

Ceramic Crystal Structure (Ionic Packing Theory)

- For those ceramics in which atomic bonding is predominantly ionic, it is assumed that they are made up of electrically charged ions instead of atoms. These ions are assumed to be hard and spherical.
- Packing of these ions in the ceramic crystal is governed by the following two criteria.

Magnitude of Electrical Charge on Ions

The crystal should be electrically neutral i.e., all cation positive charges should be balanced by an equal number of anion negative charges. In CaF2, for example, for stable crystal structure there should be 2 F^- compared to Ca^{z+} ions.

Relative Size of Cations and Anions

- For a stable structure, all the anions surrounding a cation should be in direct contact with that cation. This cation–anion contact is established for a specific value of $\left(\frac{\mu_c}{\mu_a}\right)$.
- Here coordination number and $\left(\frac{\mu_c}{\mu_a}\right)$ are related to each other.

  μ_a = Radius of anion

  μ_c = Radius of cation

$\left(\frac{\mu_c}{\mu_a}\right)$	Coordination
<0.155	2
0.155–0.225	3
0.225–0.414	4
0.414–0.732	6
0.732–1	8

Table 7.1

Solved Examples

1. In a stable ceramic crystal structure, a cation is surrounded by three anions in the form of a planar equilateral triangle. The ratio of the cation–anion radius for the crystal is nearly:

1. 0.16
2. 0.24
3. 0.32
4. 0.41

Solution:

A stable ceramic crystal structure with a cation is surrounded by three anions in the form of a planar equilateral triangle that can be represented by.

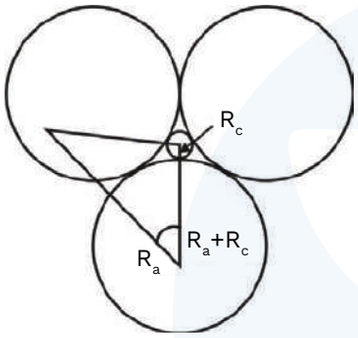


Fig. 7.1

As the triangle is equilateral, it can be written as:

$$\frac{R_a}{R_a + R_c} = \cos 30°$$

$$\frac{R_a + R_c}{R_a} = \frac{1}{\cos 30°}$$

$$1 + \frac{R_c}{R_a} = 1.155$$

$$\frac{R_c}{R_a} = 0.155 \approx 0.16$$

Silicate Ceramics

- The majority of soils, rocks, clays, and sand are classified as silicates because they are comprised mostly of silicon and oxygen, the two most plentiful elements in the earth's crust.
- The silicon atom occupies the core of a tetrahedron and is connected to four oxygen atoms at the tetrahedron's four corners. As this is the fundamental building block of silicates, it is often seen as a negative substance.
- The Si–O bond in the SiO_4^{4-} structure is 50 per cent covalent and 50 per cent ionic. The tetrahedral coordination of SiO_4^{4-} satisfies the directionality requirement of covalent bonding and the radius ratio requirement of ionic bonding.
- Due to the small, highly charged Si^{4+} ion, strong bonding forces are created within the SiO_4^{4-} tetrahedron and as a result, the SiO_4^{4-} units are normally joined corner-to-corner and rarely edge-to-edge.
- As each oxygen of the silicate tetrahedron has one electron available for bonding and hence, many different types of silicate structures can be produced.
- When positive ions bond with the oxygen of SiO_4^{4-} tetrahedron, island silicate structures are produced, for example, Fe^{2+} and Mg^{2+} ions combine with the SiO_4^{4-} to form olivine, which has the basic chemical formula $(MgFe)_2 SiO_4$.
- When two comers of each SiO_4^{4-} tetrahedron are bonded with the corners of other tetrahedrons, a chain as shown in Figure (a and b) or ring structure has the formula SiO_3^{2-} results. The mineral enstatite ($MgSiO_3$) has a chain silicate structure, and the mineral Beryle ($Be_3Al_2(SiO_3)_6$) has a ring silicate structure.
- Figure (a and b) below shows how silicate sheet structures are generated when three corners in the same plane of a silicate tetrahedron are bound to the corners of the three order silicate tetrahedron.

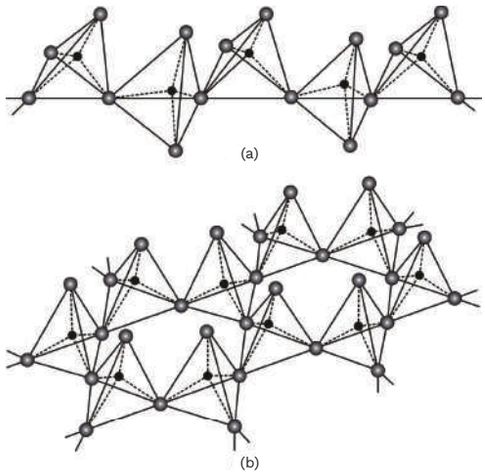


Fig. 7.2

- These silicate sheets can bond with other types of structural sheets because there is still one unbounded oxygen on the silicate tetrahedron Figure (b) above, for example, the negatively charged silicate sheet can bond with a positively charged sheet of $Al_2(OH)_4^{2+}$ to form a composite sheet of kaolinite as shown in the figure below.

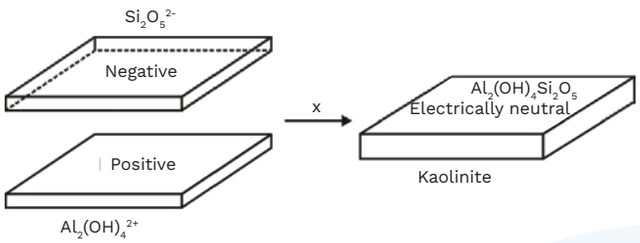


Fig. 7.3

Silica

- In silica, the four corners of tetrahedra share oxygen atoms, creating a network.
- There are two or three variations on each of the three fundamental silica structures (quartz, tridymite, and cristobalite).
- Many distinct polymorphic forms of crystalline silica are known, each one corresponding to a certain arrangement of silicate tetrahedral, in which all four corners are joined.

Feldspars

- Several naturally occurring silicates have infinite three-dimensional silicate networks. Among the industrially important networks, silicates are the feldspars which are also among the main components of traditional ceramics.
- In the feldspar silicate structural network, some Al^{3+} ions replace some Si^{4+} ions to form a network with a net negative charge. This negative charge is balanced with large ions of alkali and alkaline earth ions such as Na^+, Ca^{2+}, and Ba^{2+} which fit into interstitial positions.

Material Preparation

- Particle agglomeration is used in the production of a number of different types of ceramics. Agglomerated ceramic materials may be created in a range of states, including dry, plastic, and liquid.
- Although, hot forming technologies are utilised, cold forming techniques are more common in the ceramics sector. The following techniques are often used in the ceramics industry; processing, slip casting, and extrusion.
- Ceramic particular raw materials can be pressed in dry, plastic, or wet conditions into a die to form shaped products.

Dry Pressing

- The dry pressing method is used commonly for products such as structural refractories (high heat resistance materials) and electronic ceramic components.
- This method may be defined as the simultaneous uniaxial compaction and shaping of granular particles along with small amounts of water and/or organic binder in a die.
- After cold processing, the parts are usually fixed (sintered) to achieve the required strength and microstructural properties.
- Dry pressing is extensively used as it can form a wide variety of shapes rapidly with uniformity and close tolerances, e.g., aluminas, titanates, and ferrites can be dry pressed into sizes from a few mils to several inches in linear dimensions at a rate of up to about 500/minute.

Isostatic Pressing

- Isostatic pressing involves applying pressure on ceramic powder that has been put into a flexible (often rubber), airtight container (called a bag). The powder is equally compacted in all directions by the applied pressure, and the resulting product has the shape of a flexible container.
- After cold isostatic pressing the part must be fired (sintered) to achieve the required properties and microstructure.
- Ceramic parts manufactured by this process include refractories, bricks, and shapes, spark plug insulator radomes, carbide tools, crucibles, and bearings.

Hot Pressing

When the pressing and firing processes of ceramics are combined, the resulting components have a

higher density and better mechanical qualities. Uniaxial and isostatic techniques are used.

Slip Casting

- By using the slip casting process, ceramic shapes can be cast as illustrated in the figure below.

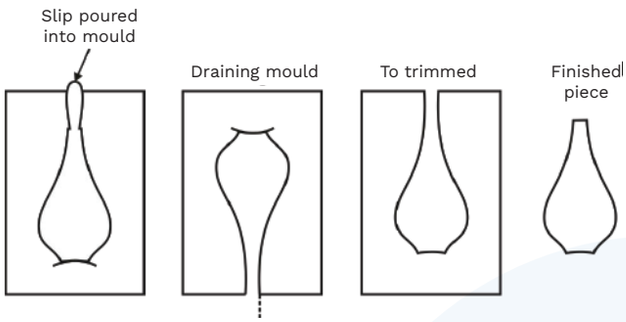


Fig. 7.4

- In this process, the main steps are as follows:
  - Slip is made by combining powdered ceramic material with a liquid (often clay and water) and mixing the resulting suspension until it is stable.
  - Slip casting is pouring the liquid component of the slip into a porous mould (often constructed of plaster of Paris) and allowing the mould to absorb some of the slip.
  - When the liquid in the slip is evaporated, a semisolid layer forms against the mould surface. When the wall thickness is right, the casting process is halted and the extra slip is drained out of the mould. 'Drain casting' describes this process.
  - For a solid object, the casting process may be allowed to continue, until the whole mould cavity is filled. The solid casting is the term for this specific kind of slip casting.
  - Allowing the material in the mould to cure for a sufficient amount of time ensures that it will be strong enough to be handled and then removed from the mould.
  - The last step in producing a cast item that has the desired microstructure and characteristics is firing.
  - Slip casting is advantageous for forming twin walled and complex shapes of uniform thickness. This process is especially economical for development parts and short production runs.

Extrusion

- One can produce the single cross-sections and hollow shapes of ceramic materials by extruding these materials in the plastic state through a forming die.
- This method is generally used to produce, for example, refractory brick, sewer pipe, hollow tiles, technical ceramics, and electrical insulators.
- The most commonly used means is the vacuum auger type extrusion machine, in which the plastic ceramic material (clay and water) is forced through hard steel or alloy die by a motor driven auger. Using a piston extrusion under high pressure special technical ceramics are frequently produced so that close tolerances can be attained.

Thermal Treatments

This is an essential step in the manufacturing of ceramics. Consider drying, sintering, and vitrification under thermal treatments.

Drying and Binder Removal

- The main purpose of drying ceramics is to remove water from the plastic ceramic body before it is fixed at higher temperatures. Usually, drying to remove water is carried out at, or below 100 degrees and can take about 24 hours for a large ceramic plant.
- By heating to the range of 200–300°C, the bulk of organic binders can be removed from ceramic parts, although some hydrocarbon residues may require heating to too high temperatures.

Sintering

- Solid-state diffusion is the process by which tiny particles of a substance link together. The porous compact is transformed into a dense, coherent product by this heating process in ceramic production. Ceramics like alumina, ferrites, beryllia, and titanates may be formed with this method.

- Particles in this process are coalesced by solid-state diffusion at extremely high temperatures, but below the melting point of the compound being sintered, for example, the alumina spark plug insulator is sintered at 1600 degrees, the melting point of alumina being around 2050 degrees.
- To establish a chemical link between the particles, atomic diffusion occurs between their contacting surfaces in this process. When this process advances, smaller particles are lost and bigger ones are created.
- The process is driven by the system's decreasing energy level. When a material is sintered, the high surface energy of the original, microscopic particles is replaced by the lower energy of the grain boundary surfaces.

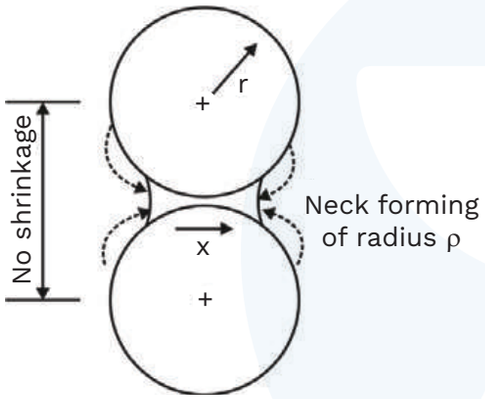


Fig. 7.5

Vitrification

- Glass phases may be found in certain electrical components and ceramic items, including porcelain and structural clay products.
- To facilitate diffusion at lower temperatures than in the remainder of the ceramic solid material, the glass phase acts as a reaction medium. The glass phase liquefies and fills the pore spaces in the material during the fire; this process is called vitrification and is unique to certain forms of ceramics.

Electrical Properties of Ceramics

- The electrical and mechanical properties of ceramics make them especially suitable for many insulator applications in the electrical and electronic industries.
- The ionic and covalent bonding in these materials restricts electron and ion mobility and therefore, these materials are quite suitable as good insulators. These bonding make most ceramic materials strong but relatively brittle.
- Here, some aspects of the structure and properties of a few insulator ceramic materials will be discussed.

Electrical Porcelain

- Typical electrical porcelain consists of approximately 50 per cent clay ($Al_2O_3 \cdot 2SiO_2 \cdot 2H_2O$), 25 per cent silica, and 25 per cent feldspar ($K_2O \cdot Al_2O_3 \cdot 6SiO_2$).
- This composition makes a material with good green body plasticity and a wide firing range of temperature at a relatively low cost.
- However, this material has a high power loss factor compared to other electrical insulator materials due to highly mobile alkali ions.

Steatite

- Steatite porcelains have low powerless factors, low moisture absorption, and exceptional impact strength, making them excellent electrical insulators. The electronic and appliance manufacturing sectors rely heavily on them.
- Industrial steatite compositions are based on 90 per cent talc ($3MgO.4SiO_2.H_2O$) and 10 per cent clay. The microstructure of fired steatite consists of enstatite $MgSiO_3$ crystals, which are bonded together by a glassy matrix.

Forsterite

Forsterite (Mg_2SiO_2) has no alkali ions in a vitreous phase, and thus it has a higher resistivity and lower electrical loss with increasing temperature than the steatite insulators. At higher frequencies, forsterite has lower loss dielectric properties.

Alumina

- These ceramics have alumnium oxide (Al_2O_3) as a crystalline phase bonded with a glassy matrix.
- The glass phase, which is normally alkali free, is compounded from mixtures of clay, talc, and alkaline earth fluxes and is usually alkali free.

- These ceramics have relatively high dielectric strengths and low dielectric losses along with relatively high strengths.
- Sintered alumina (99 per cent Al_2O_3) is widely used as a substrate for electronic device applications because of its low dielectric losses and smooth surface.
- Alumina is also used for ultralow loss applications where a large energy transfer through a ceramic window is necessary, for example, radomes.

Mechanical Properties of Ceramics

- As a class of materials, ceramics are relatively brittle. The observed tensile strength of these materials varies greatly, ranging from very low values 0.69 MPa to about 7×10^3 MPa for whiskers of ceramics such as Al_2O_3 prepared under carefully controlled conditions.
- Crystalline ceramics are rigid because of the ionic and covalent bonds that make them up. So, bonding between atoms in covalent crystals and covalently bound ceramics is directed to be strong because it involves the transfer of electron pairs. Covalently bound ceramics are brittle in both the single crystal and polycrystalline phases, as is seen when the crystals undergo brittle fracture under stress.
- The deformation of primarily bonded ceramics is different. Single crystals of ionically bonded solids, for example, magnesium oxide and NaCl exhibit considerable plastic deformation under compressive room temperature. Polycrystalline ionically bonded ceramics, however, are brittle, with cracks forming at the grain boundaries.
- In polycrystalline ceramics, adjacent grains must change shape during deformation as there are limited slip systems in ionically bonded solids, cracking occurs at the grain boundaries, and subsequent brittle fracture occurs. As most industrially important ceramics are polycrystalline, most ceramic materials tend to be brittle.
- The high hardness of some ceramic materials makes them useful as abrasive materials for cutting, grinding, and polishing other materials have a lower hardness.
- Examples of the two most commonly used manufactured ceramic abrasives are fused alumina (Al_2O_3) and silicon carbide (SiC).
- Abrasive products such as sheets and wheels are made by bonding individual ceramic particles together.
- Examples of bonding materials include fired ceramics, organic resins, and rubbers.
- The ceramic particles must be hard with sharp edges. Also, the abrasive product must have a certain amount of porosity to provide channels for air, or liquid to flow through the structure.
- Al_2O_3 (Aluminium oxide) grains are tougher than SiC (Silicon carbide) ones but are not as hard, and therefore SiC is normally used for the harder materials.
- Improved abrasives have been developed by combining ZrO_2 with Al_2O_3. This has higher strength, hardness, and sharpness than Al_2O_3 alone.
- One of these ceramic alloys contains 25 per cent ZrO_2, 75 per cent Al_2O_3 another 40 per cent ZrO_2, and 60 per cent Al_2O_3.
- Cubic boron nitride having the trade name 'borazon' is another important ceramic abrasive. This material is almost as hard as diamond but has better heat stability than diamond.

Thermal Properties of Ceramics

- Due to strong ionic covalent bonding, most ceramic materials have low thermal conductivities. Ceramics are good thermal insulators.
- Ceramic materials, due to their high heat resistance, are used as refractories, which are materials that resist the action of hot environments, both liquid and gaseous. Refractories are used widely by the metals, chemical, ceramic, and glass industries.
- There are many pure ceramic compounds with high melting points, for example, Al_2O_3, which could be used as industrial oxide, and magnesium oxide, which could be used as industrial refractory materials. However, these materials are expensive and difficult to form into shapes. This is why most industrial refractories are made of mixtures of ceramic compounds.

Glass

- Glass is a ceramic that is a mixture of substances with the principal constituent of silica that has solidified from the liquid state without crystallisation. Glass is an amorphous substance. It is a vitreous silicate with a random arrangement of silicon–oxygen tetrahedron SiO_4^{4-} units as shown in the figure.

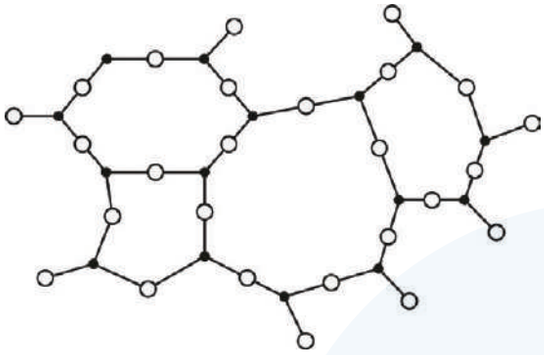


Fig. 7.6

- A glass made of silica alone has many desirable properties, but it requires a high temperature to manufacture, thereby making its manufacturing difficult and expensive. To reduce the temperature, modifying ions are added to silicates, which consist of sodium, potassium, and calcium. The modifying ions help in loosening the Si–O bonds.
- The constituents of glass include:
  - silica is the principal constituent
  - sodium or potassium carbonate reduces the melting point of silica and imparts
  - viscosity to the molten glass
  - lime to impart durability
  - manganese dioxide to correct the colour of the glass
  - colouring substance
- The glasses can be classified as follow.
  - Soda lime (crown) glass has a composition of 75 per cent sand, 12.5 per cent lime, and soda. It is the cheapest quality glass and is used for window and container glass.
  - Flint glass contains besides silica, varying proportions of lead oxide providing brilliance and high polish.
  - Pyrex (heat resistant) glass. Pyrex glass contains 80 per cent silica and 14 per cent boron oxide. It is used for cooking utensils and laboratory wares.
- The properties of glasses are given below.
  - It has no definite silicate structure, no sharp melting point, and can absorb, refract, and transmit light.
  - It is an excellent insulator at high temperatures, extremely brittle and hard, unaffected by air and water.
  - It can be produced in different colours and resistant to corrosion by chemical reagents.
  - It is weak in tension and can withstand thermal shock without cracking.
- The applications and uses of glass include:
  - the fibreglass reinforced with plastics can be used in the making of furniture, bathroom fittings, and parts of automobiles.
  - the optical glass is used for devices of science, astronomy, and bacteriology.
  - the glass linings are used to provide corrosion resistance to valves, pipes, and pumps.
  - many parts of guided missiles, noses of deep driving vehicles, and elevator cabins are made of glass.
  - the glass is used for the windows of the buildings. Colour changing glasses are developed which can store solar energy during the day and can give light at night.

Carbon

- There are many different polymorphs of carbon, and carbon may also exist in an amorphous state. This class of substances cannot be placed in the more familiar categories of metals, ceramics, or plastics.
- As graphite, one of the polymorphic forms is often regarded as a ceramic, and as the crystal structure of diamond, another polymorph, is similar to that of zinc blende, these materials will be explored in this chapter.

Graphite

- Graphite is another polymorph of carbon; its crystal structure is quite different from diamond's, and it is more stable at room temperature and pressure. Graphite has a

hexagonal lattice structure, with each carbon atom connected to three coplanar neighbours through strong covalent connections.

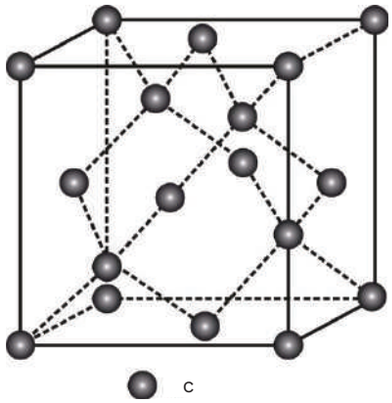


Fig. 7.7

- For example, in a Van der Waals kind of interlayer connection, the fourth bonding electron plays a little role.
- The remarkable lubricative characteristics of graphite are because its interplanar bonds are weak, making interplanar cleavage easy. Furthermore, crystallographic axes parallel to the hexagonal sheets have relatively high electrical conductivity.
- Graphite is used as a heating element for electric furnaces, as an electrode for arc welding, in metallurgical crucibles, in casting moulds for metal alloys, and in ceramic production. Among many other applications, it also has a low coefficient of thermal expansion, high resistance to thermal shock, high adsorption of gases, and is easily machined.

Diamond

- At standard conditions (room temperature and atmospheric pressure), diamond exists as a metastable form of carbon. Carbon atoms fill all available sites in the crystal structure, which is a variation of the zinc blende (both Zn and S).
- Therefore, each carbon forms four covalent connections with other carbons. This form, present in other Group IV A elements such as germanium, silicon, and grey tin (below), is fittingly named the diamond cubic crystal structure.
- Diamond's physical qualities make it a desirable material. It is exceedingly hard (the hardest known material) and has very poor electrical conductivity; these qualities are attributable to its crystal structure and strong interatomic covalent bonds.
- In addition, diamond has a high index of refraction and is optically clear in the visible and infrared parts of the electromagnetic spectrum, making it useful for cutting and polishing gemstones, even when it is rather big in size. In the manufacturing sector, diamonds are used for cutting and grinding softer materials.
- Beginning in the mid 1950s, methods were discovered to make synthetic diamonds, which have since been perfected to the point that now a significant fraction of industrial grade materials, as well as, some gem level materials, are man-made.

Fullerenes

- A new polymorphic form of carbon, fullerenes, was discovered in 1985. There is a single molecule of it, and it looks like a hollow spherical cluster of 60 carbon atoms.
- Each molecule consists of clusters of carbon atoms bound together in various geometric shapes, including hexagons (consisting of six carbon atoms) and pentagons (five carbon atoms).

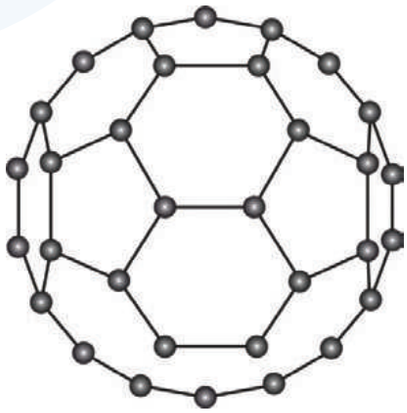


Fig. 7.8

- The figure depicts such a molecule, which is composed of 20 hexagons and 12 pentagons

arranged such that no two pentagons share a common side, and the molecular surface displays soccer ball symmetry.

- Buckminsterfullerene is the molecular substance named after R Buckminster Fuller. All of the carbon atoms in diamond and graphite make primary bonds with neighbouring atoms, making them what may be called network solids. Instead, the carbon atoms in buckminsterfullerene link together to create these spherical molecules.
- In the solid state, the units crystallise and pack in a cubic array with the faces facing each other.
- This substance is crystalline in nature, making it an excellent electrical insulator. However, it is possible to make it extremely conductive and semiconductive by adding the right kinds of impurities to it.

Practice Exercise

1. Above 1710°C, silica is:
(A) liquid
(B) gas
(C) solid
(D) plasma

2. The process by which small particles of a material are bonded together by solid-state diffusion is called:
(A) fusion
(B) vitrification
(C) sintering
(D) extrusion

3. When a dielectric fills the space between the plates, the capacitance of the capacitor is increased by a factor K, which is called:
(A) permeability
(B) permittivity
(C) dielectric constant
(D) insulation

4. Ceramic materials made from solid solutions of lead zirconate ($PbZrO_3$) and lead titanate ($PbTiO_3$) are called:
(A) ferroelectric ceramics
(B) PZT ceramics
(C) steatite
(D) forsterite

5. Because of high heat resistance, ceramic materials are used as:
(A) abrasive materials
(B) refractories
(C) semiconducting materials
(D) none of these

6. Oxides that break up the glass network are known as:
(A) network modifiers
(B) network insulators
(C) network multiplier
(D) network broker

7. The temperature at which a ferroelectric material on cooling undergoes a crystal structure change that produces spontaneous polarisation in the material is called:
(A) neel temperature
(B) curie temperature
(C) weiss temperature
(D) none of these

8. Most commercial glasses consist of:
(A) lime
(B) soda
(C) silica
(D) all the above

9. Not a characteristic property of ceramic materials:
(A) high temperature stability
(B) high mechanical strength
(C) low elongation
(D) low hardness

10. Hot isostatic pressing is not a viable option if the chief criterion is:
(A) strength without grain growth
(B) low cost
(C) zero porosity
(D) processing refractory ceramics

Answer Key

1. (A) **2.** (C) **3.** (C) **4.** (B) **5.** (B) **6.** (A) **7.** (B) **8.** (D) **9.** (D) **10.** (B)

8 Polymers

Introduction

- The word 'polymer' literally means 'many parts,' i.e., the basic makeup of polymers consists of smaller units (mers) joined together either naturally, or synthetically to produce polymers, or macromolecules. The polymeric solid material may be considered to contain many chemically bounded parts or units that themselves are bonded together to form a solid.
- Polymers are materials consisting of giant, or macromolecules, chain like molecules having average molecular weights from 10000 to more than 1000000 g/mole built by joining many mers, or units through chemical bonding.
- Molecular weights are defined as the sum of atomic masses in each molecule.
- Nature has given us several polymers like proteins, carbohydrates, silk, wool, cotton, rubber, leather, etc.
- Man has made a variety of polymers with wide ranging properties such as softness like silk and wool and strong like steel.
- Most polymers, solids, or liquids, are carbon based organic; however, they can be inorganic (e.g., silicones based on the Si–O network).

Basic Definitions

Polymers

- Polymers are the materials made from giant (or macromolecular), chain like molecules having average molecular weights from 10000 to more than 1000000 g/mol built by the joining of many mers or units by chemical bonds, i.e., a polymer is a large molecule built by the repetition of a small, simple chemical unit called monomer.
- The repeating chemical units are covalently linked to each other in a macromolecule. Polymers are usually, but not always, carbon based.
- The properties of a polymer are entirely different from those of the chemicals (or monomers) from which it is formed.

Mer

A unit group of atoms and molecules that defines a characteristic arrangement for a polymer.

Monomer

- To produce macromolecules, several simple molecules join together through covalent bonds; 'monomers.'
- Polymers cannot be made without the component parts called monomers. It is interesting to note that even the simplest molecules, such as ammonia, water, ethanol, etc., cannot act as monomers.
- Alkenes, vinyl chloride, adipic acid, and glycol with two bonding sites are all examples of compounds that have the potential to serve as monomers.

Polymerisation

- The conversion of a monomer into a polymer by a chemical reaction under controlled circumstances is the definition of this term.
- However, polymerisation cannot occur with only a monomer; an additional chemical is needed, known as an initiator.
- A monomer and an initiator are needed under the applied pressure gives us PVC polymer (polymer).

Degree of Polymerisation

A polymer's overall mass is proportional to the number of its constituent monomers that it is made up of in a repeating sequence. Polymerisation of anything may be calculated by dividing its molecular weight by the sum of its mer weights.

$$\text{Degree of polymerisation} = \frac{\text{Molecular weight of the polymer}}{\text{Molecular weight of monomer}}$$

General Characteristics of Polymer

- The general characteristics of polymers can be summarised as follows.
  - Polymers have long chain structures. The individual molecule of a polymer is very large i.e., it may consist of thousands of similar small molecules, all bonded together covalently.
  - All polymers have one thing in common, i.e., carbon, which further bonds with hydrogen, nitrogen, halogens, or other organic, or inorganic substances.
  - Although, a polymer's structure may be crystalline in simple materials generally they are nanocrystalline solids at room temperatures. No doubt, polymers pass through a vicious stage during formation.
  - Polymers have lightweight and they can be easily fabricated and shaped. They are poor conductors of electricity and their thermal conductivity is also low. Moreover, the polymers are resistant to chemical attack and decay.
- Polymers are widely used in different fields. Polymers are used in an indefinite number of forms, for example, as telephone sets, paints, radio, and television cabinets, coatings, adhesives, and other countless objects. A few notable developments in polymer materials are given below.
  - Polymer materials are strong than steel according to the weight-to-weight ratio.
  - Polymer materials are suitable for repairing damaged kidneys or hearts.
  - Polymer materials act as an adhesive, which are so strong that the use of nails in wood and leather can be avoided.
  - Lining for spacecraft to withstand high temperature, especially during re-entering into the earth's atmosphere. We know that there is an increase in temperature due to friction when the spacecraft reenters the earth's atmosphere.
  - Silicon film does not permit water to enter but allows dissolved oxygen, in water, to pass through it. Using polymers, underwater tents have been prepared, which allows a man to live and work.

Molecular Structure of Polymers

The polymers can have structures that include:

- linear chain structure
- branched chain structure
- cross linked structure

Linear Chain Structure

Polymers, or linear chains, are formed when simple molecules, or mers, link together in a chain. The figure depicts the linear polymer. Units are bound together by relatively weak secondary bonds, and the overall structure is straightforward and consistent. Polyethylene, a thermoplastic, is one such material. Softening and unusability at high temperatures characterise thermoplastics.

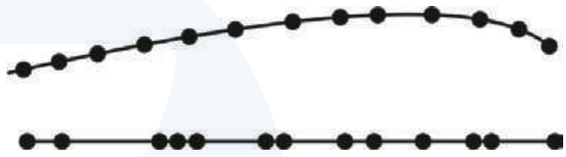


Fig. 8.1

Branched Chain Structure

The polymer has additionally branched chains besides linear chains as seen in the figure. Due to the intertwining of the chains, the polymer is both more robust and less malleable. To create a branch, C–C bonding is usually substituted for one of the side atoms on the main chain. The chain packing efficiency is diminished owing to the branching chains, consequently reducing the polymer density.

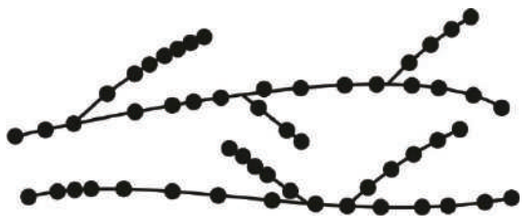


Fig. 8.2

Cross Linked Structure

- The polymer contains interlinking chains joining neighbouring linear chains as illustrated in the figure. Because of the interlocking anchors,

nearby molecular chains are held together, limiting the freedom of motion of any one chain.

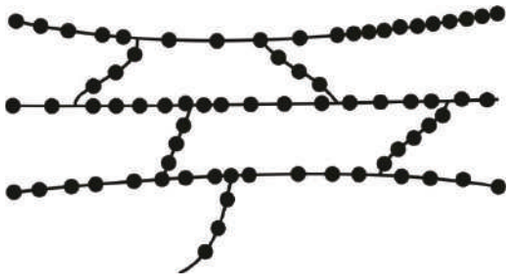


Fig. 8.3

- Incorporating cross links into a polymer structure enhances the material's strength and makes it less pliable. To harden, thermosetting polymers must have their chain ends joined together. There is no way to soften a thermosetting plastic after it has hardened and set. Hence, they may be employed at high temperatures.

Types of Polymerisations Reactions

Addition Polymerisation or Chain Reaction Polymerisation

- It is a process by which monomer units are attached one-by-one in a chain like structure. The resultant product molecule atomic weight is an exact multiple of monomer atomic weight.
- In addition to polymerisation, there is no by-product in this process. There are three stages: initiation, propagation, and termination. The polymer formed by this process is a thermoplastic type.

Condensation Polymerisation (or) Step Reaction Polymerisation

- When a polymer is formed by two or more monomers (similar or different) with the elimination of small molecules like water, ammonia, alcohol, HCl, etc. is termed condensation polymerisation.
- Monomers involved in condensation polymerisation are different then monomers involved in addition polymerisation.
- The main characteristics of monomers for condensation polymerisation are given below.
  - These monomers have functional groups such as alcohol, amine, –COOH, –CHO groups, etc, instead of a double bond.
  - Each monomer has at least two reactive sites, which usually means that there should be two functional groups.
  - This reaction is slower than addition polymerisation. Chain growth can be observed at two active centres. The polymer formed by this process is a thermosetting type.

Classification of Plastics

Thermoplastics

A plastic that can be melted repeatedly by heating and which can be molded again and again into different shapes with repeatedly melting is called thermoplastic. Hence, a thermoplastic becomes soft on heating and this property allows it to set in the desired shape again and again which it retains on cooling. For example, polyethylene, polyvinyl chloride (PVC), polystyrene, polycarbonate, and nylon.

Thermosetting Plastics

It is a plastic substance that once set or moulded does not become soft on further heating. Instead of melting, it degrades on further heating. Thermosetting plastics are irreversible plastics that cannot be melted or softened once set or moulded into a given shape. For example, bakelite, melamine, and urea-formaldehyde resin.

Thermoplastics	Thermosetting Plastics
They have long chains of polymers that are not cross linked.	They have long chains of polymers that are cross linked.
They become soft on heating. They can be molded on heating and under pressure.	They become soft on heating. They can be moulded on heating and under pressure similar to thermoplastics.
They can be remoulded on reheating.	They cannot be remoulded on heating.
They can be reused or recycled.	They cannot be reused or recycled.

Table 8.1

Thermoplastic Materials

Polyethylene

- Polyethylene can be classified as following.
  - High density polyethylene (HDPE) is produced at low pressure.
  - Low density polyethylene (LDPE) is produced at high pressure.
- LDPE has low density levels and only a small amount of branching. It is very flexible and easy to clean. It is often used to make plastic film wrap and plastic bags. It can be easily moulded.
- HDPE has higher density levels. It has a linear structure with no branching. It is stronger and resistant to chemicals. It is suitable for blow moulding.
- The properties of polyethylene include lighter plastic, chemically resistant, does not absorb moisture, and relatively low cost.
- It is used for making pipes, fans, blower casing, packaging, and buckets.

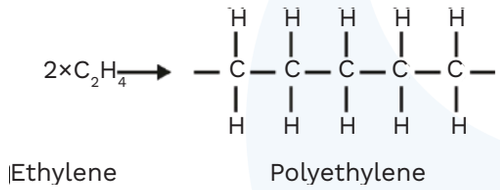


Fig. 8.4

Polyvinyl Chloride (PVC)

- PVC plastic is produced by the addition polymerisation of vinyl chloride. It is different from polyethylene as one of four hydrogen atoms attached to carbon is replaced by a heavier chlorine atom, which provides rigidity.

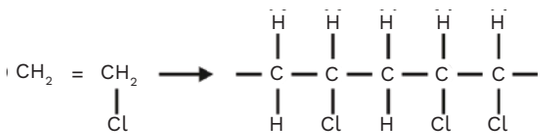


Fig. 8.5

- Plastic is a self extinguishing type and it is used widely for the making of cable jackets, lead wire insulation, fibre coating, pipes and fitting, and flooring and ceiling paneling.

Teflon

- Teflon is polytetrafluoroethylene (PTEE) and it is obtained by addition polymerisation technique from tetrafluoroethylene (C_2F_4).

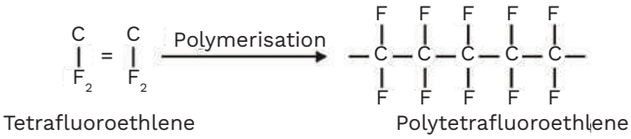


Fig. 8.6

- PTEE is completely inert chemically due to the presence of a strong carbon–fluorine covalent bond. It is non flammable and a good electrical insulator. It has high corrosion resistance and a low coefficient of friction.
- It is used in bearing bushes, piston rings, anti-corrosion seals, gaskets for IC engines, and chemical pipes.

Polystyrene

- It is obtained from the addition polymerisation of monomers styrene, a liquid petrochemical as shown below.

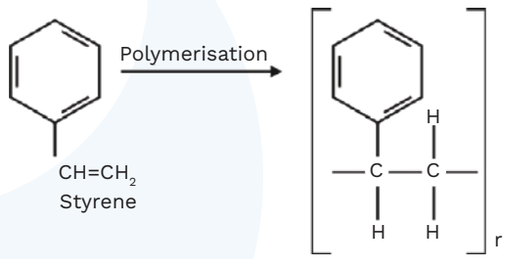


Fig. 8.7

- The polymer is hard, brittle, transparent, free from taste and harm to foodstuff, corrosion resistant, and low dielectric constant.
- It is used in electrical insulation, food containers, housewares, packaging, and lenses.

Thermosetting Materials

Phenol-Formaldehyde (PF) or Bakelite

- It is made by condensation polymerisation of phenol and formaldehyde as shown below.

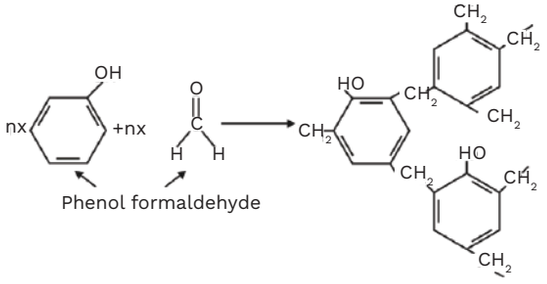


Fig. 8.8

- Bakelite is strong, rigid, and dimensionally stable. It is resistant to heat, chemicals, and solvents.
- It is used for making electrical parts, handles, and knobs for utensils and appliances, paints and adhesives, lavatory seats, and thermal insulators.

Polyester

- It is made by condensation polymerisation of a dicarboxylic acid and dihydric alcohol.
- Polyesters have good resistance to heat and chemicals.
- They are used in components of automobiles, helmets, paints, borders for glass fibres, jointing, and repair work.

Melamine

- It is obtained by condensation of ureas (melamine) with formaldehyde as shown below.

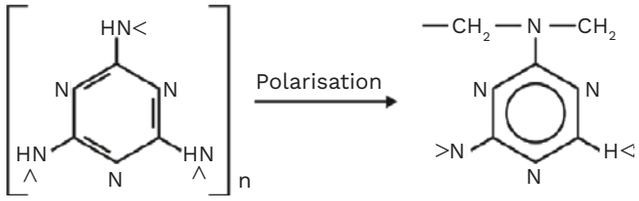


Fig. 8.9

- Melamine has high resistance to chemicals, heat, and moisture. It is also hard and scratch free. It has good electrical properties.
- It is used in surface coating, paints, crockery items, doors, plugs, switches, and buttons.

Epoxies

- They are obtained by condensation polymerisation of epichlorohydrin and poly hydroxyl compound as shown below.

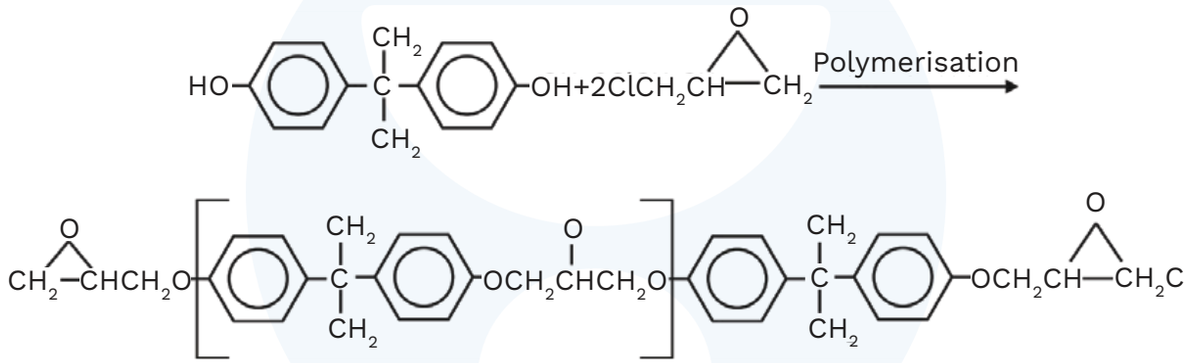


Fig. 8.10

- Epoxies are transparent, tough materials, and chemical resistant. They are used as insulating material in cable box jointing for cable and transformers. They are also used as adhesives for the jointing of various types of components.

Mechanical Behaviour of Plastics

- Melting point: the plastics have a low melting point and they are unsuitable for high temperature applications above 100°C.
- Dimensional stability and water effect: the plastics have poor dimensional stability. They are susceptible to weather conditions, i.e., they lose color and rigidity, thereby distorting when exposed to continuous sunlight.
- Impact strength: they have low impact strength, which further reduces when material softens at high load temperatures.
- Hardness: they have a lower hardness than metals and ceramics.
- Fatigue strength: they have poor resistance to fatigue failure.
- Tear strength: tear strength is the ability to resist tearing. Plastic films have moderate tear strength pose R and are commonly used for packaging.
- Elasticity: plastics have low elasticity but they display linear elasticity.
- Service behaviour: plastics develop cracks in the regions of localised stress concentrations. They fail due to the formation and propagation of cracks.

- Fracture mode: the thermosetting plastics are hard and rigid. They fail in the brittle mode. However, thermoplastics fail in ductile brittle transition mode.
- Thermal and electrical conductivity: they are poor conductors of heat and electricity. They are used as insulators.

Compounding Materials

- Compounding ingredients are added to polymers to improve the material's mechanical qualities and bring down the price. In most cases, the following are added as compounding materials.
- Binders: it is the binder's primary function to stabilise the plastic's other components. Binders may be either synthetic or natural resins. The resins used in its production determine whether the finished product is thermoplastic or thermosetting.
- Fillers: fillers for plastics increase their durability and toughness. Materials may either have a fibrous structure, like glass, or asbestos, or not have a fibrous structure, like clay or carbon black. A maximum of 50 per cent by weight of the plastic may be filler.
- Materials with a low molecular weight (about 300) are mixed with polymers as plasticisers. To increase the plastics' malleability and processing capacity and to lower the required moulding temperature and pressure, plasticisers are added.
- Additions of plasticisers, which weaken the material's structure and increase its pliability, should be kept to a minimum. To improve flexibility, plasticisers like polyesters and epoxies are often blended in. A plasticiser might make up as much as 60 per cent of the material.
- Blending is the process of blending polymer granules of various types to create a new product with novel properties.
- To reduce the potentially damaging effects of heat, sunshine, and ozone, stabilising agents are often used. It is stabilised with barium, cadmium laurate, and white lead.
- Pigments are applied as colouring agents to plastics to get the desired hue. Pigments such as zinc oxide, white lead, and titanium dioxide are often utilised.

Metals	Ceramics	Polymers
Materials are composed of one or more metallic elements.	Compounds between metallic and non metallic elements (oxides, nitrides, and carbides).	A large molecule composed of repeating 'mers.' It is an organic compound based on carbon, hydrogen, and other non metallic elements.
Distinguishing characteristics such as stiff, strong, ductile, and resistance to fracture.	Distinguishing characteristics such as stiff and strong as metals, very hard, brittle, resistance to high temperatures, and harsh environments.	Distinguishing characteristics such as low density, not stiff and strong, extremely ductile and pliable, relatively inert, and tendency to soften.
They are opaque.	They may be transparent, translucent, or opaque.	They may be transparent, translucent, or opaque.

Table 8.2 Comparison of Polymers with Ceramics and Metals

Vulcanisation

- The process of crosslinking in elastomers is known as vulcanisation. It can be achieved by a non reversible chemical reaction. Generally, it is earned out at an elevated temperature.

- In most of the vulcanisation processes, sulphur compounds are added to the heated elastomer. In this process, chains of sulphur atoms bond with adjacent polymer backbone chains and it is used for crosslinking them.

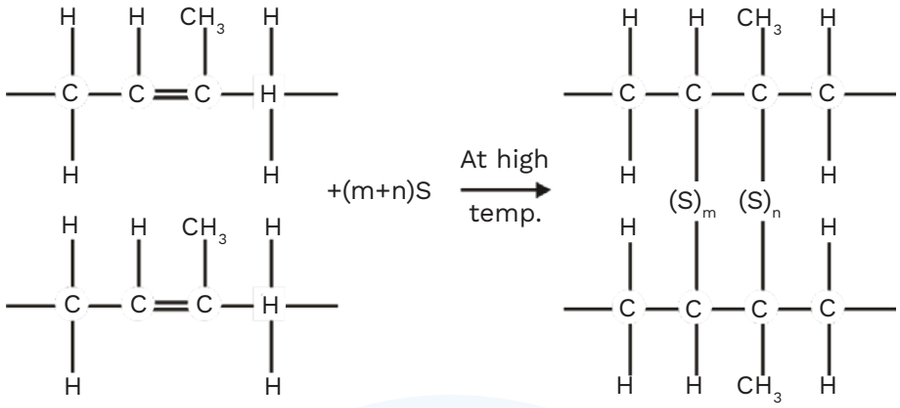


Fig. 8.11

- Before vulcanisation, crosslinking main sites are doubly bonded carbon atoms. But after the vulcanisation process, these bonds become singly bonded.
- With the increase in the extent of vulcanisation hardness, abrasion resistance improved. Modulus of elasticity, tensile strength, and resistance to degradation are all improved by vulcanisation.
- With the increase in sulphur content above a limit, hardness increases but the limit of extensibility decreases.
- Because vulcanised materials are crosslinked, these materials are thermosetting in nature.

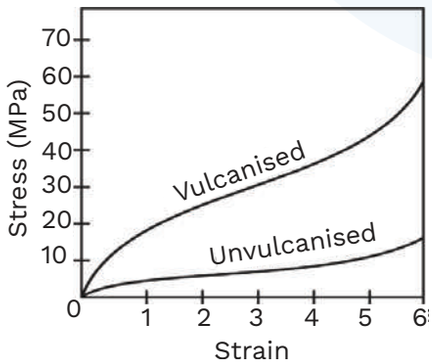


Fig. 8.12

Stress and strain curves for vulcanised and unvulcanised natural rubber.

Solved Examples

1. How much sulphur is required per 100 kg of a final rubber product to completely crosslink butadiene rubber?

(A) 17 kg
(B) 27 kg
(C) 37 kg
(D) 47 kg

Solution: (C)

For two monomer units of 1, 3 butadiene,

$$\text{molecular} = 8 \times \frac{12 \text{ gm}}{\text{mol}} + 12 \times \frac{1 \text{ gm}}{\text{mol}} = 108$$

For two monomer units, two sulphur atoms are taken to form cross linking.

Atomic weight of sulphur = 32 g/mol

2 sulphur atoms are used

now,

fraction of sulphur used

$$= \frac{2 \times 32 \text{ g/mol}}{2 \times 32 \text{ g/mol} + 108 \text{ g/mol}}$$

$$\text{fraction of sulphur used} = \frac{64}{64 + 108} = 0.37$$

For 100 kg, 37 kg of sulphur is used.

Practice Exercise

1. The side group in polyvinyl chloride is:
(A) H, Cl, Cl, Cl
(B) H, H, H, Cl
(C) H, H, H, H
(D) H, H, CH_3, Cl

2. The factors that promote non crystallinity in polymers are:
(A) branching and addition of plasticisers
(B) copolymerisation
(C) large random side groups
(D) all of these

3. Apart from three Hatoms, the fourth side group in the monomer of polypropylene is:
(A) C_6H_6
(B) CH_3
(C) C_6H_5
(D) $COOCH_3$

4. The number of double bonds required in the monomer for crosslinking is:
(A) 2
(B) 3
(C) 4
(D) 1

5. Plasticisers are added to plastics to:
(A) reduce hardness
(B) improve flexibility
(C) improve their strength
(D) reduce glass transition temperature (T_g)

6. Which one is not a characteristic of plastics?
(A) Low density
(B) Machinability
(C) High strength
(D) Large plastic deformation

7. In teflon, i.e., PTFE monomer, the four side groups are:
(A) H, H, H, CH_3
(B) F, F, F, F
(C) H, H, H, H
(D) H, H, Cl, Cl

8. An ebonite is:
(A) copolymer
(B) natural wood
(C) natural rubber
(D) none of these

9. Plastics are used in automobiles because of their:
(A) low melting point
(B) high strength
(C) capacity to be moulded with a good finish
(D) none of these

10. Consider the following statements.
I. Asphalt is a naturally occurring product having a non crystalline structure.
II. Rock asphalt occurs in limestones or sandstones.
III. Tar is a residue left after the distillation of crude oil.
IV. Resins in asphalt provide stickiness to the product.
Which of the above statements are correct?
(A) I, II, and III only
(B) III and IV only
(C) I, II, and IV only
(D) I, II, III, and IV

Answer Key

1. (B) **2.** (D) **3.** (B) **4.** (A) **5.** (B) **6.** (C) **7.** (B) **8.** (C) **9.** (C) **10.** (D)

9 Composites

Introduction

- Composites are multiphase materials that have desirable characteristics with both their component phases. It is the 'principle of coupled action' to explain this phenomenon. Using this technique, one can create new and interesting things by skillfully combining two, or more different kinds of stuff.
- An excellent example of composite material is reinforced concrete. When steel rebar is used to reinforce concrete, the concrete acts as a matrix to protect and support the rebar.
- All composites consist of a matrix or binder and a reinforcing substance as this is the fundamental building block of composites. One or more phases of different materials are spread across a matrix material to create a composite. Generally, a material is considered to be an advanced composite if its fibres are both continuous and in a certain orientation.
- When constructing composite materials, scientists and engineers have cleverly blended different metals, ceramics, and polymers to create a new generation of amazing materials.
- Fibreglass fishing rods, for instance, are made with a reinforcing material. A thermosetting resin matrix is used to fix glass fibres. The end result is a fishing pole that is both lightweight and flexible despite its strength.
- It is possible to organise composites using the straightforward system shown in the image. Particle reinforced composites, fibre reinforced composites, and structural composites are the three primary categories. Specifically, it is found that every major category may be further broken down into two subcategories.

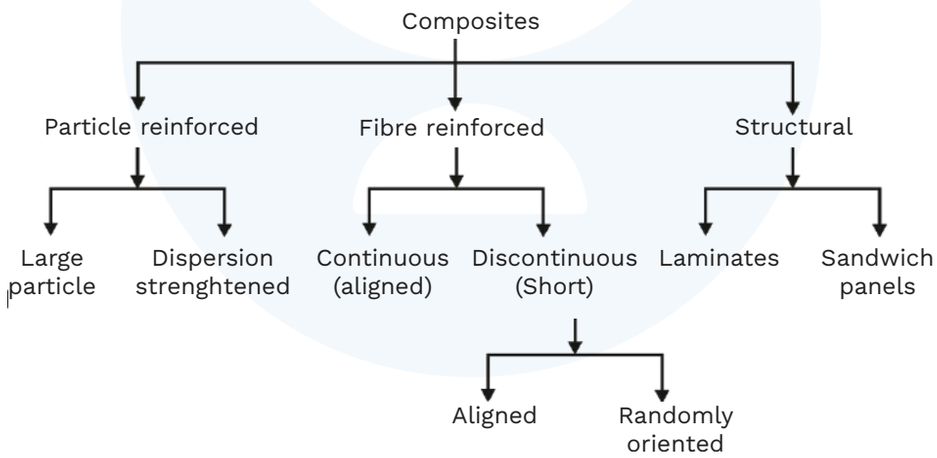


Fig. 9.1

General Characteristics of Composites

- Composites are complex materials with components that are chemically incompatible or only minimally soluble with one another, and that are separated by clear boundaries.
- When compared to traditional building materials, composites excel in every metric of structural performance, including specific strength, high temperature strength, stiffness, fatigue strength, and more.
- One may design a material with a certain set of characteristics and then include those specs into the finished product. In addition, this method allows for the material to be moulded as closely to the final product, or structural unit form as is feasible.
- The physical and mechanical characteristics of the constituent materials and the quality of the intermolecular bonds are the primary determinants of the final composite's performance.

- Composites are distinguished by maximising the benefits of their individual parts. In certain cases, composite materials may develop advantageous qualities beyond those of their individual parts. Composites are optimised by selecting constituent materials with opposing yet mutually reinforcing characteristics.
- Natural processes have been used as a model for composite production. Natural composites include plant branches and trunks, as well as, human and animal skeletons. Bones are made up of thin, strong phosphate fibres that are connected together by plastic collagen, while wood is made up of cellulose fibres that are held together by plastic lignin.
- Composites may have a polymer, or metal matrix, or a metal foundation (metallic composites), compounds of carbon and ceramics (non metallic composites).
- When it comes to composites, the matrix plays a crucial role as the binding and shaping agent. Working temperature, fatigue strength, resistance to environmental influences, density, and specific strength are all examples of significant operational parameters that are heavily influenced by their qualities throughout the production of composite materials. Composites may have a mixed matrix with two, or more compositionally distinct layers.
- Matrix combination composites are also referred to as multimatrix, or multilayer composites. The variety of desirable qualities shown by multimatrix composites is broader. It has been suggested that combining titanium and aluminium may improve the composite material's strength in directions perpendicular to the fibres. Matrix materials with aluminium layers have a lower density than those without.

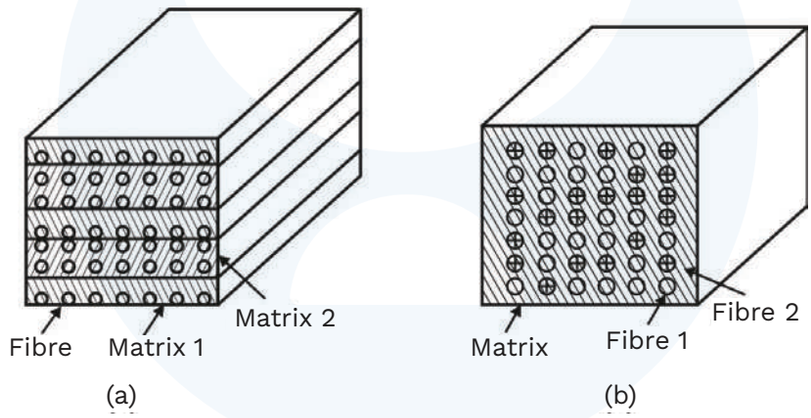


Fig. 9.2

- Other components, or 'fillers,' are spread out evenly across a matrix. They are crucial to the process of strengthening composites, which is why they are referred to as 'strengtheners.' High levels of hardness, strength, and elastic modulus are desirable in fillers. These traits need to be far more advanced than the matrix.
- If the filler's elastic modulus, or ultimate strength is increased, the composite material's corresponding qualities will also grow, but they will not approach the value of the filler. Reinforcing components are another term for fillers. The word 'filler' encompasses a wider range of materials than the more narrowly defined 'strengthener,' and it does not dictate that the filler be employed just for strengthening purposes.
- Composite material qualities may also be affected by filler geometry (form), dimensions (size), concentration (amount), and distribution (reinforcement pattern).
- Fillers may be broken down into three distinct categories based on their dimensionality: zero, one, and two dimensional.
- Composites may be broken down into three categories according to the reinforcing pattern: uniaxial, biaxial, and triaxial.

- To acquire a broader complex of qualities or to enhance a single attribute of composite material, several fillers of varying shapes may be utilised. A zero-dimensional filler, for instance, may improve the binding between a polymer matrix and a filler element with a single dimension (such as glass or carbon fibres).
- Similar results may be obtained by using fillers of the same form but a different composition to reinforce a composite material. Additional reinforcement using boron fibres, for instance, may improve the modulus of elasticity of composite materials having a polymer matrix reinforced by glass fibres.
- Complex reinforced composites are composites with two or more types of fibres.

Particle Reinforced Composites

- Large particle composites and dispersion strengthened composites are two subcategories here.
- Most of these composites have a particulate phase that is tougher and stiffer than the matrix. These reinforcing particles tend to slow down the matrix phase's motion close to each individual particle.
- Differences between them may be seen in their respective techniques of reinforcement, or strengthening. When dealing with particle-matrix interactions of this size, continuum mechanics is used as atomic or molecular scale treatments are inadequate.
- Typically, the particles used in dispersion enhanced composites are substantially smaller (diameter between 0.01 and 0.1 μm). Atomic and molecular level particle matrix interactions result in reinforcement.
- Particles take on a portion of the tension imparted to the matrix because of their ability to disperse the force. To what extent the mechanical behaviour is reinforced, or enhanced relies on the strength of the bonds between the matrix particles.
- It's interesting to compare the strengthening process to that of raindrops. The matrix takes the brunt of the stress, while the scattered particle slows down the dislocations.
- Limiting plastic deformation increases hardness, yield strength, and tensile strength.

Large Particle Composites

- There are large particle composites made when fillers are added to certain polymeric materials. Fillers replace a portion of the polymer volume with cheaper filler material and/or alter the material's characteristics.
- Although, particles' morphologies might be fairly different from one another, they should all have about the same dimensions (equiaxed). For efficient reinforcement, particles should be tiny and dispersed uniformly throughout the matrix.
- Besides plastics, concrete is a common large particle composite. Cement (the matrix) and aggregates like sand and gravel are what make up concrete (the particulates).
- Ceramic metal composites, or cermets, are one such material. Cemented carbide is made of refractory carbide ceramic particles like tungsten carbide or titanium carbide, which are exceedingly hard (TiC). The most prevalent kind of cermet is one in which a ceramic material is encased in a metal matrix, often nickel, or cobalt. Cutting tools made from these composites are often used on hardened steel.
- Metals, polymers, and ceramics are all used in large particle composite applications.

Concrete

- The matrix and dispersion phases of this typical large particle composite are both ceramics. Generally, concrete refers to a composite material made up of an aggregation of particles bonded together in a solid body by a binding substance, such as cement. Portland and asphaltic cement concrete, with their aggregates of gravel and sand, are the most common types of concrete used in the world.
- The primary use of asphaltic cement is as a paving material, whereas portland cement concrete is widely employed for structural purposes.

Portland Cement Concrete

- Portland cement, sand, gravel, and water make up portland cement concrete's three aggregate types and a liquid binder.

- Aggregate particles serve as a filler and help to keep concrete costs down. However, cement is expensive.
- As a primary building material, this is essential. Portland cement concrete sets up at room temperature and can even withstand being immersed in water. However, it is shown to be very weak and exceedingly brittle as a structural material, with tensile strength around 10 to 15 times lower than its compressive strength.
- The optimal strength and workability of a concrete mixture can only be achieved if the materials are introduced in precise amounts.
- Also, massive concrete buildings may expand and compress significantly when the weather changes. External pores are also penetrated by water, which may lead to significant cracking when exposed to freezing temperatures.

Concrete Reinforcement

- The strength of portland cement concrete may be increased by using reinforcement. Concrete may be reinforced with steel rods, wires, bars (rebar), or mesh before it sets.
- Steel is an excellent reinforcing material as its thermal expansion coefficient is very close to that of concrete. Not only does steel not rust quickly in a cement environment, but it also forms a reasonably strong adhesive connection with the curved concrete. Adding curves to the steel member's surface improves its ability to adhere to other surfaces. This allows for a tighter interlocking structure.
- Increased tensile, compressive, and shear stress tolerance is achieved by reinforcement in a hardened structure. If concrete breaks, the reinforcing is still there.
- To further strengthen portland cement concrete, one may include high modulus materials like glass, nylon, steel, and polyethylene into the new concrete fibres and mix them in. It is important to use caution when using this sort of reinforcement, however, as certain fibre materials rapidly deteriorate when exposed to a cement environment.

Dispersion Strengthened Composites

- This type of composite contains small particulates or dispersants, which increase the strengths of the composite by blocking the movement of dislocations. The dispersoid is typically a stable oxide of the original material.
- A common example is sintered aluminium powder (SAP). SAP has an aluminium matrix, which contains up to 14 per cent aluminium oxide (Al_2O_3). This composite is produced with the powder metallurgy process, where the powders are mixed, compacted at high pressures, and sintered together.
- Sintering involves heating a material until the particles of the material fuse together. Only the edges of the particles are generally bonded together so the particle does not melt.
- In terms of sintered ceramics, the product is a strong, rigid, and brittle product that exhibits good compressive strength, high melting points, and good heat resistance.
- Examples of dispersion strengthened composites include Ag–CdO, used as electrical contact material, Pb–PbO, used in battery plates, and Be–BeO, used in nuclear reactor and aerospace components.

Fibre Reinforced Composites

- These are strong fibres embedded in a softer matrix to produce products with high strength to weight ratios. The matrix material transmits the load to fibres, which absorb the stress. Under applied stress, the fibre matrix bond ceases at the fibre ends, yielding a matrix deformation pattern as shown in the figure.

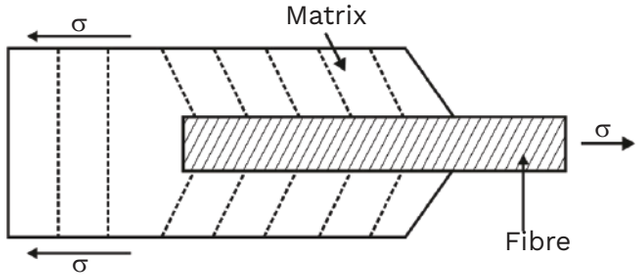


Fig. 9.3

- To have effective strengthening and stiffening of the composite material, some critical fibre length is essential. This critical length l_c is dependent on the fibre diameter d and its ultimate (or tensile) strength σ_f^* on the fibre matrix bond strength (or the shear yield strength of the matrix, whichever is smaller) τ_c according to the following relation.

  $l_c = \sigma_f^* d/2\tau_c$

- The reinforcing fibres' aspect ratio (length/diameter) affects the composite's final characteristics. If the aspect ratio is high enough, the composite will be quite sturdy.
- The composites' strength comes from the interlocking structure of the reinforcing fibres and the matrix.
- That is why it is preferable to use long and uninterrupted fibres in composites rather than shorter ones. Yet, it is more challenging to manufacture and embed continuous fibres in the matrix. Reinforcement is diminished, however, by the fact that shorter fibres are simpler to embed in the matrix.
- One may introduce the reinforcing fibres into the matrix in many orientations. Note, short and randomly oriented fibres having a small aspect ratio—typical of fibre glass—are easily introduced into the matrix and exhibit relatively isotropic behaviour in the composite.
- Long, or even continuous unidirectional arrangements of fibres yield anisotropic properties, with particularly good strength and stiffness parallel to the fibres. Often, these fibres are designated as 0° piles, indicating that all of the fibres are aligned with the direction of the applied stress. However, unidirectional orientations provide poor properties, when the load is perpendicular to the fibres.

Continuous and Aligned Fibre Composites

- Stress-strain behaviours of the fibre and matrix phases are crucial to the mechanical behaviour of this composite. Both the tension or load direction and the phase volume fractions have a role.
- A composite with aligned fibres has highly anisotropic characteristics, meaning that the measured values vary significantly concerning the direction of measurement.

Discontinuous and Aligned Fibre Composites

- Discontinuous fibres indeed have lesser reinforcing effectiveness than continuous fibres, yet composites made from discontinuous and aligned fibres are in higher demand.
- Glass cullet is the most common kind of discontinuous fibre utilised, although carbon and aramid strands are also common.
- These short fibre composites may be manufactured with moduli of elasticity and tensile strengths approaching 90 per cent and 50 per cent of their continuous fibre equivalents, respectively.

Laminar Composites

- Aligned layers that are cemented together one on top of the other at multiple orientations are often used in laminar composites, which are used when multidirectional loads are exerted within a single plane. So, they are called Laminar composites.

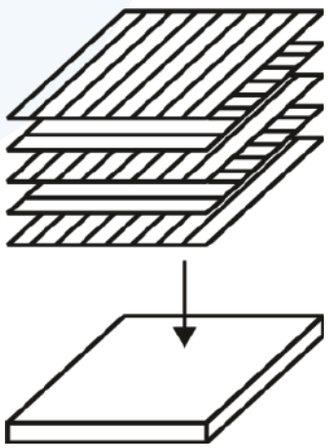


Fig. 9.4

- Laminar composites are well-known in the form of plywood, where the veneers are adhered together using adhesives (usually phenolic or amine resins) to achieve high strength and cheap cost at a lower weight.

- To ensure that the grain in each layer travels perpendicular to the layers above and below it, the piles are staked in odd numbers. Using this method, one may get plywood, that is, both durable and affordable.
- Some safety glass has a laminated structure; when shattered, the pieces stay in place thanks to an adhesive like polyvinyl butyral sandwiched between two layers of glass. Laminate countertops may also be seen in Formica's widespread usage.
- Laminar composites have great strength in many directions on a flat plane, but their strength in any one direction is lower than it would be if all the fibres were orientated in that direction.

Sandwich Structures

- These feature a low-density material or comb core covered by thin facing layers; examples include polymer foam and expanded metal. Corrugated cardboard is a common example of a sandwich style composite.
- Corrugated paper is constructed with two layers of paper that sandwich a corrugated paper core. The face material in such a building secures the inner core, which is the structurally important part. Aluminium alloys, fibre reinforced polymers, and plywoods are all examples of common materials for face panels.
- The core's structural roles are two fold: it separates the faces and resists deformations in planes perpendicular to the faces, and it offers some shear stiffness in these same planes.
- Cores may be made from a variety of materials and architectures, including foamed polymers, inorganic cement, synthetic rubbers, balsa wood, etc. The core is weaker and less rigid than the rest of the structure.
- Honeycomb structures, which combine increased strength with reduced weight, are another common kind of core used in fields like the aerospace industry.
- Thin foils have been fashioned into interlocking hexagonal cells, with the axis orientated perpendicular to the face panels, to create the honeycomb structure. The material might be the same as the front.
- These structures are lightweight, stiff, and strong and can be filled to provide sound and vibration damping. The honeycomb structure is shown in the figure.

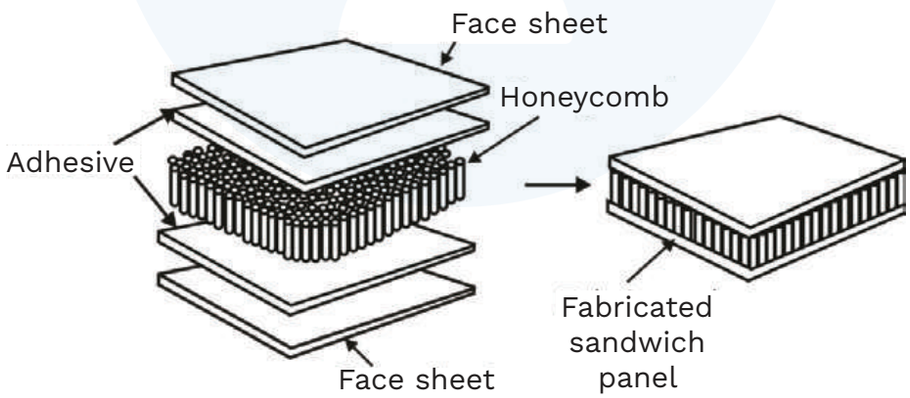


Fig. 9.5

Polymer Matrix Composites (PMCs)

- The matrix of these materials is a polymer resin (here resin means a high molecular weight reinforcing plastic), and the reinforcement is provided by fibres.
- Due to their wide range of useful qualities at room temperature, low cost, and simple production, these PMCs are employed in a wide variety of composite applications.
- Classifications of PMCs, their uses, and the polymer resins that are often used vary depending on whether the reinforcement is made of glass, carbon, or aramid.

Glass Fibre Reinforced Polymer (GFRP) Composites

- Fibre glass is a composite consisting of glass fibres, which may be continuous or discontinuous and contained within a polymer matrix.
- These are commonly produced in 'E' glass (E for electrical) because it draws well and has good strength and stiffness. A typical composition (Wt. per cent) would be $52SiO_2$, 17CaO, $14Al_2O_3$, $10Ba_2O_3$ with some oxides of Mg, Na, and K, and molten glass is gravity fed into a series of platinum bushings each of which has several hundred holes in its base.
- At rates of several thousand metres per minute, fine glass filaments are dragged mechanically as the glass drips from the perforations and coiled onto drums.
- As a group, they have extensive experience with the wide range of uses for fibre glass, including but not limited to automobile and marine body construction, plastic pipe installation, plastic flooring in industrial settings, and container manufacturing.
- Glass fibre reinforced polymers are being extensively used in the transportation industry to reduce vehicle weight and increase fuel efficiency. The automobile industry is actively exploring several potential new uses.

Carbon Fibre Reinforced Polymers (CFRPs)

- These are made of tiny graphite crystals. Strong covalent connections hold atoms in the basal planes together, whereas Van der Waals forces between layers keep them apart.
- The modulus of carbon fibres is proportional to the degree of perfection in the alignment of atom planes, and this is essential if one desires a high modulus and strength from the fibre.
- The specific modulus and specific strength of carbon fibres are the greatest of any reinforcing fibre material. The tensile modulus and strength of carbon fibres are unaffected by temperature increases. High temperature oxidation, however, might be an issue.
- At room temperature, carbon fibres are resistant to a broad range of solvents, acids, and bases.
- There is a wide range of physical and mechanical properties that carbon fibres may display, which can be used to tailor the qualities of composites that include carbon fibres to meet a variety of engineering needs.
- Carbon reinforced polymer composites are widely employed in a wide variety of applications, including fishing rods, golf clubs, filament wound rocket motor casings, pressure containers, and structural components of aircraft of all types (military, commercial, fixed-wing, and rotorcraft).

Aramid Fibre–Reinforced Polymer Composites

- The exceptional strength to weight ratios of aramid fibres make them more appealing than metals in many applications. Poly para phenylene terephthalamide is the chemical name for this class of compounds.
- Numerous aramid materials exist. Kevlar and Nomex are two brand names for these fabrics that are widely used. Mechanical properties vary over the range of Kevlar grades (Kevlar 29, 49, and 149). This image depicts the chemical structure of mer and the manner of chain alignment in aramids like Kevlar.
- When compared to other polymeric fibre materials, these fibres exhibit greater longitudinal tensile strengths and tensile moduli, but they are very weak when compressed.
- Toughness, impact resistance, and resistance to creep and fatigue failure are other hallmarks of this material.

Practice Exercise

1. The cermets are examples of:
(A) ceramic metal composites
(B) carbon–carbon composites
(C) metal matrix composites
(D) none of these

2. Whiskers are:
(A) very thin steel wires
(B) very thin copper wires
(C) very thin single crystals
(D) none of these

3. Consider the following statements.
I. Baking soda is used in fire extinguishers.
II. Quick lime is used in the manufacture of glass.
Which of the above statements is/are correct?
(A) I only
(B) II only
(C) Both I and II
(D) Neither I nor II

4. The matrix phase of fibrous composites may be:
(A) a metal only
(B) a polymer only
(C) a ceramic only
(D) a metal, polymer, or ceramic

5. For metal matrix composites the matrix is a:
(A) ductile metal
(B) polymer
(C) ceramic
(D) none of these

6. A laminar composite is composed of:
(A) one-dimensional sheets or panels
(B) two-dimensional sheets or panels
(C) three-dimensional sheets or panels
(D) none of these

7. Most advanced and promising composite is:
(A) carbon–carbon
(B) metal matrix
(C) polymer matrix
(D) none of these

8. Trade names for the two most common aramid materials are:
(A) silicon carbide, silicon nitride
(B) e-glass, aluminium oxide
(C) Kevlar, Nomex
(D) high strength steel, aluminium oxide

9. Mechanical properties of fibre reinforced composites depend on:
(A) properties of constituents
(B) interface strength
(C) fibre length, orientation, and volume fraction
(D) all the above

10. Not an example of laminar composites:
(A) wood
(B) bimetallic
(C) coatings/paints
(D) claddings

Answer Key

1. (A) **2.** (C) **3.** (C) **4.** (D) **5.** (A) **6.** (B) **7.** (A) **8.** (C) **9.** (D) **10.** (A)

10 Ferrous Metals

Introduction

- The ferrous metals are iron based and they include all varieties of irons and steels. Iron, cast iron, and steel are ferrous metals in which constituent iron is the major element. Pig iron is a crude and impure form of iron that is extracted from iron ore by melting the ore in a blast furnace. Wrought iron is the purest form of iron that contains 99.5–99.8 per cent of pure iron.
- As wrought iron does not flow like cast iron; hence, it is not suitable for casting. It has to be forged or rolled into the desired shape. Cast iron is obtained from pig iron by re-melting pig iron in a cupola furnace with a definite amount of limestone and steel scrap. Cast iron is highly suitable for casting in the foundry. The content of carbon in cast iron varies from two to 4.3 per cent. Carbon steels differ from cast iron in the amount of carbon content in it. Steel has carbon content varying from 0.5 to 1.5 per cent. Alloy steel is carbon steel to which one or more elements are added to improve its properties.

Pig Iron

- Iron is quite reactive and it is not found in a free state in nature. Iron is found in a combined state in the form of oxide, carbonate, and sulphide.
- In the combined state, iron is mined out from the earth. This is called iron ore that can be haematite (Fe_2O_3), magnetite (Fe_3O_4), limonite ($Fe_2O_3\ 3H_2O$), and siderite ($FeCO_3$).
- Most of the iron metal is produced from haematite. Haematite has iron with iron oxide and sand (silicon dioxide) as an impurity. Therefore, one has to carry out two things to get iron metal from the ore which, includes using the following.
  - A reducing agent to reduce haematite
  - A suitable flux (solvent) to remove the sand present in the ore
- The reducing agent used is carbon monoxide (CO), which is obtained by burning coke (carbon) in the blast furnace, whereas sand is removed by using calcium oxide (CaO) obtained by heating limestone. The reactions are as follow.

$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$
$2C + O_2 \rightarrow 2CO$
$CaCO_3 \rightarrow CaO + CO_2$
$CaO + SiO_2 \rightarrow CaSiO_3$(slag)

- The blast furnace is used for obtaining iron from the ore. The furnace is named a blast furnace as a very high temperature (1300°C) is generated inside it using forcing a blast of hot air and burning of coke. The molten iron is called pig iron when it is extracted from the blast furnace. Pig iron is a crude and impure form of iron (90 per cent iron).

Cast Iron

- The pig iron is unsuitable for the casting process as it contains a high percentage of impurities. To make it suitable for casting and other manufacturing processes, it is refined in a furnace known as a cupola. The refined iron with carbon content varying from 2 to 5 per cent is called cast iron.
- The carbon in cast iron is found in two forms that are given below.
  - Carbon or graphite
  - Combined form with iron, i.e., iron carbide (Fe_2C), which is called cementite
- Free carbon (graphite) imparts a grey colour to cast iron. It exists in carbon in the form of small flakes. It is comparatively softer and less brittle. However, cementite is a combined form of iron and carbon (iron carbide) and it is a very hard and brittle substance. Cementite makes cast iron unsuitable for machining jobs. Cementite is formed when cast iron is cooled rapidly after melting.

Grey Cast Iron

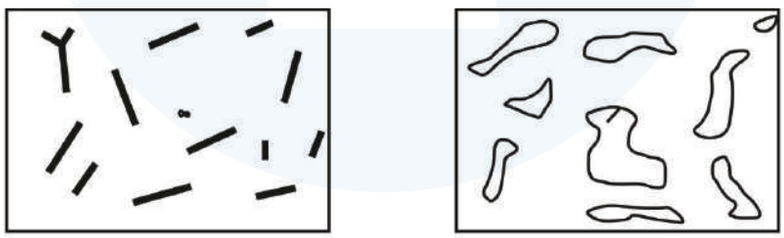
- Grey cast iron is a cast iron in which most of the carbon is present in the form of free

graphite. Graphite is greyish which provides greyish colour to the cast iron. Graphite exists in the form of small flakes. The graphite content varies from 2.5 to four per cent. The properties of grey cast iron are as follows.

- The presence of graphite provides a sort of natural lubrication. Hence, grey cast iron is suitable for sub applications where sliding friction is desired.
- It has high compressive strength but low tensile strength. High compressive strength makes cast iron highly suitable for making such parts, which are subjected mainly to compressive loadings, such as legs and bases of heavy machinery.
- It is extensive and brittle. Hence, it is unsuitable for parts that are subjected to sudden and impact loading.
- It has poor ductility and plasticity. Hence, it is unsuitable for forging and metal forming processes.
- It has high fluidity in liquid form, which makes it very suitable for casting processes.
- It has excellent damping properties. Hence, it is extensively used for making structures of machinery.
- Grey cast iron is mainly used for making legs and bases of heavy machinery, cast iron pipes, pipe fittings, manhole covers, cylinder blocks, and flywheels.

White Cast Iron

- White cast iron is cast iron in which carbon content is about 1.75–3.6 per cent and that too in the combined form of iron carbide (cementite). Due to the presence of cementite, the white cast iron shows white colour at fracture. The cementite is extremely hard and brittle (white cementite). The properties of cast iron are follows.
  - It is extremely hard and brittle due to the presence of cementite. Hence, it is unsuitable for machining.
  - It has high wear resistance and low fluidity. Hence, it is unsuitable for casting processes.
  - It has a great tensile but poor compressive strength.
- White cast iron is used in producing crushing equipment, cast wheels, brake blocks, rollers for the paper industry, hammers, and as raw material for the manufacturing of malleable cast iron.



(a) Grey cast iron [graphite flakes (black) and ferrite (white)]

(b) White cast iron [cementite (white) in pearlite (dark)]

Fig. 10.1

White Cast Iron	Grey Cast Iron
Carbon is present in a combined form called cementite.	Carbon is present in the free form as graphite.
It is obtained on rapid cooling of molten cast iron.	It is obtained by slow cooling of molten cast iron.
It has low fluidity and it cannot be used in foundry work.	It has high fluidity and is used for foundry work.

White Cast Iron	Grey Cast Iron
It is hard and brittle. It can be machined by grinding.	It can be easily machined.
It has high wear resistance and it is used for ball mills, brakes, follers, and hammers.	It has self lubricating properties and is used in applications subjected to sliding.
It has no damping capacity.	It has excellent damping capacity.

Table 10.1

Malleable Cast Iron

- White cast iron is generally used for producing malleable cast iron. Malleable cast iron is produced from white cast iron by converting cementite of white cast iron gradually into free graphite, i.e., $Fe_3C \rightarrow 3Fe + C$, and reducing the presence of graphite to about 2–2.5 per cent.
- The removal of carbon in malleable cast iron makes it more or less similar in composition to that of wrought iron. To achieve this, white cast iron is melted in a cupola, and then it is cast in sand moulds.
- These castings are kept with any material rich in oxygen at a temperature of about 800ºC–1000ºC in an annealing furnace. The liberated oxygen combines with the carbon of the castings of white cast iron, thereby reducing the proportion of carbon present in the casting.
- The castings so obtained are quite tough and they have high resistance to shock and impact loading. The properties of malleable cast iron are as follows.
  - It has high ductility and it can be easily bent without breaking.
  - It has high yield strength.
  - It has lower hardness and brittleness.
  - It has high toughness, good wear, and shock resistance.
  - It has excellent machinability due to its low hardness.
  - The malleable cast iron is used in making hardware, hinges, hand levers, spanners, gear wheels, surgical instruments, conveyor chains, and crankshafts.

Nodular Cast Iron

- The production of nodular cast iron involves the changing of graphite of cast iron from flake form to modular form by the addition of magnesium to the molten cast iron. The carbon content is the same as in grey cast iron (2.5–4 per cent), but there is lower brittleness graphite is present in modular form, thereby no sharp point or condition of stress raiser is existing.
- The properties of nodular cast iron are follows.
  - It has high fluidity and it is castable.
  - It has graphite that helps in providing high machinability and high surface finish.
  - It can be easily welded and galvanised.
  - It can be used as a replacement to great extent for both malleable cast iron and steel.
  - It has high toughness and ductility.
- It is commonly used in pipes, pipe fittings, the housing of gears, railways and locomotive rolling stocks, lifting jack blocks, flywheels, gears, hand tools, and bodies of electric motors.

Alloy Cast Iron

- A large variety of alloy cast irons have been developed by the suitable addition of varying proportions of different alloying elements in cast iron. Alloy cast irons are developed largely to overcome the inherent drawbacks of cast iron. The main alloying element in most of these alloy cast irons is nickel.
- The effects of alloying various elements with cast iron are given below.
  - **Nickel:** it improves the carbon structure present in cast iron, thereby improving physical properties and machinability.
  - **Silicon:** it promotes the formation of free graphite and decreases the combined

carbon (cementite). Hence, silicon is used as a softener in cast iron.
    - **Chromium:** it acts as a carbide stabiliser, thereby improving hardness, strength, and corrosion resistance.
    - **Copper:** it promotes graphite formation, thereby improving machinability, castability, and toughness.
    - **Sulphur:** it promotes the formation of cementite. Hence, it increases hardness and brittleness but decreases fluidity.
    - **Phosphorus:** it improves the castability of cast iron.
    - **Molybdenum:** it improves machinability, toughness, and fatigue strength.
    - **Manganese:** it reduces the effect of sulphur on cast iron, thereby reducing hardness and brittleness.
- Alloy cast irons are used in the cylinder head, piston rings, grinding, and crushing machinery.

Wrought Iron

- Wrought iron is the purest form of iron which contains 99.5–99.8 per cent of pure iron, it cannot flow like cast iron when melted at high temperature and hence it is unsuitable for the casting process; however, it has high plasticity at high temperatures, thereby enabling it to be hammered into a thin sheet, or two of its pieces can be joined by hammering. It is capable to be forged or rolled into desired shapes when it is red hot.
- Wrought iron cannot be hardened by heating and quenching as it does not have carbon content. It is very ductile and malleable and hence it is suitable for making sheets and wire drawings.
- The properties of wrought iron are as follows.
    - It has high resistance to corrosion, ductility, and malleability.
    - Its ultimate strength can be increased by cold working.
    - It is never cast and all manufacturing processes from this material are accomplished by hammering, pressing, and rolling. It has high resistance to fatigue failure.
- The maximum applications of wrought iron are for forged articles, especially are made by forge welding and parts which are likely to be subjected to conditions encouraging rusting.
- Wrought iron is commonly used in shipbuilding, pipes for steam, oil, and water, nuts and bolts, agriculture implements, crane chokes, and railway couplings.

Steel

- Steel differs from cast iron in the amount of carbon content in it. Secondly, it is not only the quality of carbon content but also the form in which it is present in metals. Steel has a carbon content of up to 1.5 per cent and it is completely in the combined form. As steel is an alloy of iron with a small amount of carbon, it can be produced by either extracting carbon from cast iron or adding a small amount of carbon to wrought iron.
- Generally speaking, the higher the carbon content in steel (not, above 1.5 per cent), the harder and stronger the steel. In fact, steel is an alloy of iron and carbon with a carbon content not exceeding 1.5 per cent. These steels having 1.5 per cent carbon are called plain carbon steels because their properties are dependent on the percentage of their carbon content.
- The plain steels can be classified as follow.
    - Dead mild steels with carbon content less than 0.15 per cent
    - Mild steel with carbon content varies from 0.15 to 0.30 per cent
    - Medium carbon steels with carbon content varying from 0.30 to 0.80 per cent
    - High carbon steel with carbon content varying from 0.80 to 1.5 per cent

Steel	Cast Iron
Carbon content is low (up to 1.5 per cent) and exists in a combined form.	Carbon content is high (1.5-4.5 per cent) and exists in the free form of graphite.
Besides carbon, other alloying elements are also generally present.	Generally, no alloying element is present except carbon.
It has a high value of strength, hardness, and ductility.	It is hard and brittle.
It is commonly used for forging, rolling, and welding in making parts.	It is commonly used for casting in making parts.
The structure is ferrite and pearlite.	The structure is cementite and ludeburite.
Bessemer converter is used with pig iron to manufacture.	A cupola is used with pig iron to manufacture.

Table 10.2 Comparison between Steel with Cast Iron

Alloy Steel

- Alloy steel is carbon steel to which one, or more elements are added to improve its properties. The commonly added elements to improve the properties are nickel, chromium, molybdenum, tungsten, manganese, vanadium, copper, aluminium, boron, silicon, and cobalt.
- The elements are added to the alloyed steel for the following reasons.
  - To improve tensile strength while maintaining ductility
  - To improve toughness, resistance to corrosion, wear resistance, and hardenability
  - To improve cutting ability as well as the ability to resist distortion at elevated temperatures
  - To improve case hardening properties and promote fine grain size

Effect of Alloying Elements on Carbon Steels

- The various alloying elements that are added to improve the properties of steel are given below.
  - Nickel: it improves, toughness, tensile strength, ductility, and corrosion resistance.
  - Chromium: it improves tensile strength, thermal resistance, hardness, toughness, and magnetic properties.
  - Cobalt: it improves tensile strength, thermal resistance, hardness, toughness, and magnetic properties.
  - Manganese: it improves strength, toughness, and hardness. It minimises the bad effects of sulphur.
  - Silicon: it improves elasticity and magnetic permeability. It also decreases hysteresis losses.
  - Molybdenum: it increases wear resistance, thermal resistance, and hardness.
  - Tungsten: it improves toughness, hardness, shock resistance, wear resistance, and red hot hardness.
  - Vanadium: it improves tensile strength, elastic limit, density, and shock resistance.
  - Niobium: it improves ductility and impact strength. It promotes fine grain growth and decreases hardenability.

Special Alloys Steels

- Special alloy steels have been developed for special applications and specific requirements by modifying the properties of the steels.

- The special alloy steels are given below.
  - Stainless steel
  - Magnetic steels
  - High resistance steels
  - High speed steels

Stainless Steels

- Stainless steels are also called corrosion resistant steels. The main alloying element in this steel is chromium while some other elements like nickel and manganese can also be added in small amounts. About 12 per cent chromium metal is added to the steel, which forms chromium oxide with atmospheric oxygen. The chromium oxide is highly corrosion resistant and it protects the items made of this steel.
- Stainless steel is used in utensils, cutlery, surgical instruments, and furnace parts

High Speed Steels

- These steels are meant for the manufacturing of cutting tools. As the amount of heat generated during metal cutting is very high, these tools are required to retain their hardiness even at elevated temperatures.
- When tools fail to retain their hardness at elevated temperatures, they tend to become soft and fail to perform any cutting. This is the reason why high carbon steels are not used at high cutting speeds and this has necessitated the need for special alloy steels called high speed steels. The property of retaining hardness and toughness at elevated temperatures is called red hot hardness.
- High speed steel has tungsten 18 per cent, chromium four per cent, and vanadium one per cent.
- High speed steel with an 18:4:1 composition is the most commonly used tool material as it high hardness, corrosion resistance, and toughness at a high temperature.

High Temperatures Resisting Alloys

These are also called superalloys. Superalloys have high nickel based steels, which have high red hot hardness.

Magnetic Steels

Carbon, chromium, tungsten, and cobalt steels are used for permanent magnets as these steels have high magnetic retentive power. A typical alloy may have 60 per cent iron, 20 per cent nickel, 8 per cent cobalt, and 20 per cent aluminium.

Cryogenic Steels

Cryogenic steels are nickel based steels. Cryogenic steels are used to store and transport liquefied gases below the temperature of –157.5°C.

To serve this purpose, various nickel based cryogenic steels are used. As the nickel percentage increases, the nickel based steels can handle gasified liquid at a lower temperature. For example, 5 per cent nickel based steel can work suitably at –120°C while nine per cent nickel is suitable at –190°C.

Phases of Iron

α-Iron or α-Ferrite

- It is the purest form of iron at room temperature.
- It has a BCC structure. It is magnetic below 768°C and magnetic properties disappear and become para magnetic above 768°C.
- α-Ferrite is a solid solution of iron and carbon with a maximum solubility of carbon 0.02 per cent in α-iron.

Austenite or γ-Iron

γ-Iron has an FCC structure. It is a soft, tough, and ductile form of iron. It is para magnetic. The maximum solubility of carbon in γ-iron is 2.10 per cent.

δ-Iron or δ-Ferrite

It has a BCC structure. The solubility of carbon in δ-ferrite is less but it is more than a-ferrite. The heat treatment process is not used for δ-ferrite.

Pearlite

It has a plate like structure of α-ferrite and cementite. For pure pearlite form per cent of carbon must be equal to 0.83 per cent in

α-ferrite (hypo eutectoid steel has less than 0.83 per cent of carbon and hyper eutectoid steel has carbon per cent in between 0.83 and 2.1 per cent). It is relatively hard and stronger than α-ferrite.

Cementite or Iron Carbide

It is very hard and brittle. It has 6.67 per cent of carbon in iron in its purest form. The chemical formula of cementite is Fe_3C. Cementite has an orthorhombic structure. It is very strong in compression and weak in tension.

Practice Exercise

1. Mild steel belongs to the following category.
(A) Low carbon steel
(B) Medium carbon steel
(C) High carbon steel
(D) Alloy steel

2. The ultimate tensile strength of low carbon steel by working at a high strain rate will:
(A) decrease
(B) increase
(C) remain constant
(D) first, increase and then decrease

3. Mild steel is an alloy of iron and carbon with a percentage of carbon ranging from:
(A) up to 0.2 per cent
(B) 0.15–0.3 per cent
(C) 0.3–0.5 per cent
(D) above 0.5 per cent

4. The object of 'normalising' a steel specimen is:
(A) to reduce hardness
(B) to relieve stresses
(C) to refine structure
(D) to improve ductility

5. The stainless steels owe its resistance to the presence of:
(A) carbon
(B) sulphur
(C) manganese
(D) chromium

6. Which of the following pairs regarding the effect of alloying elements in steel are correctly matched?
I. Molybdenum: forms abrasion resisting particles.
II. Phosphorous: improves machinability in free cutting steels.
III. Cobalt: contributes to red hardness by hardening ferrite.
IV. Silicon reduces oxidation resistance.
Which of these statements are correct?
(A) II, III, and IV
(B) I, III, and IV
(C) I, II, and III
(D) I, II, and IV

7. Which of the following is the purest form of iron?
(A) Pig iron
(B) Cast-iron
(C) Wrought iron
(D) Steel

8. Assertion (A): cast iron is generally hard, brittle, and wear resistant.
Reason (R): cast iron contains more than 20 per cent carbon and as such the percentage of cementite in it is higher.
(A) Both A and R are true and R is the correct explanation of A.
(B) Both A and R are true but R is not a correct explanation of A.
(C) A is true but R is false.
(D) A is false but R is true.

9. Statement I: manganese is always added to steels as since it combines with the sulphur content to form manganese sulphide.
Statement II: if manganese is not added, iron sulphide which is not harmful to steel, would form.
(A) Both Statement I and Statement II are individually true and Statement II is the correct explanation of Statement I.

(B) Both Statement I and Statement II are individually true but Statement II is not the correct explanation of Statement I.

(C) Statement I is true but Statement II is false.

(D) Statement I is false but Statement II is true.

Answer Key

1. (A) **2.** (B) **3.** (B) **4.** (C) **5.** (D) **6.** (C) **7.** (C) **8.** (C) **9.** (C)

11 Non-ferrous Metals and Alloys

Introduction

- Non-ferrous metal is any metal including alloys that do not contain iron. Non-ferrous metals are more expensive than ferrous metals as they are not abundantly available. However, they have many desirable properties such as low weight, high conductivity, non magnetic properties, low melting points, and resistance to corrosion.
- Important non-ferrous metals include aluminium, lead, copper, zinc, nickel, tin, titanium, and alloys, such as brass, bronze, duralumin, and hindalium.

Aluminium

- Aluminium is abundant on the earth, but it is not found in a free state. It is obtained in pure form by processing its ore called bauxite, which is a mixture of monohydrate and trihydrate aluminium oxide $Al_2O_3 \cdot H_2O$ and $Al_2O_3 \cdot 3H_2O$.
- Bauxite is first purified and then dissolved in fused cryolite (double fluoride of aluminium and sodium). Thereafter, aluminium is separated from the solution by electrolysis at 910°C.
- The physical and mechanical properties of aluminium are as follows: it has silvery colour and lustre with a melting point of 658°C, a lightweight metal having a specific gravity of 2.7.
- It is a ductile metal, thereby it can be drawn into thin wires. It is extensively used in cables and conductors.
- Its ultimate strength is sufficiently high (50–70 MPa) and an excellent conductor of heat and electricity.
- It forms a protective layer of aluminium oxide (Al_2O_3) when exposed to the atmosphere. This protective layer makes it corrosion resistant.
- Pure aluminium is soft and it can be made hard when it is alloyed with elements like Ni, Mg, Si, and Co.
- Aluminium hardened by cold working can be amended by heating it to about 350°C and then cooling it in air.
- Based on characteristics the uses of aluminium are given below.
  - **Packaging:** one of the most common uses of aluminium is packaging including drinks cans foil wrappings, bottle tops, and foil containers. Aluminium's natural resistance to corrosion aids it in its role in packaging. It is completely impermeable and also it does not let out the aroma of the packed food.
  - **Strength to weight ratio:** aluminium has unbeatable strength to weight ratio. It can be used for bodies of trains, boats, and cars due to its lightweight property.
  - **Corrosion resistance:** there is no need to paint aluminium as it is corrosion resistant, thereby saving the requirement of painting the outer surface of planes. Aluminium is also used in the cladding of walls of buildings.
  - **Electrical conductivity:** weight is also important in aluminium's use despite its lower electrical conductivity, as compared to copper (63 per cent). The tact is that 1 kg of aluminium can conduct almost twice as much electricity as 1 kg of copper. This is the reason why aluminium is being used in high voltage electrical transmission in place of copper as it is the most cost-efficient power line material. With copper, many heavy and expensive support structures (pylons) have to be used while with aluminium, fewer lighter, and cheaper supports are needed. Aluminium is also more ductile than copper so it is easier to draw it into wires to produce aluminium power lines. Aluminium also has a better property of corrosion resistance. Hence, aluminium is a perfect choice for long distance electricity distribution.
  - **Low melting point:** it helps aluminium to be easily melted for die casting.
  - **Kitchen utensils:** aluminium alloy is ductile and corrosion resistant. It has about one third the cost of stainless steel. Hence,

it is extensively used for manufacturing utensils in place of stainless steel.

- **Pigment in paint:** aluminium paints are used for providing an effective protective coating.

Limitation of Aluminium in Power Cables

- Copper cables carrying the electricity are buried in the ground while aluminium transmission lines or conductors are suspended from pylons.
- The aluminium has a low density and it can safely be suspended from inexpensive pylons. Pylons have the disadvantage that they look ugly on the landscaped but have the advantage of easy access to the transmission lines for maintenance and repair.

Aluminium Alloys

- Aluminium can be alloyed with elements, such as copper, silicon, magnesium, zinc, manganese, and nickel to improve its tensile strength and hardness.
- Copper is the main hardening element, which is added to aluminium while magnesium in small quantities can improve further the hardness and strength of the copper–aluminium alloy. Strength can also be improved with the addition of a small amount of manganese and nickel in the copper–aluminium alloy. Manganese acts as a strengthening element, while silicon if added can improve the hardness as it forms magnesium silicide.

Duralumin

- It is an age hard enable aluminium alloy. It is based on the discovery that all aluminium alloys containing about four per cent copper would slowly harden when left at room temperature for several days.
- The main constituents of duralumin are aluminium 93.5 per cent, copper 4.4 per cent, magnesium 1.5 per cent, and manganese 0.6 per cent.
- Its first and main purpose was to use in rigid aircraft frames. Before final heat treatment, the alloy is ductile and malleable. After heat treatment (quenching), increased hardness and strength are achieved when it is left at room temperature for several days.
- It can be cast, forged, and stamped as it is ductile and malleable before heat treatment.
- It has high hardness and strength when it is heat treated after fabrication and left at room temperature for several days. The strength of duralumin is equal to that of mild steel. But density is one third of steel.
- It has high electrical conductivity, a high strength to weight ratio, and is non magnetic.
- It is widely used in structures of aircraft and other machinery and parts of automobiles as it has a high strength to weight ratio, surgical, and orthopaedic work as it is non magnetic and instruments needing non magnetic properties.

Y alloy

- It is an aluminium alloy containing nickel elements. It has been developed as an aluminium alloy, which can retain its strength at high temperatures as strength at high temperature is necessary for the pistons of engines.
- Y alloy has a composition of aluminium 92.5 per cent, copper four per cent, nickel two per cent, and magnesium 1.5 per cent.
- The strength and hardness like duralumin are achieved in it by age hardening, that is, heating to a temperature of 500ºC–520ºC and then quenching in boiling water. The ageing process is carried out at room temperature for about five days.
- It is extensively used for components, which have to retain strength at high temperatures such as pistons, cylinder heads, and crankcases of IC engines.
- It can be die-cast to manufacture components, such as piston rods and sparking chisels.

Hindalium

- It is an alloy of aluminium, magnesium, manganese, and silicon and it is produced by Hindustan Aluminium Corporation in India.
- It is manufactured in the form of sheets of 16 gauge, which are further mainly used

for manufacturing utensils as the cost of hindalium is one third of stainless steel.

- The properties of hindalium are strong, hard, non reactive with food acids, good surface finish, and are not easily scratchable.
- Its cost is lower than stainless steel. It is mainly used for cooking utensils.

Magnalium

- It is an alloy of aluminium with the composition of aluminium 95 per cent, magnesium 1–5.5 per cent, copper 0–2.5 per cent, nickel 0–1.2 per cent, manganese 0.2–0.6 per cent, and tin 0–3 per cent.
- It is a lightweight alloy, which is highly ductile (elongation 30 per cent), has poor castability and excellent machinability.
- Magnalium is used in gearbox housing, door handles, luggage racks, grinder parts, and fixtures.

Copper

- Copper is not found in pure form, but it is found in a mineral form, such as copper glance (Cu_2S), copper pyrites (CuFeS), malachite ($CuCO_3$, CuO_2H_2), and azurite ($2CuCO_3$. CuO_2H_2).
- Copper is obtained from ores in the 'blister stage,' which is 99 per cent pure. It is further refined in a furnace through an oxidation process to remove sulphur and other impurities. The excess oxygen is removed by the 'poling' process. The copper is now called 'tough pitch copper.'
- The physical and mechanical properties of copper are follows: it is a reddish brown metal with a melting point of about 1084ºC.
- It has a specific gravity of 8.9, i.e., denser than iron with a tensile strength of about 300–470 MPa which makes it a strong material.
- It is a ductile metal having a ductility of more than 15 per cent. It can be easily drawn into wires. Hence, it is useful for making cables, strands and conductors.
- It is one of the best conductors of heat and electricity. However, conducting is affected by impurities.
- It can be easily joined by brazing (a kind of welding). It forms a protective layer of copper oxide (CuO) when exposed to the atmosphere. This layer is highly corrosion resistant.

Brasses

- Brasses are alloys of copper and zinc with the content of zine varying from five to 45 per cent.
- The important characteristics and uses of brass are as follows.
  - Malleability and acoustic properties: these properties have made brass the metal of choice for musical instruments.
  - Low melting point and flow characteristics: the brass has a relatively low melting point of about 900ºC–940ºC as well as, excellent flow characteristics. Hence, these characteristics make it a highly suitable material, which can be easily cast.
  - Bright gold like appearance: brass has a bright gold like appearance. Therefore, it is used for decoration purposes.
  - Low friction: it has low friction and it is used for applications where low friction is required such as locks, gears, bearings, doorknobs, ammunition cartridges, and valves.
  - Machinability: brasses set the standard by which the machinability of other materials is judged.
  - Thermal conductivity: this property makes brass metal of choice for the making of heat exchangers.
  - Anti-microbial: brass plays a leading role in the fight against hospital acquired infections.
  - Non sparking: no spark is generated when it strikes any surface, thereby it is used where ammunition and explosives are stored.
- Different types of Brass are available based on the alloying element added.
  - **Manganese brass:** manganese (0.75–2.5 per cent) is added as deoxidiser and also to harden and strengthen the alloy.
  - **Iron brass:** iron (up to three per cent) is added to make brass hard, strong, and tough.
  - **Tin brass:** tin (up to two per cent) is added to increase hardness, strength, and

corrosion resistance. Tin also increases the tenacity of the alloy.

- **Lead brass:** lead is added to soften the brass, thereby increasing the machinability of brass. Lead is added about one-four per cent in the brass.
- **Aluminium brass:** aluminium is added up to 3 per cent. It decreases the ductility of brass. The alloy can be cast, forged, or rolled. This type of brass has also corrosion resistance.

Bronze

- Bronze is a metal alloy consisting primarily of copper with tin as the main additive; it generally contains copper (75–95 per cent) and tin (5–25 per cent).
- The tensile strength of bronze increases gradually with the content of tin, reaching a maximum of about 20 per cent of tin. Bronze is most ductile when it contains about five per cent of tin. The ductility gradually decreases and ductility almost disappears with a tin content of about 20 per cent.
- The bronze with 8–11 per cent of tin has both strength and toughness and these bronzes are called gun metal as guns were manufactured from these bronzes.

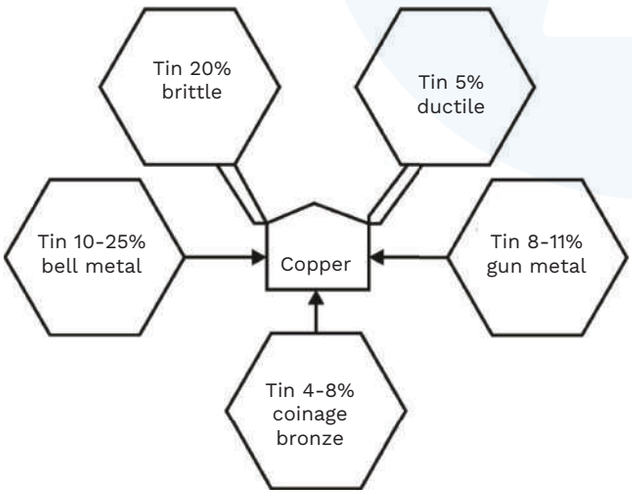


Fig. 11.1

- The bronzes containing four to eight of tin are called 'coinage bronze' as these bronzes were used for making coins.
- The bronzes containing 15–25 per cent of tin are called 'bell metals' as these metals are sonorous and used in making bells.
- Bronze is less brittle than iron and it forms an oxide layer to protect the underlying metal. It can be easily cast as it has a lower melting point.
- Bronze has better fatigue resistance and it resists corrosion under seawater. It conducts better heat and electricity.
- The properties of bronzes are modified with different alloying elements as given below.
  - **Zinc bronze:** the addition of zinc in bronze increases its fluidity in a molten state, thereby preventing the occurrence of gas flaws. It consists of 88 per cent copper, 8–10 per cent tin, and 2–4 per cent tin.
  - **Phosphore bronze:** the addition of phosphore (0.1–0.3 per cent) increases the qualities of tensile strength, elasticity, and fatigue resistance. It can be forged, drawn, cold rolled, and cast. It is used for making propeller blades as it has improved corrosion resistance in seawater.
  - **Silicon bronze:** it contains 96 per cent copper, three per cent silicon, 0.5 per cent iron, and 0.5 per cent manganese. It is used for making screws, bolts, pumps, boilers, and pipes.
  - **Aluminium bronze:** it contains three per cent aluminium, 88 per cent copper, three per cent iron and 0.5 per cent tin. It is highly corrosion resistant and possesses good strength. It is used in the making of heat exchangers, rollers, gears, etc.
- **Beryllium bronze:** it contains about two per cent beryllium. It possesses high yield strength and fatigue endurance. It is used for making bearings, springs, etc.

Brass	Bronze
It is an alloy of copper and zinc.	It is an alloy of copper and tin.
It is soft, ductile, and malleable and does not resist corrosion.	It is hard and brittle. Hardness depends upon the content of tin. More tin more hardness. Gun metal is a type of bronze used for making guns. Resists corrosion, especially from seawater. Used in boat and ship fittings.
Bright gold like appearance and used for decorative purposes.	Reddish brown in colour.
Lowering point (900ºC) and can be easily cast.	Melts at 950ºC but melting depends upon the content of tin.
It has good acoustic properties and it is used for making trumpets and homes.	It is sonorous and it is used for making bells.
It has less friction and is used for applications where low friction is needed, such as locks, gears, bearings, doorknobs, ammunition (cartridges), and valves.	Widely used for cast moulding of sculptures as it expands on the setting, thereby it can fill up the intricate shape of the mould. It is also used for springs, bearings, bushings, and so on.
It does not produce a spark when it strikes a hard surface. Used where ammunition and explosives and stored.	More fatigue resistance. It is also a better conductor of heat and electricity.

Table 11.1 Differences between Brass and Bronze

Lead

- Lead is a non-ferrous metal. It is a bluish grey metal, which has a high metallic lustre when freshly cut. The main ore is lead sulphide (PbS), which is called galena.
- The properties and uses of lead are given below.
  - It is very soft and highly malleable. It is used for making flexible pipes, which are used in the drainage system of households and industry. It is also used as a sheath or covering for electric cables.
  - It is poisonous, which makes its applications restricted and low melting point of about 327°C. Hence, the metal can be easily welded, soldered, and used in die casting.
  - It has electropositive characteristics. Hence, it is widely used in storage batteries.
  - It has good corrosion resistance. Hence, the lead lining is used to prevent any corrosion of other metals. Lead sheets are used for roofs and gutters as they are corrosion resistant. It is also used as pigments in paints.
  - It has a high density. Hence, it is used as explosive balls in cannons.

Tin

- It is a brightly shining white metal. It is extracted from an ore such as tin stone (SnO_2). The ore is melted in a furnace using anthracite coal and sand to obtain crude tin.
- The crude tin is then refined in a reverberatory furnace. The electrolysis process is used for obtaining pure tin from crude tin.
- The properties and uses of tin are given below.
  - It is highly soft and malleable. It can be easily hammered into tin foil. It is, therefore, used for moisture proof packing.
  - It has high corrosion resistance characteristics. Its coating is, therefore, used on other materials for providing corrosion resistance.
  - It is highly workable and it can be, therefore, rolled into very thin sheets.
  - It has a low melting point of 232ºC. Hence, it is used in making cable jointing and solder.

- It is used for alloying with copper to make bronze and bell metals.

Nickel

- It is a silvery white metal having a good response to polish. The main source of nickel is iron sulphide which contains about three per cent of nickel. The ore is roasted with limestone and quartz. The above mixture is then melted in a blast furnace.
- The mixture is treated in a Bessemer converter and it is afterwards treated with sulphuric acid to obtain nickel.
- The properties and uses of nickel are given below.
  - Its melting point is 1452ºC and it, therefore, provides good heat resistance.
  - It has good corrosion resistance. Hence, it is used in the making of cooking utensils, laundry, dairy machinery, and plating on iron and brass as a protective coating.
  - It is paramagnetic at normal temperature. It is used for its high potential field, high permeability, and coercive or retentive power.
  - It is generally used as an alloying element in some types of steel and cast iron, monel metal, permalloy, cupro nickel, and nickel silver.
  - It is almost as hard as steel, quite malleable, lesser ductile than soft steel but the addition of magnesium improves its ductility.

Magnesium

- It is an alkaline earth metal and it is an abundant element in the earth's crust and seawater. The free metal burns with characteristics of brilliant white light, making it a useful ingredient in flares. The metal is obtained by the electrolysis of magnesium salts.
- The chief use of magnesium is as an alloying agent in aluminium, thereby making alloys have relative superior lightness and strength.
- The physical and mechanical properties of magnesium are given below.
  - It is the lightest metal, harder than aluminium.
  - It can take high polish on buffing, is easily machined, and ignites easily.
  - Magnesium is used in automotive and truck components due to its low weight, in mobile phones, laptops, computers, cameras, and other components as it has low weight, and good mechanical, and electrical properties.
  - It is one of the main aerospace construction metals, used with aluminium for making beverage cans. It is used with zinc to produce photo engraving plates and dry cell battery cases.

Titanium

- It is a strong, lustrous, corrosion resistant (even in seawater) transition metal with silvery colour.
- The properties and uses of titanium are given below.
  - It has high corrosion resistance and can produce strong lightweight alloys for aerospace jet engines, missiles, spacecraft, military applications, automotive, medical prostheses, orthopaedic implants, dental and endodontic instruments, jewellery, mobile phones, and industrial process (pump and paper).
  - It has the highest strength to weight ratio. It is 45 per cent lighter than steel for the same strength.
  - It is a fairly hard, non magnetic, poor conductor of heat and electricity used as an alloying element in steel, copper, vanadium, and molybdenum.

Tungsten

- Tungsten is a transition metal which is hard. It has also a high density comparable to uranium and gold but higher than lead.
- The properties and uses of tungsten are given below.
  - It is a hard steel grey metal. Sintering is a common method to produce its components.
  - It has the highest melting point of 3422ºC, highest tensile strength, and lowest coefficient of thermal expansion.
  - The toughness of steel can be increased by alloying with a small quantity of tungsten mainly used for making tungsten carbide and alloyed steels.

Practice Exercise

1. Copper is used for making electrical conductors because it is:
(A) ductile
(B) resists corrosion
(C) has low resistance
(D) cheap

2. Brass is an alloy of:
(A) copper and zinc
(B) tin and zinc
(C) copper and tin
(D) copper and Al

3. A small amount of phosphorous is present in:
(A) all bronzes
(B) phosphor bronze
(C) tin bronze
(D) beryllium bronze

4. Aluminium alloys find use in the aircraft industry because of:
(A) high strength
(B) low specific gravity
(C) good corrosion resistance
(D) good weldability

5. Match List I (alloying element in steel) with List II (property conferred on steel by the element) and select the correct answer using the codes given below.

List I	List II
A. Nickel	I. Corrosion resistance
B. Chromium	II. Magnetic permeability
C. Tungsten	III. Heat resistance
D. Silicon	IV. Hardenability

Codes

	A	B	C	D
(A)	IV	I	III	II
(B)	IV	I	II	III
(C)	I	IV	III	II
(D)	I	IV	II	III

6. Lead is widely used in:
(A) transformers
(B) switchgear
(C) galvanised pipes
(D) batteries

7. Gunmetal, which is used in journal bearings contains the following.
(A) 88% Cu, 10% Sn, 2% Zn
(B) 80% Cu, 10% Zn, 10% Al
(C) 85% Cu, 5% Mg, 10% Al
(D) 85% Cu, 5% Sn, 10% Pb

8. Monel metal is an alloy of:
(A) iron and carbon
(B) copper and zinc
(C) aluminium and copper
(D) copper and nickel

9. Which of the following elements given below determine(s) the maximum attainable, hardness in steel?
I. Chromium
II. Manganese
III. Carbon
IV. Molybdenum
Select the correct answer using the code given below.
(A) I only
(B) I and II
(C) III only
(D) II and IV

10. Match List I (alloy) with List II (major constituents) and select the correct answer using the codes given below the lists.

List I	List II
A. Babbit	I. Nickel
B. Invar	II. Tin and lead
C. Gun metal	III. Aluminium
D. Duralumin	IV. Copper

Codes

	A	B	C	D
(A)	II	IV	I	III
(B)	III	I	IV	II
(C)	II	I	IV	III
(D)	III	IV	I	II

Answer Key

1. (C) **2.** (A) **3.** (A) **4.** (B) **5.** (A) **6.** (D) **7.** (A) **8.** (D) **9.** (D) **10.** (C)